ECO2011 MIDTERM REVIEW

Microeconomics

Choices, markets, and welfare

Victor Chen 陈绪成

October 2026

The review at a glance

01 Principles and economic models
Scarcity, decisions, and the economist's toolkit

02 Comparative advantage
Opportunity cost, specialization, and trade

03 Supply and demand
Equilibrium and changes in markets

04 Elasticity
Responsiveness and total revenue

05 Welfare
Consumer surplus, producer surplus, and efficiency

06 Price controls and output policies
Ceilings, floors, supports, and quotas

07 Taxes and subsidies
Price wedges, incidence, and welfare

08 International trade
Imports, tariffs, and quota rents

01
Principles and models

How people make decisions, how they interact, and how the economy works as a whole

Ten principles: the framework

Our route. The highlighted item is our current focus.

1 **How people make decisions** →
P1 trade-offs; P2 opportunity cost; P3 rationality and marginal changes; P4 incentives

2 **How people interact**
P5 trade; P6 markets; P7 government

3 **How the economy as a whole works**
P8 productivity; P9 money and inflation; P10 the short-run trade-off

Scarcity and economics

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2. **Economics**
   The study of how society manages its scarce resources.
3. **Three levels of decisions**
   Households: what to buy, how much to work, save, and spend.
   Firms: how much to produce and how many workers to hire.
   Society: how to divide resources between national defense, consumer goods, protecting the environment, and other needs.

Principle 1 People face trade-offs

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To get something that we like, we have to give up something else that we also like.

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   Society gets the most from its scarce resources. Think about the size of the economic pie.

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3. **Equality**
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4. **The trade-off**
   Redistributing income from wealthy to poor can achieve greater equality, but reduced incentives to work and produce can shrink the economic pie.

Principle 2: The cost of something is what you give up to get it

1 **Making decisions**
Compare costs with benefits of alternatives.

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Ask what you give up, rather than looking only at money paid.

Principle 3 Rational people think at the margin

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For the additional action, compare marginal benefit with marginal cost.

Principle 4 People respond to incentives

1 **Incentive**
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2. **Why behavior changes**
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3. **Apply the principle**
   When considering a policy, ask how it changes people's incentives and therefore their decisions.

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Principle 5 Trade can make everyone better off

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Comparative advantage explains where these gains come from.

Principle 6: Markets are usually a good way to organize economic activity

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3. **Interactions**
   In a market economy, these decisions result from the interactions of many households and firms.

Prices and the invisible hand

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The interaction of buyers and sellers determines prices.

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3. **The invisible hand**
   Prices guide self-interested households and firms to make decisions that promote general economic well-being.

Principle 7: Governments can sometimes improve market outcomes

1. **Enforce property rights**
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   A market failure means that the market fails to allocate resources efficiently. The two causes here are externalities and market power.
3. **Alter market outcomes to promote equity**
   Tax or welfare policies can change how the economic pie is divided when the distribution of well-being is not desirable.

Market failure: the two causes in the notes

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   Production or consumption of a good affects bystanders.

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3. **Identify the role of government**
   Public highways: promote efficiency through wider benefits.
   Patent laws: enforce property rights.

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Distinguish output per unit of labor from the number of labor units or competition from abroad.

Principle 9: Prices rise when the government prints too much money

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The long-run time horizon is part of this principle. Short-run movements are considered separately.

Policy and overall demand

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A sufficiently strong contraction can contribute to deflation, a decline in the overall price level. The outcome also depends on production, expectations, and other shocks.

Principle 10: Society faces a short-run trade-off between inflation and unemployment

1. **The short-run horizon**
   Over a period of a year or two, many economic policies push inflation and unemployment in opposite directions.

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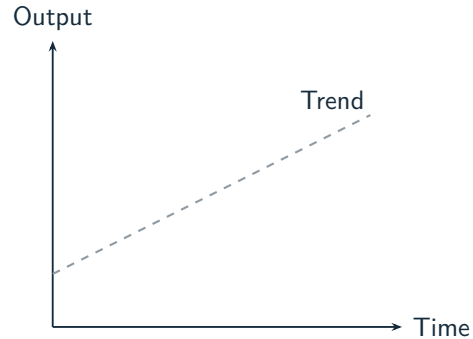
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   Other factors can make this trade-off more or less favorable.

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1. **The short-run horizon**
   Over a period of a year or two, many economic policies push inflation and unemployment in opposite directions.
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   Other factors can make this trade-off more or less favorable.
3. **Interpret the principle**
   Distinguish this short-run trade-off from Principle 9's long-run explanation of inflation.

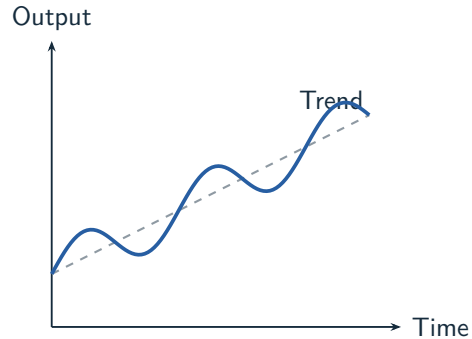
Business cycles

1. **The longer-run path**
   Economic activity has a longer-run growth path.



Business cycles

1. **The longer-run path**
   Economic activity has a longer-run growth path.
2. **Fluctuations**
   Economic activity fluctuates around that path: expansions and contractions.



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Framework complete. Connect the parts before moving on.

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Thinking like an economist: the framework

Our route. The highlighted item is our current focus.

1. **Economists as scientists** →
   Scientific method, assumptions, models, microeconomics and macroeconomics
2. **Economists as policy advisers**
   Positive and normative statements
3. **Why economists disagree**
   Different positive theories and different values

Economists play two roles

1 **Scientists**
Try to explain the world.

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Explaining what happens and recommending what should happen are different tasks.

Scientific method, assumptions, and models

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   Simplify the complex world and make it easier to understand.

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   Dispassionate development and testing of theories about how the world works.
2. **Assumptions**
   Simplify the complex world and make it easier to understand.
3. **Models**
   A model is a highly simplified representation of a more complicated reality. Economists use models to study economic issues.

Microeconomics and macroeconomics

1 **Microeconomics**
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Microeconomics and macroeconomics

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2. **Macroeconomics**
   The study of economy-wide phenomena, including inflation, unemployment, and economic growth.

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Positive and normative statements

1. **Positive statements**
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   Example: a binding ceiling creates a shortage in a competitive market.

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1. **Positive statements**
   Describe how the world is. Evidence can in principle confirm or refute them.
   Example: a binding ceiling creates a shortage in a competitive market.
2. **Normative statements**
   Describe how the world ought to be. They include value judgments and cannot be confirmed or refuted by data alone.
   Example: the government should prioritize equality over efficiency.

Thinking like an economist: the framework

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Why economists give conflicting policy advice

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Why economists give conflicting policy advice

1. **Different positive theories**
   Economists may disagree about the validity of alternative positive theories about the world.
2. **Different values**
   They may have different values and therefore different normative views about what policy should try to accomplish.

Ask whether the disagreement concerns evidence and predictions, or objectives and values.

Thinking like an economist: the framework

Framework complete. Connect the parts before moving on.

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2. **Economists as policy advisers** ✓
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3. **Why economists disagree** ✓
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02
Comparative advantage

Interdependence, the PPF, gains from trade, and the driving force of specialization

Comparative advantage: the framework

Our route. The highlighted item is our current focus.

1 **Interdependence** →
Why people and nations choose to depend on each other

2 **The production possibilities frontier**
Points, properties, movement, and shape

3 **How trade makes people better off**
Production, exchange, and consumption

4 **The driving force of specialization**
Absolute advantage versus comparative advantage

Interdependence: why trade links our decisions

1. **What interdependence means**
   People and nations rely on one another for goods and services rather than producing everything they consume themselves.

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2. **The question from Principle 5**
   "Trade can make everyone better off." Why do people and nations choose to be interdependent?

Interdependence: why trade links our decisions

1. **What interdependence means**
   People and nations rely on one another for goods and services rather than producing everything they consume themselves.
2. **The question from Principle 5**
   "Trade can make everyone better off." Why do people and nations choose to be interdependent?
3. **The explanation to build**
   Specialization changes what each produces. Exchange allows each to consume a different bundle. We compare consumption with and without trade.

Comparative advantage: the framework

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1. **Interdependence** ✓
   Why people and nations choose to depend on each other
2. **The production possibilities frontier** →
   Points, properties, movement, and shape
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The production possibilities frontier

1 **Definition**

A graph that shows the combinations of two goods the economy can possibly produce given the available resources and the available technology.

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   The PPF describes what is feasible. It does not by itself decide which combination society should choose.

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We will read it from four perspectives: points, properties, movement, and shape.

PPF: four perspectives

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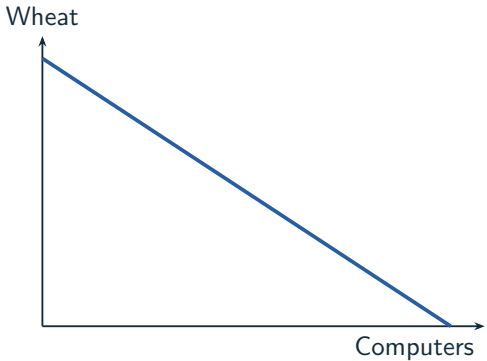
1 **Points** →
On, inside, and outside the frontier: feasibility and efficiency

2 **Properties**
Trade-offs and the opportunity cost measured by slope

3 **Movement**
Economic growth shifts the frontier; reallocation moves along it

4 **Shape**
Constant versus increasing opportunity cost

1 Points: feasibility and efficiency

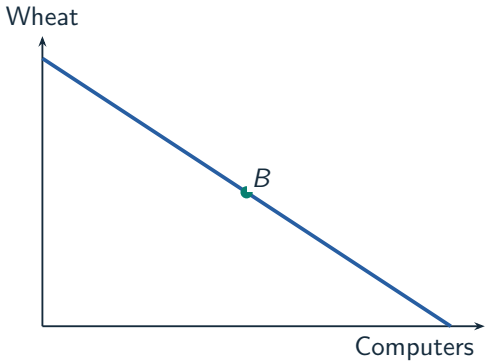


At an inside point, either output can rise without sacrificing the other.

1 The PPF

Locate each point relative to the frontier.

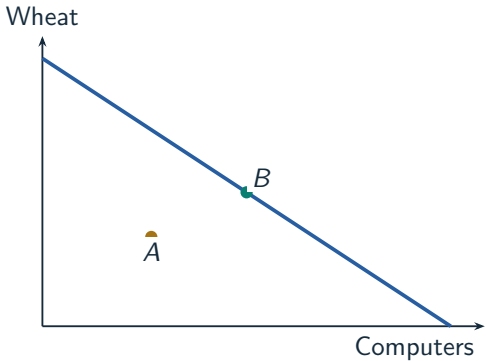
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At an inside point, either output can rise without sacrificing the other.

1. **The PPF**
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2. **On the frontier: B**
   Attainable and productively efficient.

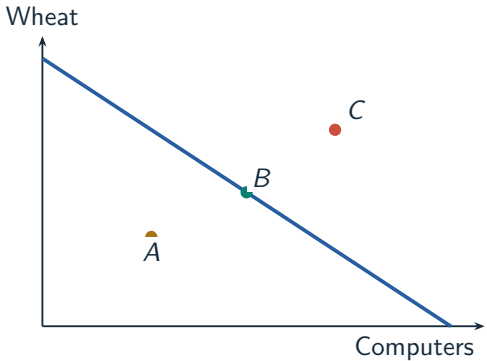
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   Attainable but inefficient: resources are underutilized, such as unemployed workers or idle factories.

1 Points: feasibility and efficiency



At an inside point, either output can rise without sacrificing the other.

1. **The PPF**
   Locate each point relative to the frontier.
2. **On the frontier: B**
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3. **Inside the frontier: A**
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4. **Outside the frontier: C**
   Not possible with current resources and technology.

PPF: four perspectives

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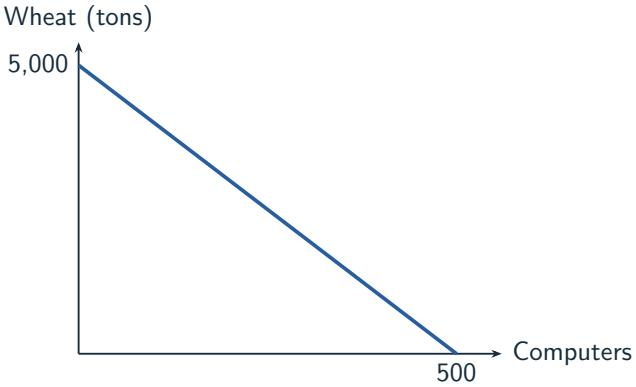
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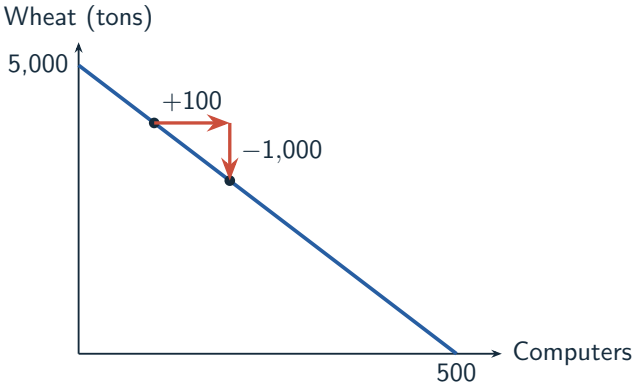
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2 Properties: trade-offs and slope



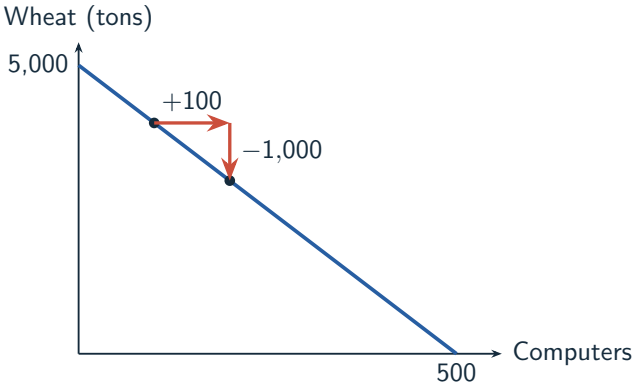
1. **Move along the PPF**
   Resources shift between industries: more computers require less wheat.

2 Properties: trade-offs and slope



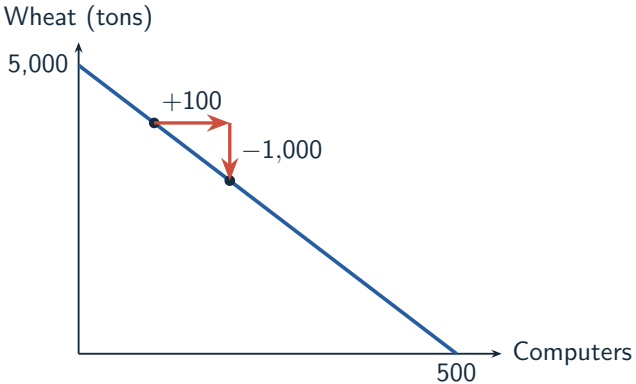
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2. **Compute the signed slope**
   $$\frac{\Delta W}{\Delta C} = \frac{-1{,}000}{100} = -10.$$

2 Properties: trade-offs and slope



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   Resources shift between industries: more computers require less wheat.
2. **Compute the signed slope**
   $\frac{\Delta W}{\Delta C} = \frac{-1{,}000}{100} = -10.$
3. **State the opportunity cost**
   1 computer costs 10 tons of wheat.
   The cost is the magnitude of the slope.

2 Properties: trade-offs and slope



1. **Move along the PPF**
   Resources shift between industries: more computers require less wheat.
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   1 computer costs 10 tons of wheat.
   The cost is the magnitude of the slope.

With A horizontal and B vertical: $OC(1A) = |\Delta B/\Delta A|$ units of B.

PPF: four perspectives

Our route. The highlighted item is our current focus.

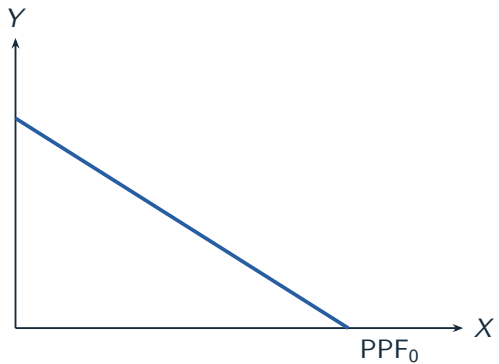
1 **Points** ✓
On, inside, and outside the frontier: feasibility and efficiency

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3 **Movement** →
Economic growth shifts the frontier; reallocation moves along it

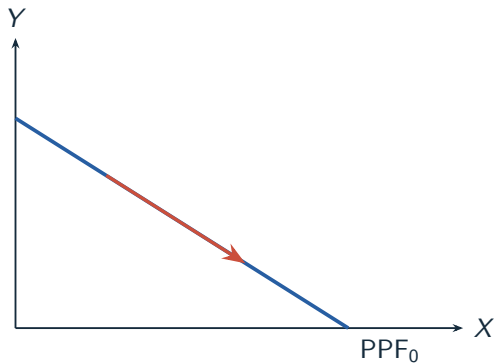
4 **Shape**
Constant versus increasing opportunity cost

3 Movement: reallocation and growth



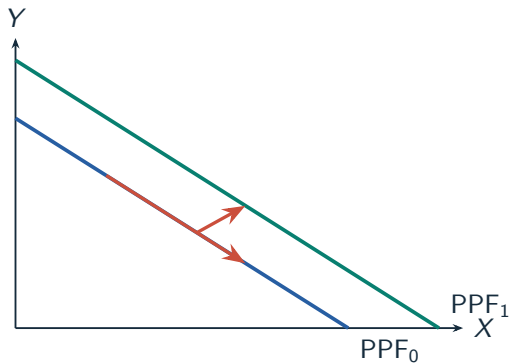
1. **Current resources and technology**
   The PPF describes current productive capacity.

3 Movement: reallocation and growth



1. **Current resources and technology**
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   More of one good requires less of the other.

3 Movement: reallocation and growth



1. **Current resources and technology**
   The PPF describes current productive capacity.
2. **A movement along it**
   More of one good requires less of the other.
3. **An outward shift**
   Additional resources or better technology allow more of either good, or a combination in between.

PPF: four perspectives

Our route. The highlighted item is our current focus.

1 **Points** ✓
On, inside, and outside the frontier: feasibility and efficiency

2 **Properties** ✓
Trade-offs and the opportunity cost measured by slope

3 **Movement** ✓
Economic growth shifts the frontier; reallocation moves along it

4 **Shape** →
Constant versus increasing opportunity cost

4 Shape: constant or increasing opportunity cost?

1. **A straight line**
   Opportunity cost remains constant as resources shift from one industry to the other. In the computer-wheat example, one computer always costs ten tons of wheat.

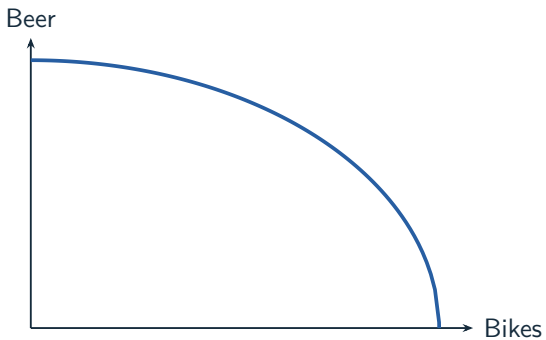
4 Shape: constant or increasing opportunity cost?

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   The opportunity cost of a good rises as the economy produces more of it.

4 Shape: constant or increasing opportunity cost?

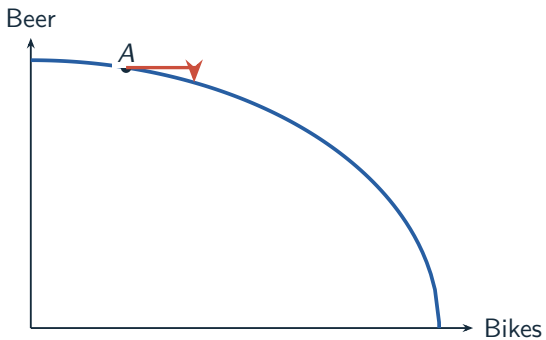
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   The opportunity cost of a good rises as the economy produces more of it.
3. **Why costs can differ**
   Different workers have different skills; different types of land suit different uses. The mix of resources has varying opportunity costs.

4 Shape: why the frontier becomes steeper



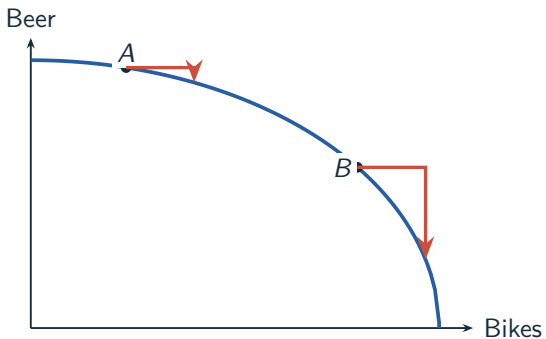
1. **Resource differences**
   Workers have different skills; land and equipment suit different uses.

4 Shape: why the frontier becomes steeper



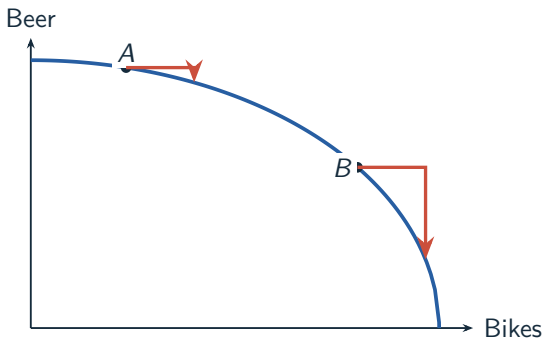
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   Further expansion takes the best brewers from beer. Opportunity cost rises.

4 Shape: why the frontier becomes steeper



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   Workers have different skills; land and equipment suit different uses.
2. **Near A**
   The first resources moved into bikes have a low opportunity cost.
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   Further expansion takes the best brewers from beer. Opportunity cost rises.

The PPF becomes steeper as bike output rises.

PPF: four perspectives

Framework complete. Connect the parts before moving on.

1 **Points** ✓
On, inside, and outside the frontier: feasibility and efficiency

2 **Properties** ✓
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Constant versus increasing opportunity cost

Comparative advantage: the framework

Our route. The highlighted item is our current focus.

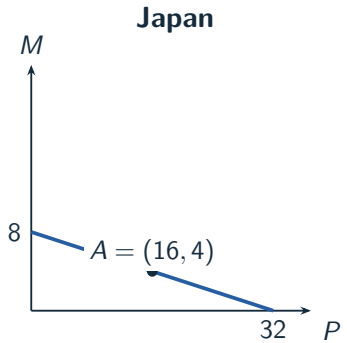
1 **Interdependence** ✓
Why people and nations choose to depend on each other

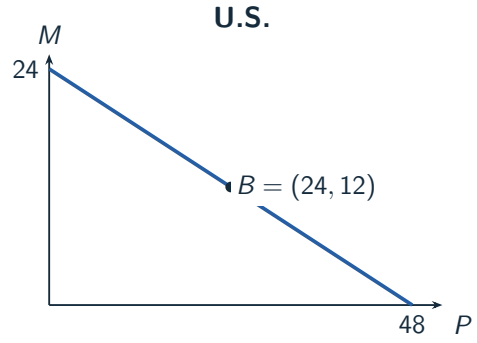
2 **The production possibilities frontier** ✓
Points, properties, movement, and shape

3 **How trade makes people better off** →
Production, exchange, and consumption

4 **The driving force of specialization**
Absolute advantage versus comparative advantage

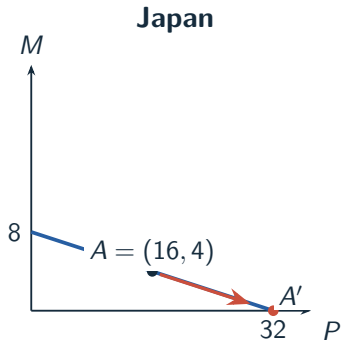
Specialization changes production

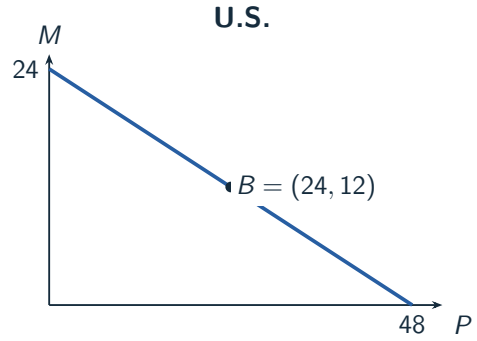




Coordinates list (potatoes, meat). Both countries reallocate toward their comparative advantage.

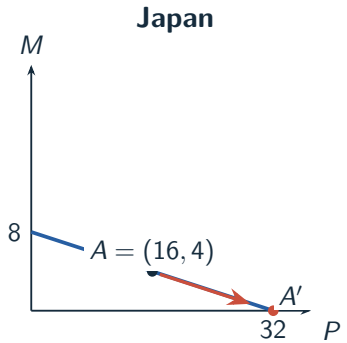
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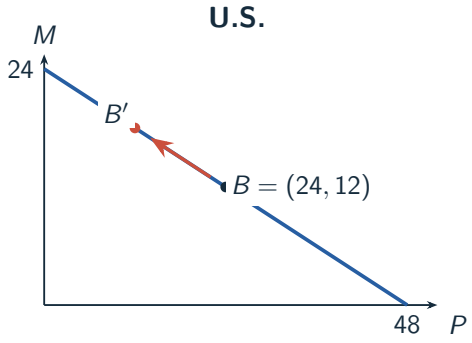




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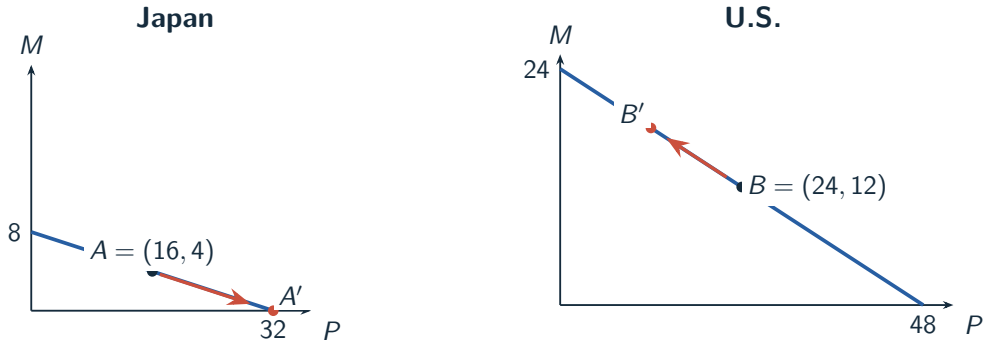
Specialization changes production





Coordinates list (potatoes, meat). Both countries reallocate toward their comparative advantage.

Specialization changes production



Combined output rises: meat $16 \to 18$**, potatoes** $40 \to 44$**.**

Coordinates list (potatoes, meat). Both countries reallocate toward their comparative advantage.

Exchange and gains from trade

1. **Terms of trade**
   Japan gives 15 potatoes to the U.S. in exchange for 5 meat.
   The exchange rate is 1 meat for 3 potatoes, between the opportunity costs 2 and 4.

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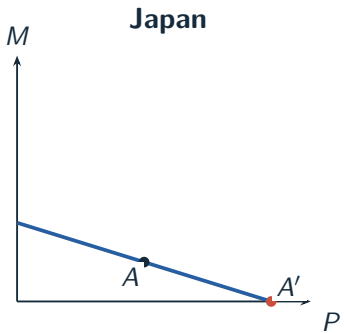
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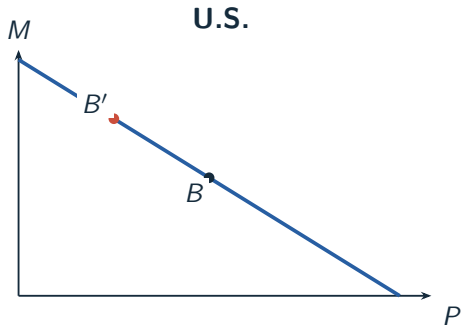
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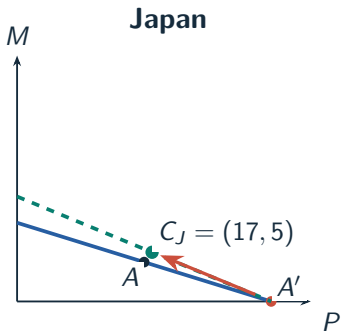
Both consume more of both goods. Total consumption equals total production: $(44, 18)$**.**

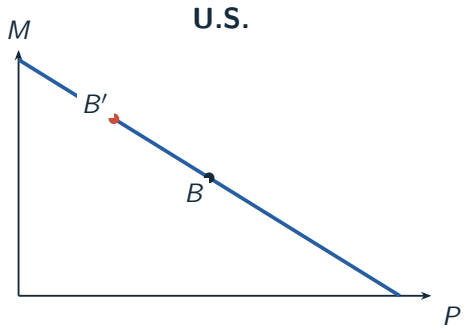
Consumption beyond the domestic PPF



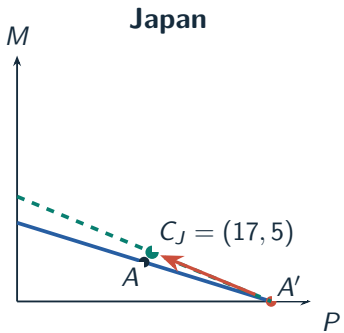


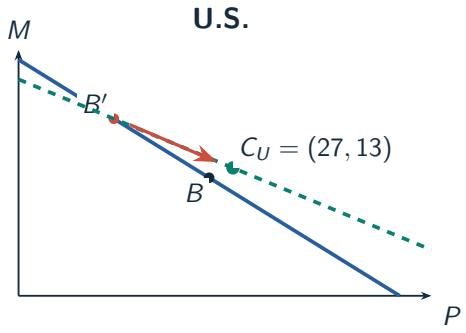
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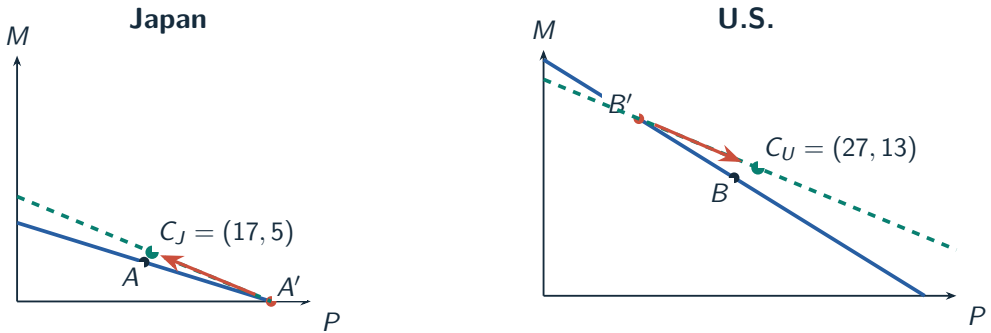


Consumption beyond the domestic PPF





Consumption beyond the domestic PPF



Trade expands consumption possibilities without moving either country's production frontier.

Comparative advantage: the framework

Our route. The highlighted item is our current focus.

1 **Interdependence** ✓
Why people and nations choose to depend on each other

2 **The production possibilities frontier** ✓
Points, properties, movement, and shape

3 **How trade makes people better off** ✓
Production, exchange, and consumption

4 **The driving force of specialization** →
Absolute advantage versus comparative advantage

Absolute and comparative advantage

Maximum output (ounces), same resources

	Meat	Potatoes
U.S.	24	48
Japan	8	32

1. **Absolute advantage**
   Produce more with the same inputs, or the same output with fewer inputs.
   The U.S. has it in both goods.

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	1 meat costs	1 potato costs
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   U.S.: meat. Japan: potatoes.

Trade depends on comparative advantage, even when one country is absolutely better at both goods.

Comparative advantage: the driving force of specialization

1 **Absolute advantage compares inputs**
The ability to produce a good using fewer inputs than another producer.

Comparative advantage: the driving force of specialization

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Comparative advantage: the driving force of specialization

1. **Absolute advantage compares inputs**
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2. **Comparative advantage compares opportunity costs**
   The ability to produce a good at a lower opportunity cost than another producer. Absolute advantage is not necessary for comparative advantage.
3. **Why gains arise**
   Gains from trade arise from differences in opportunity costs. Specialization according to comparative advantage increases total production: the world's economic pie is bigger.

Comparative advantage: the calculation sequence

1. **Read units and the resource constraint**
   Identify maximum outputs or input requirements, with consistent units.

Comparative advantage: the calculation sequence

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4. **Check exchange and consumption**
   Use an exchange rate between the opportunity costs. Check that goods balance.

Comparative advantage: the framework

Framework complete. Connect the parts before moving on.

1. **Interdependence** ✓
   Why people and nations choose to depend on each other
2. **The production possibilities frontier** ✓
   Points, properties, movement, and shape
3. **How trade makes people better off** ✓
   Production, exchange, and consumption
4. **The driving force of specialization** ✓
   Absolute advantage versus comparative advantage

03
Supply and demand

The market mechanism and a systematic way to analyze changes in equilibrium

Supply and demand: the framework

Our route. The highlighted item is our current focus.

1 **Markets and competition** →
Buyers, sellers, competitive markets, and price takers

2 **Demand**
Buyers, marginal benefit, schedules, movements, and shifters

3 **Supply**
Sellers, marginal cost, schedules, movements, and shifters

4 **Demand and supply together**
Equilibrium, adjustment, comparative statics, and price signals

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A group of buyers and sellers of a particular good or service. Buyers determine demand; sellers determine supply.

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Markets and competition

1. **Market**
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2. **Competitive market**
   Many buyers and many sellers, each with a negligible impact on the market price.
3. **Perfectly competitive market**
   All goods are exactly the same. Buyers and sellers are so numerous that no one can affect the market price: they are price takers.

Price taking and individual pricing power

1. **Price takers**
   Each buyer and seller takes the market price as given.

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$$F_n(u) \longrightarrow 0 \qquad \text{as} \quad n \longrightarrow \infty.$$

The board's notation is an informal mnemonic. Its function and argument are not specified; seller numbers alone do not establish a general theorem.

Supply and demand: the framework

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Demand: the framework

Our route. The highlighted item is our current focus.

1 **Behavior and the law of demand** →
Quantity demanded, marginal benefit, and other things equal

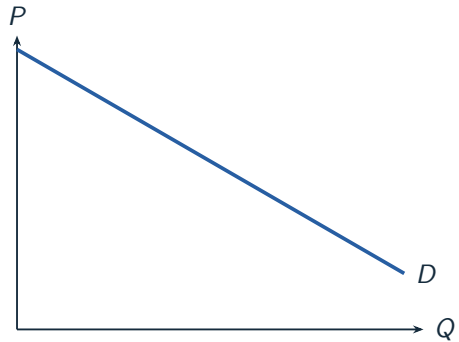
2 **Schedule and market demand**
Price-quantity pairs and horizontal aggregation

3 **Movement versus shift**
Change in quantity demanded versus change in demand

4 **Demand-curve shifters**
Income; buyers; related goods; tastes; expectations

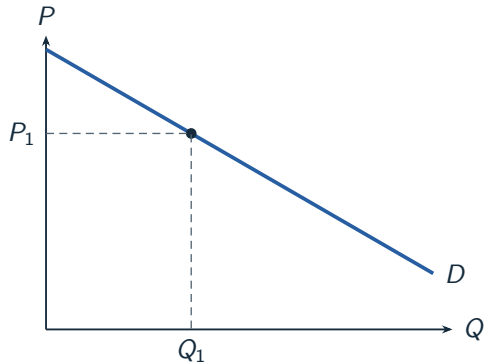
Demand: willingness and ability to buy

1 **Quantity demanded**
Demand comes from the behavior of buyers. Quantity demanded is the amount of the good that buyers are willing and able to purchase.



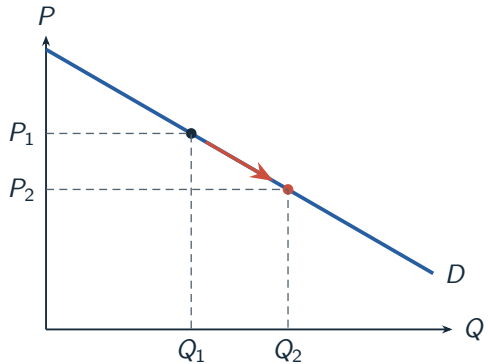
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   Quantity demanded is determined by marginal benefit. The demand curve records marginal willingness to pay.



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   Demand comes from the behavior of buyers. Quantity demanded is the amount of the good that buyers are willing and able to purchase.
2. **Marginal benefit**
   Quantity demanded is determined by marginal benefit. The demand curve records marginal willingness to pay.
3. **The law of demand**
   Other things equal, when the price of a good rises, the quantity demanded of the good falls.



Demand: the framework

Our route. The highlighted item is our current focus.

1. **Behavior and the law of demand** ✓
   Quantity demanded, marginal benefit, and other things equal
2. **Schedule and market demand** →
   Price-quantity pairs and horizontal aggregation
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Demand schedule and market demand

1. **Demand schedule**
   A table that shows the relationship between the price of a good and the quantity demanded.

Demand schedule and market demand

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A table that shows the relationship between the price of a good and the quantity demanded.

Price	Sijia	Meiling	Market
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Demand schedule and market demand

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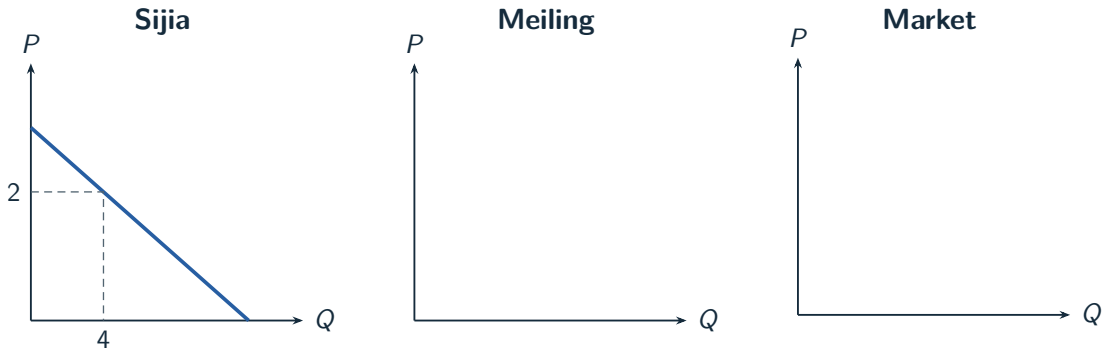
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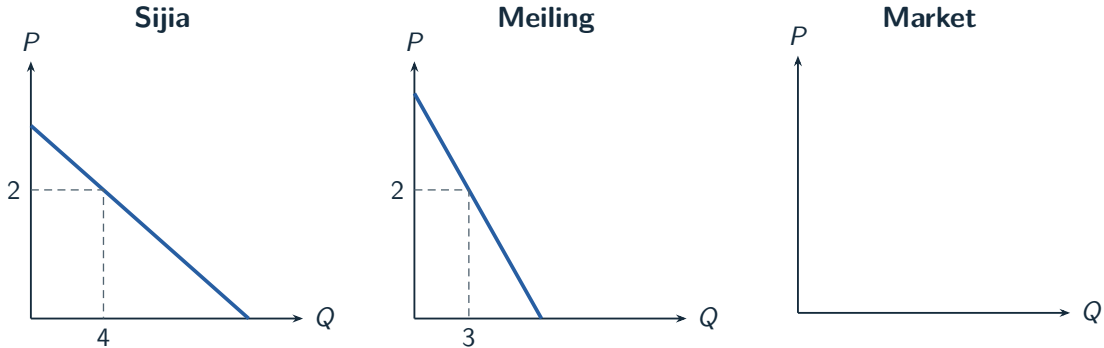
2 Market demand

The sum of all individual demands for a good or service. To obtain the market demand curve, sum the individual demand curves horizontally.

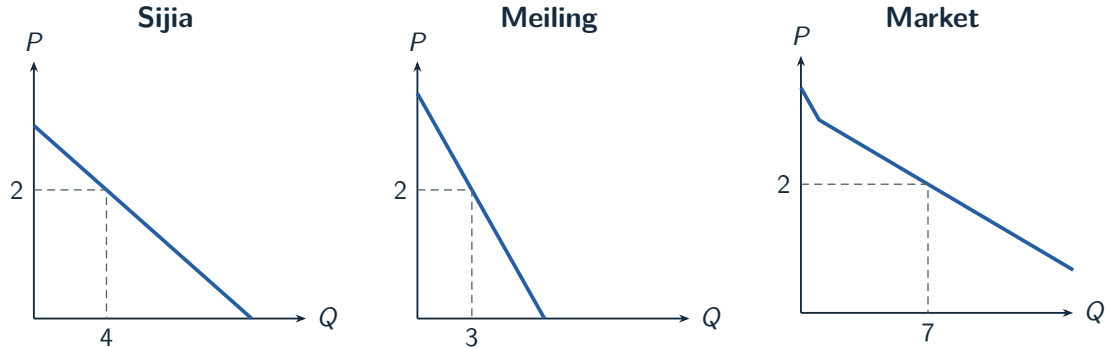
Market demand is the horizontal sum



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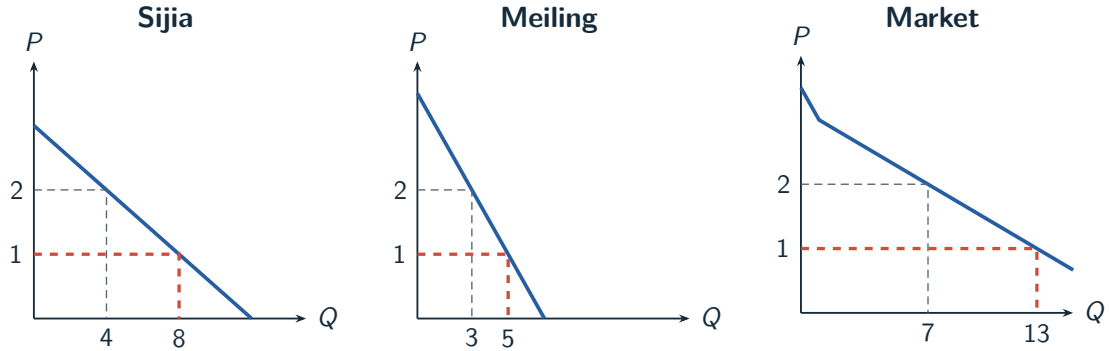


Market demand is the horizontal sum



$$Q_D(P) = Q_{D,1}(P) + Q_{D,2}(P) \qquad \text{at the same price.}$$

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At $P = 2$**:** $4 + 3 = 7$**.** **At** $P = 1$**:** $8 + 5 = 13$**.**

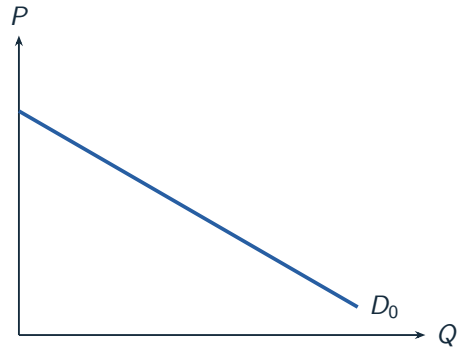
Demand: the framework

Our route. The highlighted item is our current focus.

1. **Behavior and the law of demand** ✓
   Quantity demanded, marginal benefit, and other things equal
2. **Schedule and market demand** ✓
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3. **Movement versus shift** →
   Change in quantity demanded versus change in demand
4. **Demand-curve shifters**
   Income; buyers; related goods; tastes; expectations

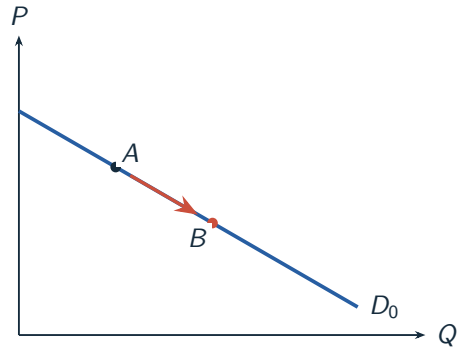
Movement along demand and a shift in demand

1. **The good's own price changes**
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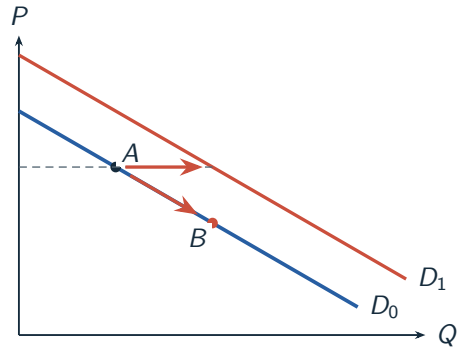
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Movement along demand and a shift in demand

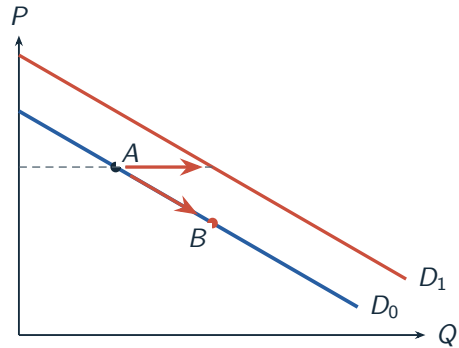
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2. **Another determinant changes**
   At each price, buyers want a different quantity. This is a change in **demand**.



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First identify what changed, then name the change correctly.



Demand: the framework

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4. **Demand-curve shifters** →
   Income; buyers; related goods; tastes; expectations

Demand shifters: five determinants

Our route. The highlighted item is our current focus.

1 Income →
Normal goods and inferior goods

2 Number of buyers
More or fewer buyers at each price

3 Prices of related goods
Substitutes and complements

4 Tastes
Preferences for the good

5 Expectations
Expected future income and prices

Demand shifter / 1 Income

1 **Normal good**
An increase in income leads to an increase in demand. At each price, buyers want more. The demand curve shifts right.

Demand shifter / 1 Income

1. **Normal good**
   An increase in income leads to an increase in demand. At each price, buyers want more. The demand curve shifts right.
2. **Inferior good**
   An increase in income leads to a decrease in demand. The demand curve shifts left.

Demand shifter / 1 Income

1. **Normal good**
   An increase in income leads to an increase in demand. At each price, buyers want more. The demand curve shifts right.
2. **Inferior good**
   An increase in income leads to a decrease in demand. The demand curve shifts left.

The direction depends on whether the good is normal or inferior.

Demand shifters: five determinants

Our route. The highlighted item is our current focus.

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Normal goods and inferior goods

2 Number of buyers →
More or fewer buyers at each price

3 Prices of related goods
Substitutes and complements

4 Tastes
Preferences for the good

5 Expectations
Expected future income and prices

Demand shifter / 2 Number of buyers

1. **More buyers**
   An increase in the number of buyers increases quantity demanded at each price. The demand curve shifts right.

Demand shifter / 2 Number of buyers

1. **More buyers**
   An increase in the number of buyers increases quantity demanded at each price. The demand curve shifts right.
2. **Fewer buyers**
   A decrease in the number of buyers decreases quantity demanded at each price. The demand curve shifts left.

Demand shifters: five determinants

Our route. The highlighted item is our current focus.

1. **Income** ✓
   Normal goods and inferior goods
2. **Number of buyers** ✓
   More or fewer buyers at each price
3. **Prices of related goods** →
   Substitutes and complements
4. **Tastes**
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5. **Expectations**
   Expected future income and prices

Demand shifter / 3 Prices of related goods

1 **Identify the market**
We are studying demand for good B. Change the price of a different good, A.

Demand shifter / 3 Prices of related goods

1. **Identify the market**
   We are studying demand for good B. Change the price of a different good, A.
2. **Substitutes**
   An increase in the price of A leads to an increase in demand for B.

Demand shifter / 3 Prices of related goods

1. **Identify the market**
   We are studying demand for good B. Change the price of a different good, A.
2. **Substitutes**
   An increase in the price of A leads to an increase in demand for B.
3. **Complements**
   An increase in the price of A leads to a decrease in demand for B.

Demand shifters: five determinants

Our route. The highlighted item is our current focus.

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2. **Number of buyers** ✓
   More or fewer buyers at each price
3. **Prices of related goods** ✓
   Substitutes and complements
4. **Tastes** →
   Preferences for the good
5. **Expectations**
   Expected future income and prices

Demand shifter / 4 Tastes

1 **A change in preferences**
Anything that causes a shift in tastes toward a good increases demand for that good.

Demand shifter / 4 Tastes

1. **A change in preferences**
   Anything that causes a shift in tastes toward a good increases demand for that good.
2. **Curve response**
   The demand curve shifts right. A shift in tastes away from the good has the opposite effect.

Demand shifters: five determinants

Our route. The highlighted item is our current focus.

1. **Income** ✓
   Normal goods and inferior goods
2. **Number of buyers** ✓
   More or fewer buyers at each price
3. **Prices of related goods** ✓
   Substitutes and complements
4. **Tastes** ✓
   Preferences for the good
5. **Expectations** →
   Expected future income and prices

Demand shifter / 5 Expectations

1 **Expected future income**
An expected increase in income can increase current demand for a normal good.

Demand shifter / 5 Expectations

1. **Expected future income**
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   Expecting a higher future price can increase current demand.

Demand shifter / 5 Expectations

1. **Expected future income**
   An expected increase in income can increase current demand for a normal good.
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   Expecting a higher future price can increase current demand.

These expectations change demand today, at each current price.

Demand shifters: five determinants

Framework complete. Connect the parts before moving on.

1 **Income** ✓
Normal goods and inferior goods

2 **Number of buyers** ✓
More or fewer buyers at each price

3 **Prices of related goods** ✓
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Preferences for the good

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Demand: the framework

Framework complete. Connect the parts before moving on.

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4 **Demand-curve shifters** ✓
Income; buyers; related goods; tastes; expectations

Supply and demand: the framework

Our route. The highlighted item is our current focus.

1 **Markets and competition** ✓
Buyers, sellers, competitive markets, and price takers

2 **Demand** ✓
Buyers, marginal benefit, schedules, movements, and shifters

3 **Supply** →
Sellers, marginal cost, schedules, movements, and shifters

4 **Demand and supply together**
Equilibrium, adjustment, comparative statics, and price signals

Supply: the framework

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Quantity supplied and marginal cost

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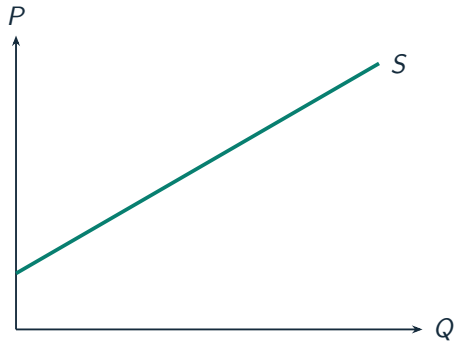
3 **Movement versus shift**
Change in quantity supplied versus change in supply

4 **Supply-curve shifters**
Input prices; technology; sellers; expectations

Supply: willingness and ability to sell

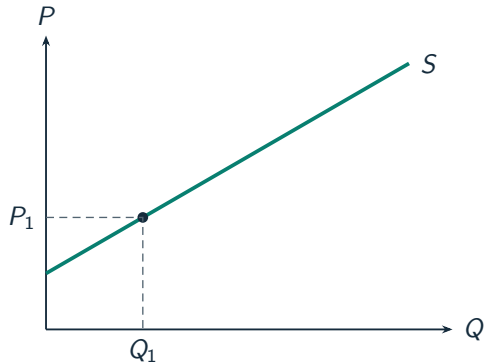
1 Quantity supplied

Supply comes from the behavior of sellers. Quantity supplied is the amount of the good that sellers are willing and able to sell.



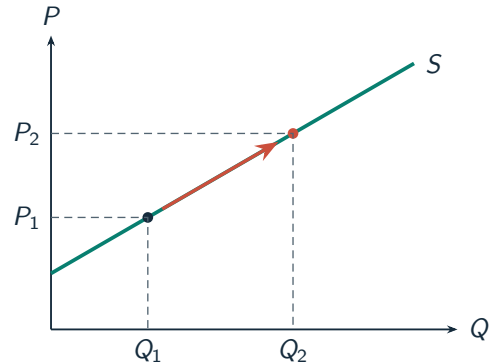
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   Supply comes from the behavior of sellers. Quantity supplied is the amount of the good that sellers are willing and able to sell.
2. **Marginal cost**
   The supply curve records marginal production cost in the competitive model.
3. **The law of supply**
   Other things equal, the quantity supplied of a good rises when the price of the good rises.



Supply: the framework

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   Quantity supplied and marginal cost
2. **Schedule and market supply** →
   Price-quantity pairs and horizontal aggregation
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   Change in quantity supplied versus change in supply
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   Input prices; technology; sellers; expectations

Supply schedule and market supply

1 **Supply schedule**
A table that shows the relationship between the price of a good and the quantity supplied.

Supply schedule and market supply

1 Supply schedule

A table that shows the relationship between the price of a good and the quantity supplied.

Price	Seller A	Seller B	Market
P_L	$q_{A,L}$	$q_{B,L}$	$q_{A,L} + q_{B,L}$
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2 **Market supply**
The sum of all individual supplies. Sum quantities horizontally at the same price: $Q_S(P) = \sum_i q_{S,i}(P)$.

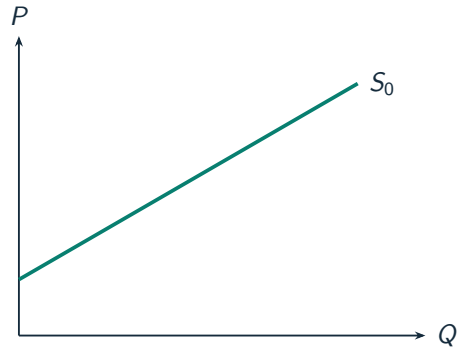
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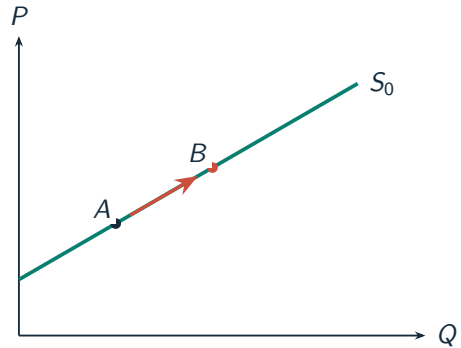
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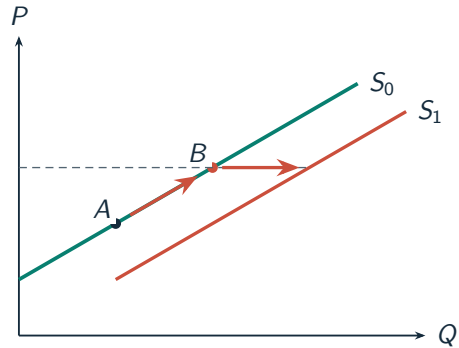
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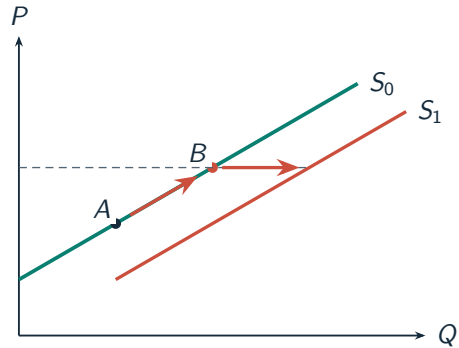
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   Input prices; technology; sellers; expectations

Supply shifters

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   Lower input prices reduce costs and shift supply right. Higher input prices shift it left.

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4. **Expectations**
   For a storable good, an expected future price rise can encourage sellers to hold inventory and reduce current supply.

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Demand and supply together: the framework

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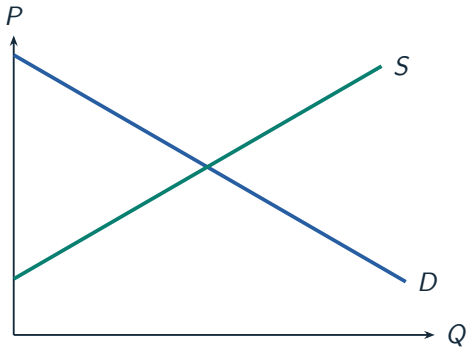
1 **Equilibrium** →
Equilibrium price and equilibrium quantity

2 **Outside equilibrium**
Surplus, shortage, and adjustment toward equilibrium

3 **Analyze a change**
Three steps; single and simultaneous shifts

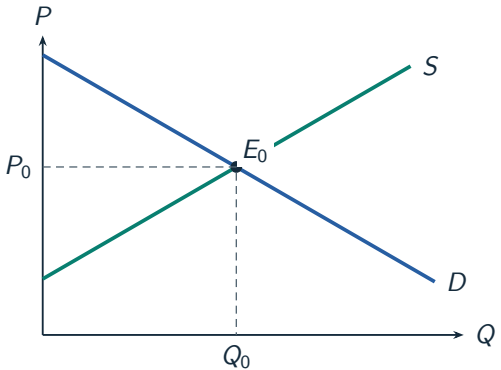
4 **Interpret the result**
Prices balance plans and guide resource allocation

Equilibrium: supply equals demand



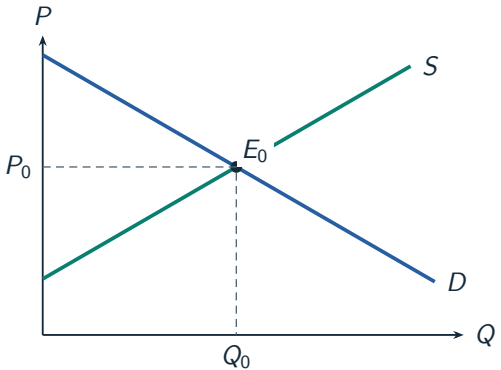
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The price reaches the level at which quantity supplied equals quantity demanded.

Equilibrium: supply equals demand



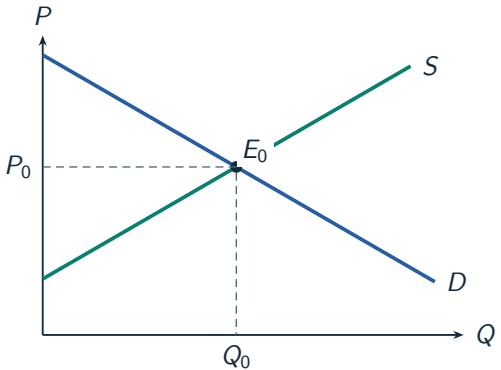
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Equilibrium: supply equals demand



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Price adjusts to balance the plans of buyers and sellers.

Demand and supply together: the framework

Our route. The highlighted item is our current focus.

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Equilibrium price and equilibrium quantity

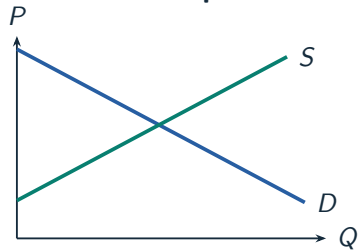
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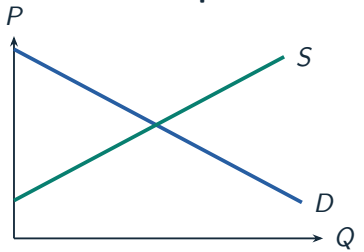
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Prices balance plans and guide resource allocation

Surplus and shortage

Price above equilibrium

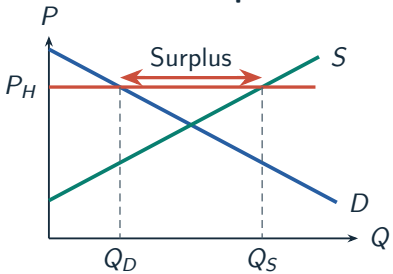


Price below equilibrium

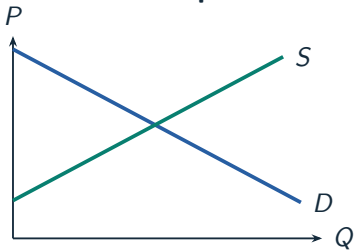


Surplus and shortage

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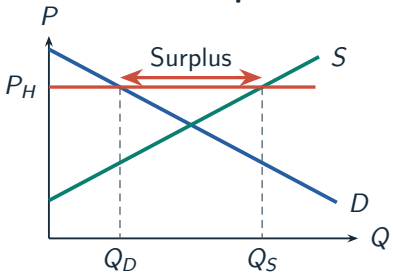


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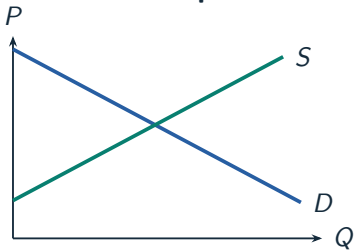
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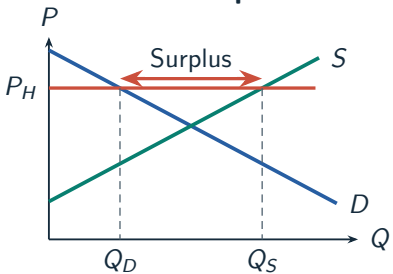
$Q_S > Q_D$. Unsold goods put downward pressure on price.

Price below equilibrium



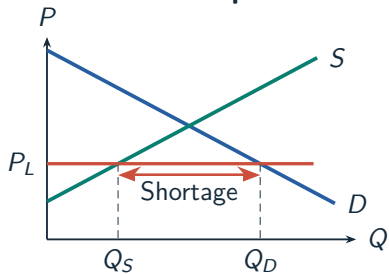
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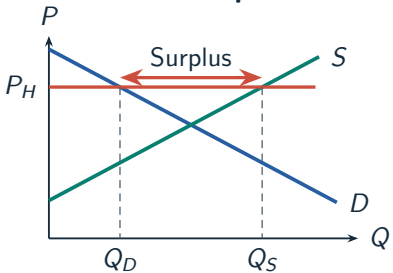
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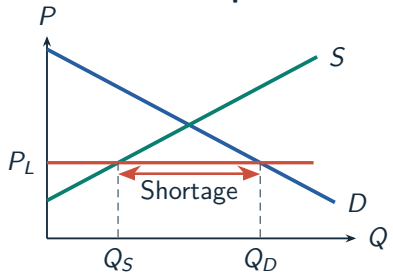
Surplus and shortage

Price above equilibrium



$Q_S > Q_D$. Unsold goods put downward pressure on price.

Price below equilibrium



$Q_D > Q_S$. Competition among buyers puts upward pressure on price.

The latte example: a schedule check

$$Q_D = 24 - 3P, \qquad Q_S = 5P.$$

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Price	Quantity demanded	Quantity supplied	Market condition
5	9	25	Surplus of 16
3	15	15	Equilibrium
1	21	5	Shortage of 16

Compare quantities at the same price. The sign of $Q_S - Q_D$ distinguishes surplus from shortage.

Demand and supply together: the framework

Our route. The highlighted item is our current focus.

1 **Equilibrium** ✓
Equilibrium price and equilibrium quantity

2 **Outside equilibrium** ✓
Surplus, shortage, and adjustment toward equilibrium

3 **Analyze a change** →
Three steps; single and simultaneous shifts

4 **Interpret the result**
Prices balance plans and guide resource allocation

The three-step method for equilibrium changes

1 **Which curve changes?**
Identify the determinant. Does the event shift demand, supply, or both?

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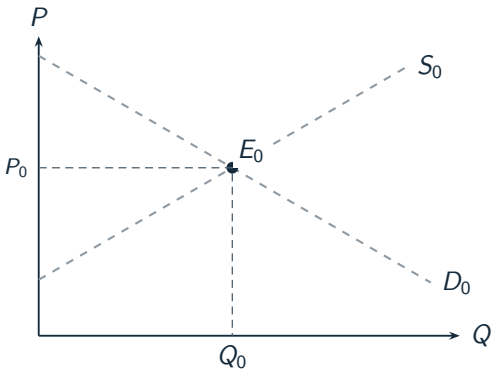
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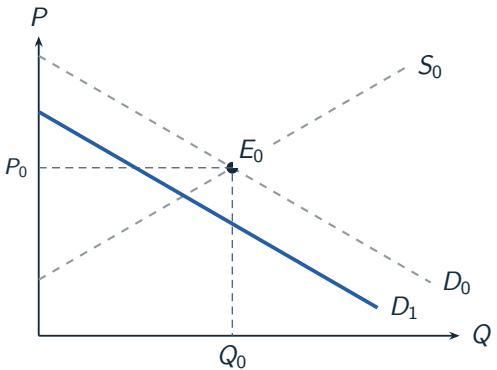
A shift of one curve creates a movement along the other curve.

Demand decreases



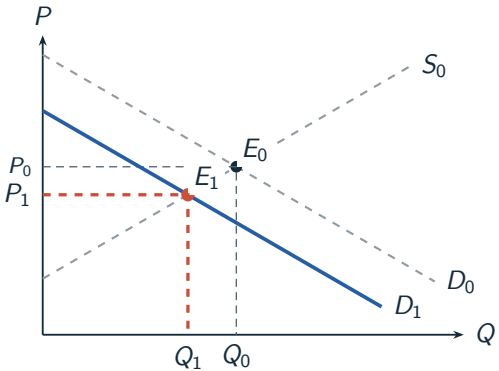
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Demand decreases



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2. **Demand shifts left**
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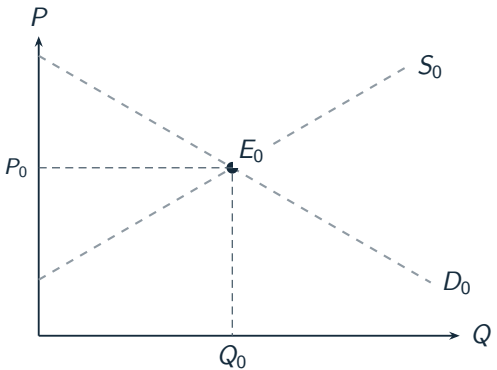
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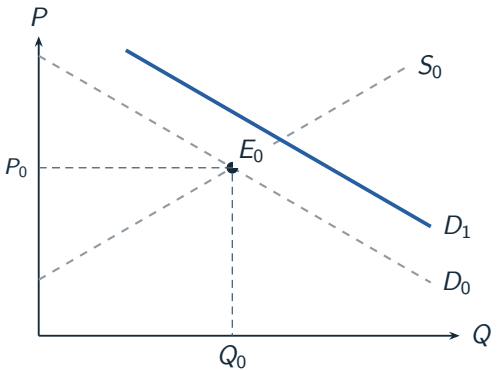
$P \downarrow, \quad Q \downarrow$

Demand increases



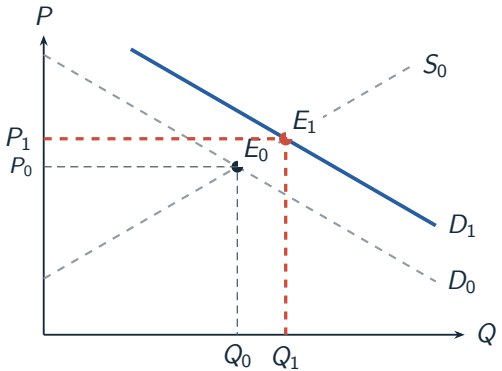
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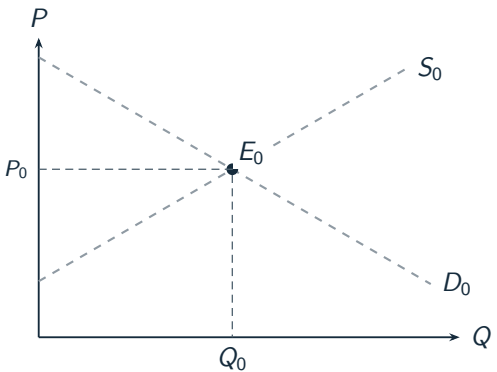
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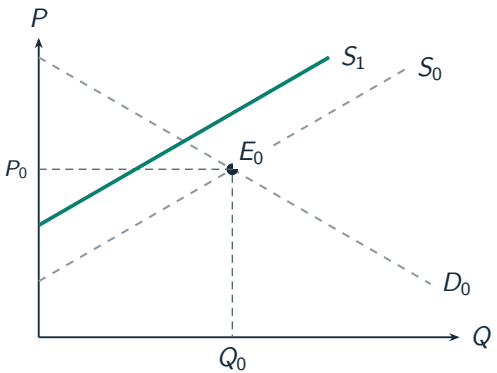
$P \uparrow, \quad Q \uparrow$

Supply decreases



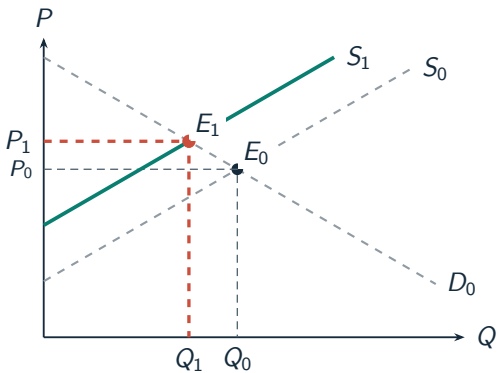
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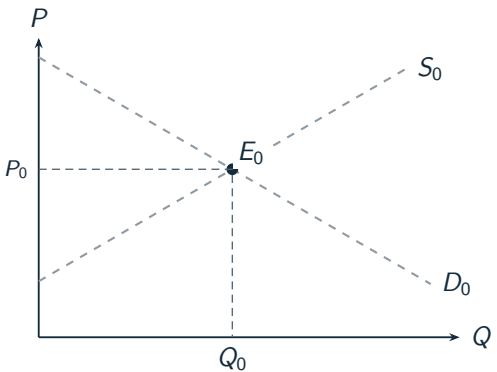
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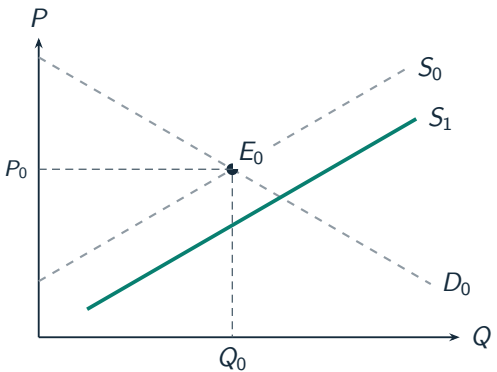
$P \uparrow, \quad Q \downarrow$

Supply increases



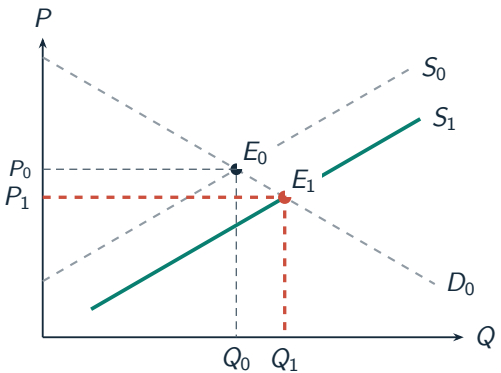
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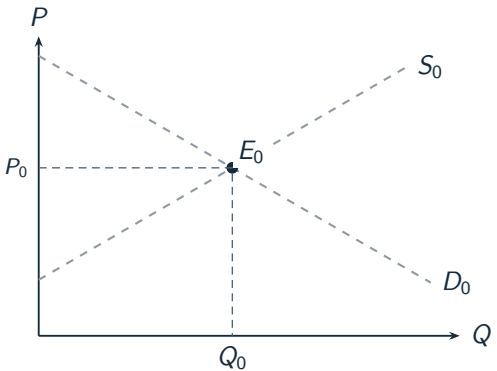
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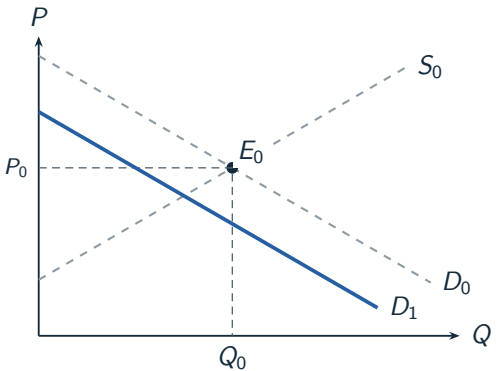
$P \downarrow, \quad Q \uparrow$

Demand and supply both decrease



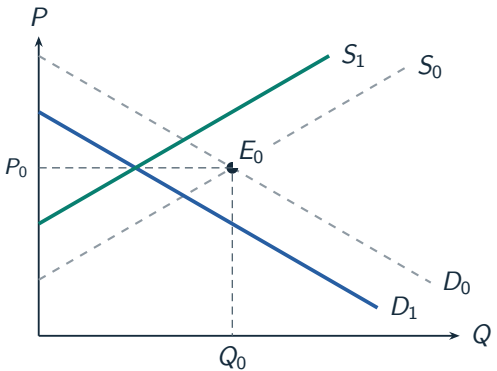
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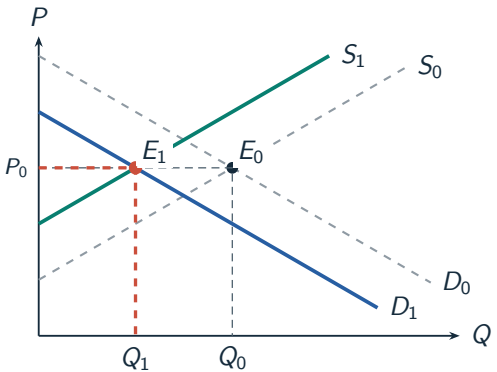
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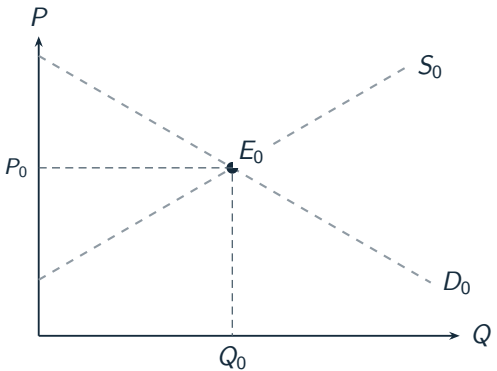


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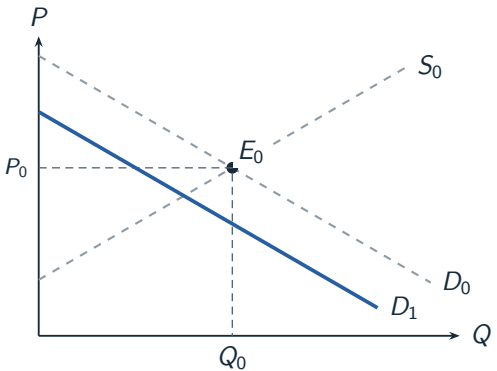
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Demand decreases and supply increases



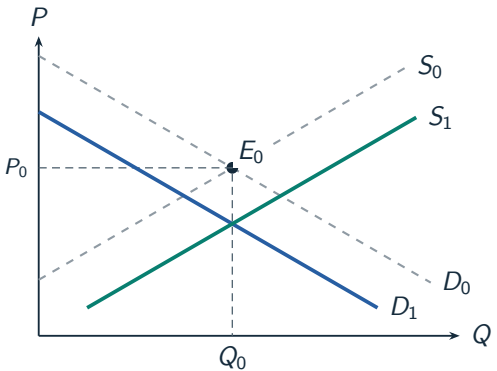
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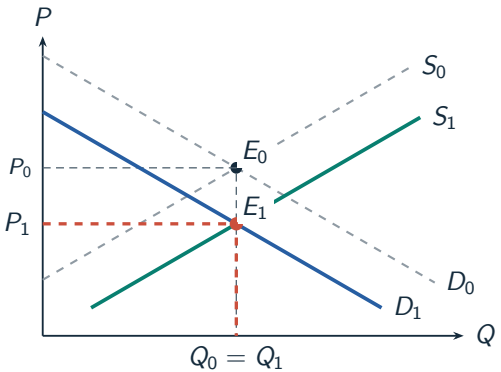
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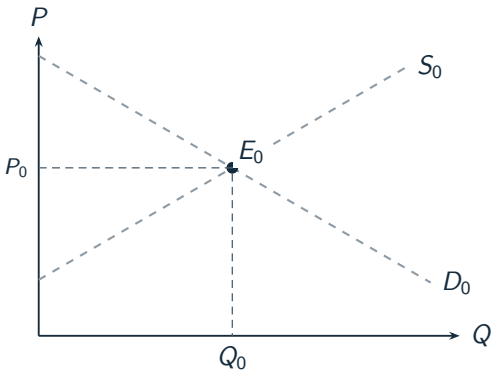


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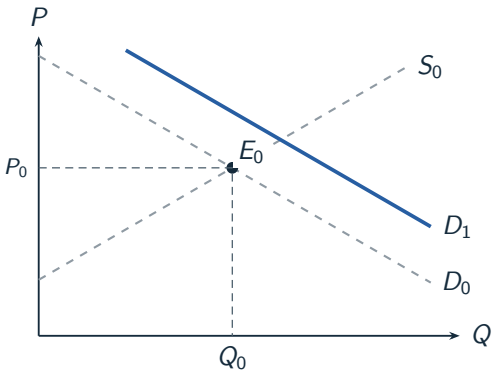
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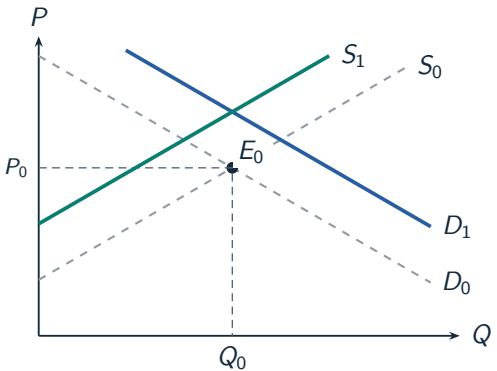
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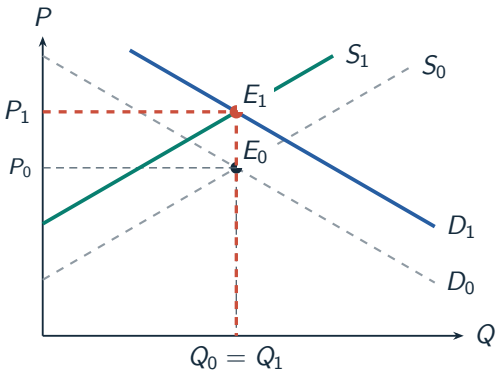
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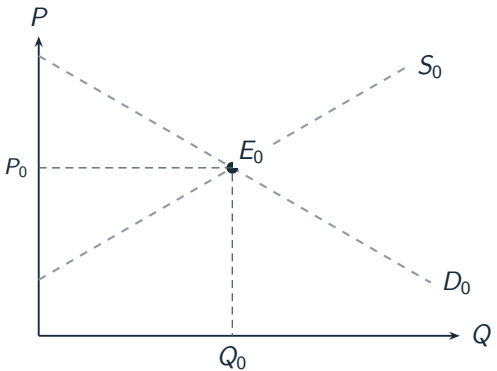


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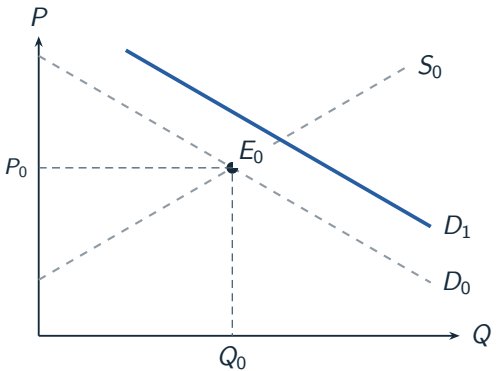
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Demand and supply both increase



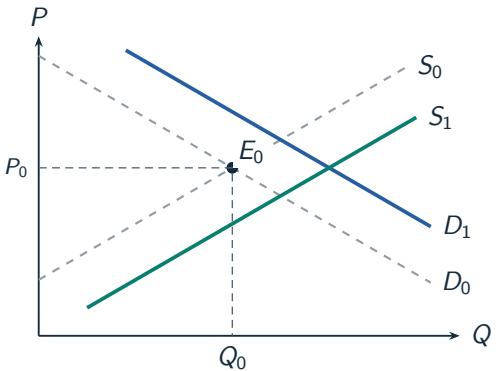
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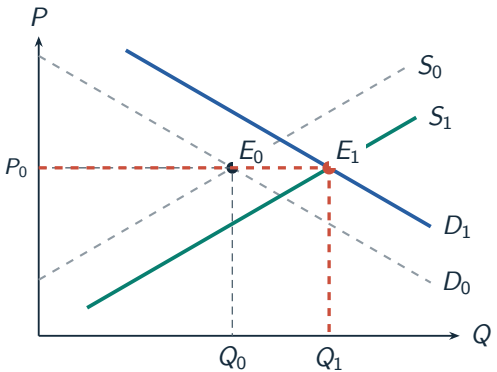
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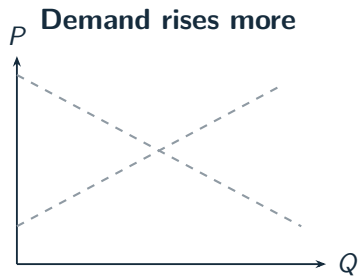


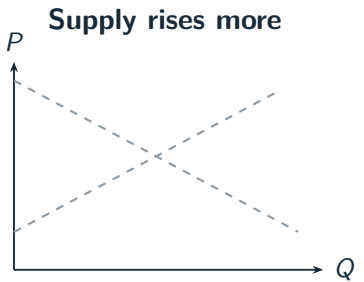
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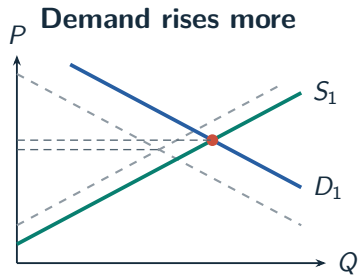
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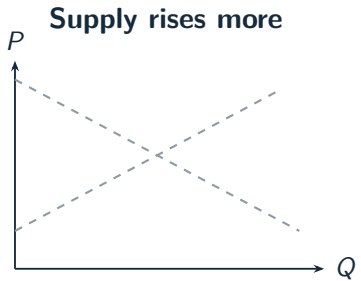
Why a simultaneous shift can be ambiguous





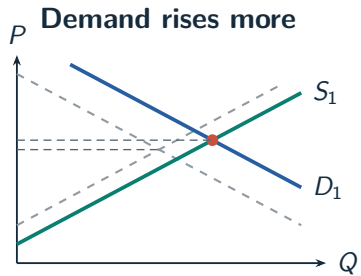
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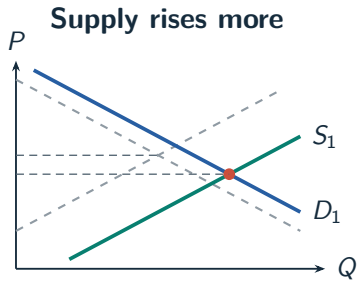


D and S both rise. Price rises.

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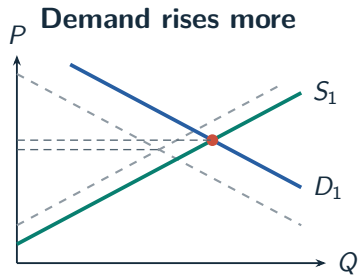


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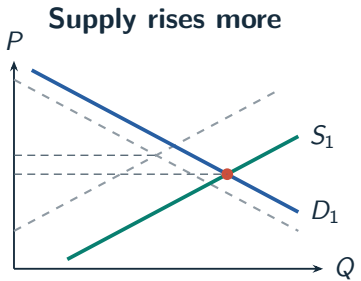


D and S both rise. Price falls.

Why a simultaneous shift can be ambiguous



D and S both rise. Price rises.



D and S both rise. Price falls.

Quantity rises in both cases. Price depends on the relative sizes of the shifts.

Supply and demand: the complete result table

	Supply unchanged	Supply increases	Supply decreases
Demand unchanged	P same, Q same	$P\downarrow$, $Q\uparrow$	$P\uparrow$, $Q\downarrow$
Demand increases	$P\uparrow$, $Q\uparrow$	P ambiguous, $Q\uparrow$	$P\uparrow$, Q ambiguous
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Prices balance supply and demand and transmit information about scarcity.

Demand and supply together: the framework

Our route. The highlighted item is our current focus.

1 **Equilibrium** ✓
Equilibrium price and equilibrium quantity

2 **Outside equilibrium** ✓
Surplus, shortage, and adjustment toward equilibrium

3 **Analyze a change** ✓
Three steps; single and simultaneous shifts

4 **Interpret the result** →
Prices balance plans and guide resource allocation

What equilibrium prices mean

1. **Balance supply and demand**
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Return to Principle 6: markets are usually a good way to organize economic activity.

Demand and supply together: the framework

Framework complete. Connect the parts before moving on.

1 **Equilibrium** ✓
Equilibrium price and equilibrium quantity

2 **Outside equilibrium** ✓
Surplus, shortage, and adjustment toward equilibrium

3 **Analyze a change** ✓
Three steps; single and simultaneous shifts

4 **Interpret the result** ✓
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Supply and demand: the framework

Framework complete. Connect the parts before moving on.

1. **Markets and competition** ✓
   Buyers, sellers, competitive markets, and price takers
2. **Demand** ✓
   Buyers, marginal benefit, schedules, movements, and shifters
3. **Supply** ✓
   Sellers, marginal cost, schedules, movements, and shifters
4. **Demand and supply together** ✓
   Equilibrium, adjustment, comparative statics, and price signals

04
Elasticity

How strongly quantities respond, and why that response matters for revenue and policy

Elasticity: the framework

Our route. The highlighted item is our current focus.

1. **Price elasticity of demand** →
   Definition, calculation, determinants, curves, and total revenue
2. **Other elasticities of demand**
   Income elasticity and cross-price elasticity
3. **Price elasticity of supply**
   Calculation, properties, determinants, and variation along a curve

Price elasticity of demand: the framework

Our route. The highlighted item is our current focus.

1 **Definition** →
Responsiveness of quantity demanded to price

2 **Calculation methods**
Standard percentage changes and the midpoint method

3 **Determinants**
Substitutes, market definition, necessities/luxuries, and time

4 **Properties and curve types**
Elastic, inelastic, unit elastic, and the two extremes

5 **Along a linear demand curve**
Constant slope, varying elasticity, and the same-point comparison

6 **Total revenue and applications**
Price and quantity effects; drugs and farming

Elasticity measures responsiveness

1 The general idea

Elasticity measures the percentage response of one variable to a percentage change in another. It is unit-free.

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$$\varepsilon_d = \left| \frac{\%\,\Delta Q_D}{\%\,\Delta P} \right|$$

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3. **Interpretation**
   If $\varepsilon_d = 2$, a small 1% price increase reduces quantity demanded by about 2%, other things equal.

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   $\%\Delta X = \dfrac{X_2 - X_1}{X_1} \times 100\%$.
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   Reverse: $25\%/16.67\% \approx 1.50$.

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$$\varepsilon_d = \frac{20/90}{2/11} = \frac{11}{9} \approx 1.22.$$

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   More close substitutes make it easier to switch away when price rises. Demand is more elastic.

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Four determinants of demand elasticity

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   Demand for luxuries tends to be more elastic. Demand for necessities tends to be less elastic.
4. **Time horizon**
   Demand is usually more elastic in the long run because buyers have more time to adjust.

Price elasticity of demand: the framework

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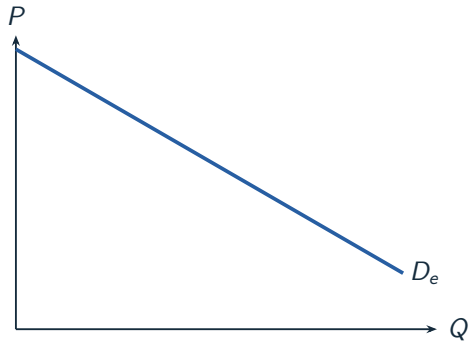
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Elastic, unit elastic, and inelastic demand

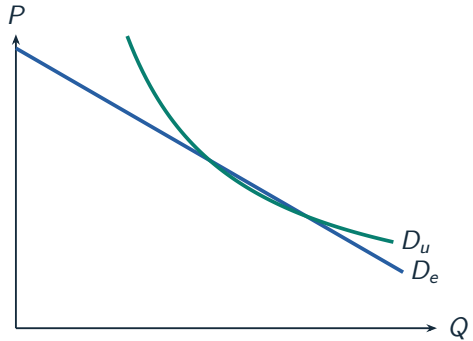
1. **Elastic demand** D_e
   $\varepsilon_d > 1$: quantity changes proportionately more than price.



The three curves pass through the same point. The labels refer to elasticity there. Their elasticity varies along each line. The hyperbola has constant unit elasticity.

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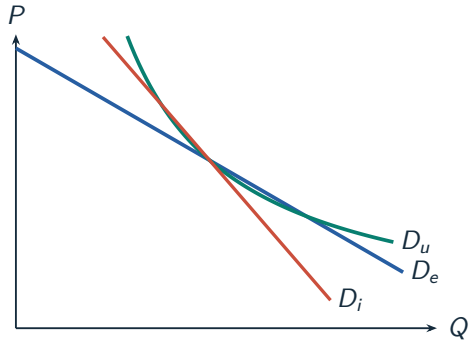
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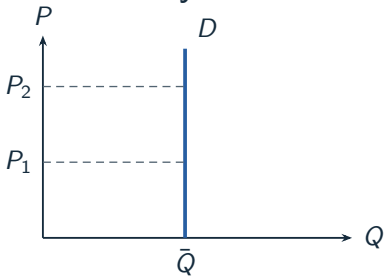
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3. **Inelastic demand** D_i
   $0 < \varepsilon_d < 1$: quantity changes proportionately less than price.



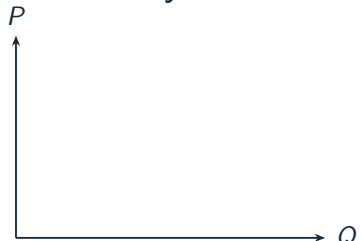
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The two extreme demand curves

Perfectly inelastic

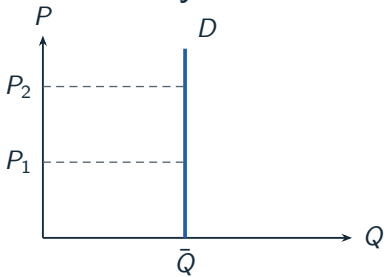


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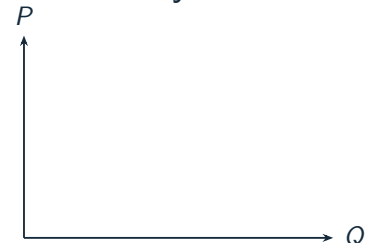


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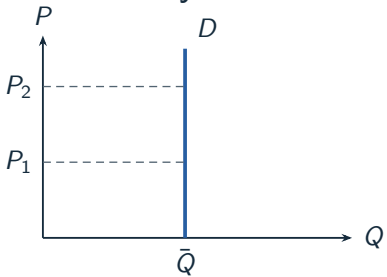
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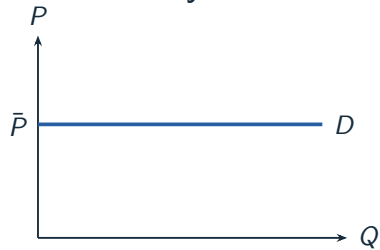
$\varepsilon_d = 0$. Quantity does not respond to price.

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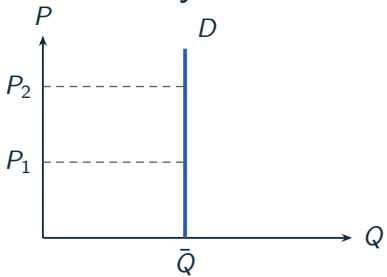
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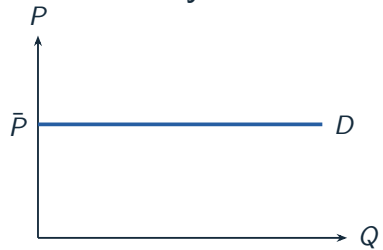
The two extreme demand curves

Perfectly inelastic



$\varepsilon_d = 0$. Quantity does not respond to price.

Perfectly elastic



$\varepsilon_d = \infty$. Demand is arbitrarily responsive at the given price.

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Point elasticity and the slope on the board

1. **Take a local change**
   For a differentiable downward-sloping demand curve, take the limit of the percentage-change ratio.

Point elasticity and the slope on the board

1 Take a local change

For a differentiable downward-sloping demand curve, take the limit of the percentage-change ratio.

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   When P is vertical and Q is horizontal, the drawn slope is $k = dP/dQ < 0$. Thus $dQ/dP = 1/k$.

Point elasticity and the slope on the board

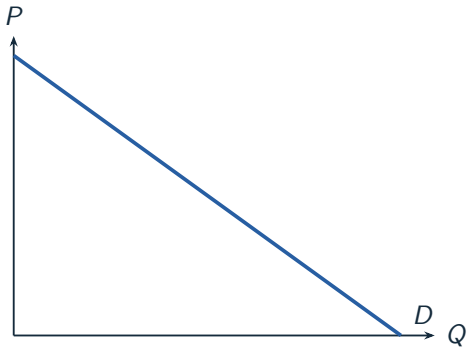
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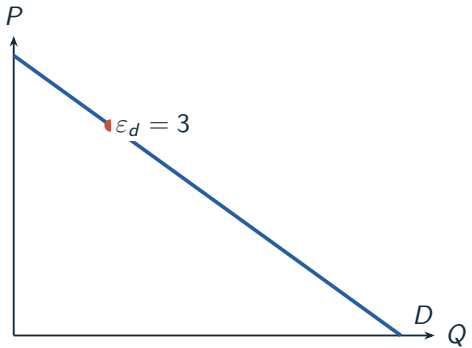
$$\boxed{\varepsilon_d = -\frac{1}{k}\frac{P}{Q} = \frac{1}{|k|}\frac{P}{Q}}$$

Elasticity along a linear demand curve



1. **A constant slope**
   For $P = a - bQ$, $dQ/dP = -1/b$.

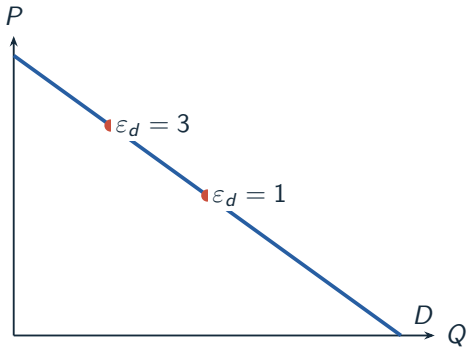
Elasticity along a linear demand curve



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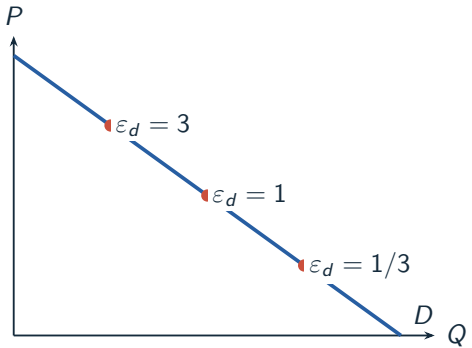
2. **Upper portion**
   High P, low Q: elastic.

Elasticity along a linear demand curve



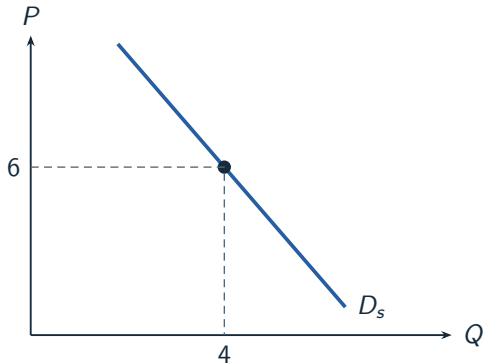
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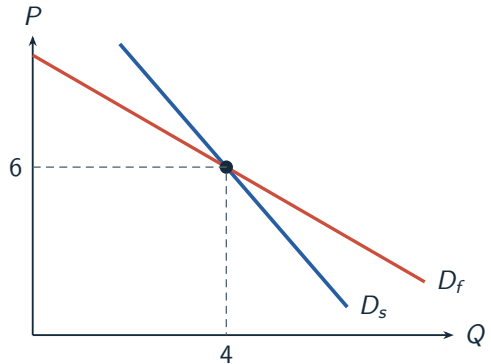
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4. **Lower portion**
   Low P, high Q: inelastic.

A flatter curve through a given point



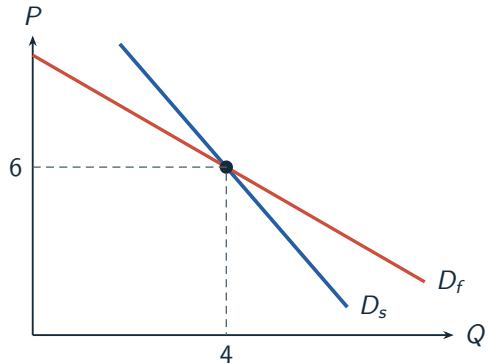
1. **Hold the comparison point fixed**
   Both curves pass through $(Q, P) = (4, 6)$ on the same axes.

A flatter curve through a given point



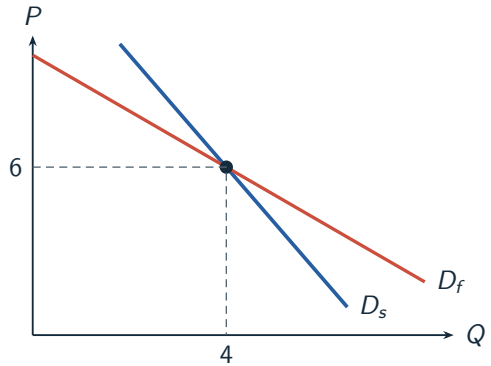
1. **Hold the comparison point fixed**
   Both curves pass through $(Q, P) = (4, 6)$ on the same axes.
2. **Compare their slopes**
   Steeper: $|dP/dQ| = 2$.
   Flatter: $|dP/dQ| = 1$.

A flatter curve through a given point



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3. **Compute the elasticities**
   $\varepsilon_{\text{steep}} = (1/2)(6/4) = 0.75$.
   $\varepsilon_{\text{flat}} = (1)(6/4) = 1.50$.

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At the same point, the flatter demand curve is more elastic.

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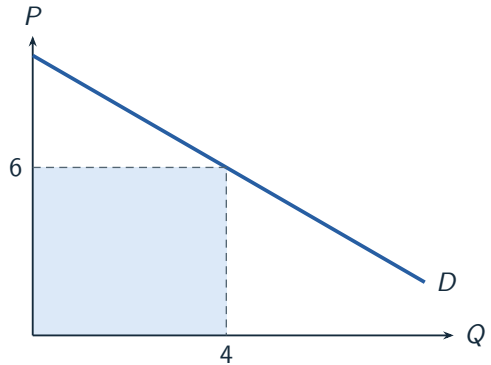
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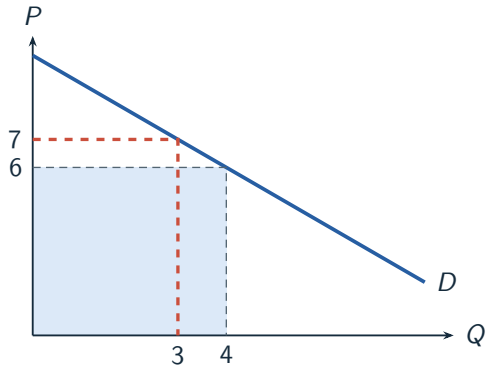
6 **Total revenue and applications** →
Price and quantity effects; drugs and farming

Total revenue is a price-quantity rectangle



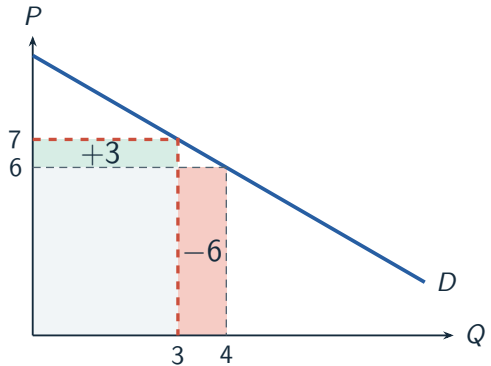
1. **Initial revenue**
   $TR = P \times Q = 6 \times 4 = 24.$

Total revenue is a price-quantity rectangle



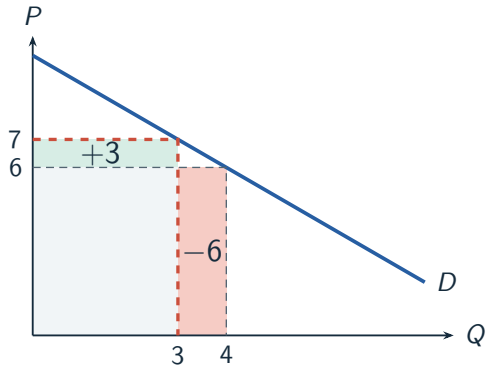
1. **Initial revenue**
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2. **A price increase**
   On $P = 10 - Q$, price rises to 7 and quantity falls to 3.

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   Gain 1 on each of 3 remaining units: $+3$.
   Lose the revenue on 1 forgone unit: -6.

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New revenue is $7 \times 3 = 21$.
The change is $3 - 6 = -3$.

Demand elasticity and total revenue

$$TR = P\,Q(P), \qquad \frac{dTR}{dP} = Q\,(1 - \varepsilon_d).$$

Demand	Price increases	Price decreases

Demand elasticity and total revenue

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Unit elastic: $\varepsilon_d = 1$	No local change	No local change

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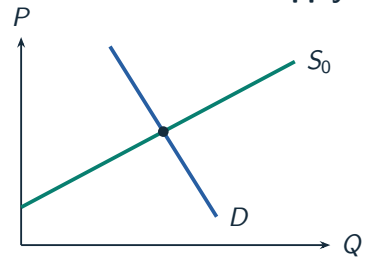
$$TR = P\,Q(P), \qquad \frac{dTR}{dP} = Q\,(1 - \varepsilon_d).$$

Demand	Price increases	Price decreases
Elastic: $\varepsilon_d > 1$	TR falls	TR rises
Unit elastic: $\varepsilon_d = 1$	No local change	No local change
Inelastic: $\varepsilon_d < 1$	TR rises	TR falls

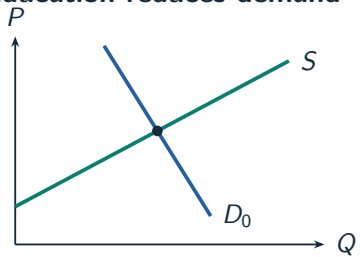
The derivative statement is local. A constant-unit-elastic demand curve has the same revenue at every point. For a finite move across a linear curve, evaluate the actual endpoint revenues.

Application: interdiction and education

Interdiction reduces supply

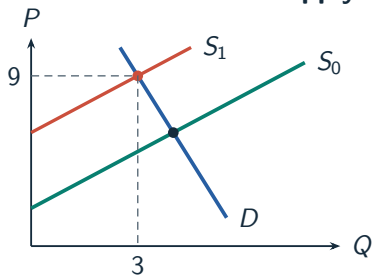


Education reduces demand



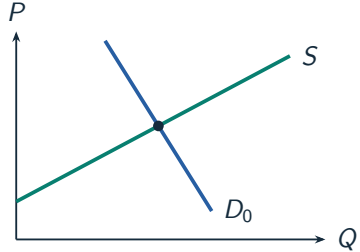
Application: interdiction and education

Interdiction reduces supply



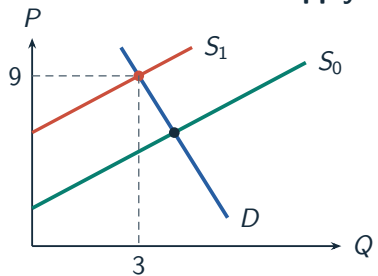
With inelastic demand, $TR : 24 \to 27$. Higher spending can increase incentives for drug-related crime.

Education reduces demand



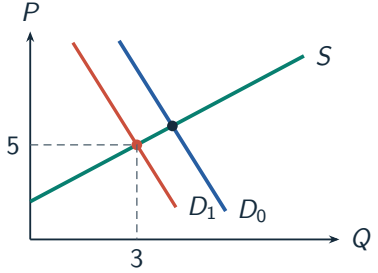
Application: interdiction and education

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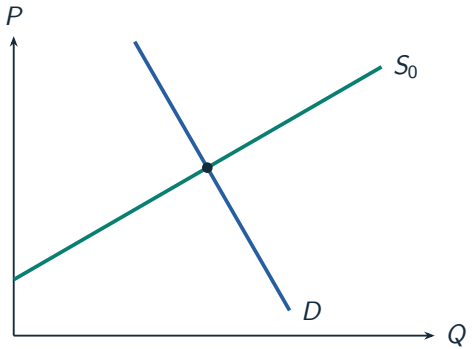
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Education reduces demand



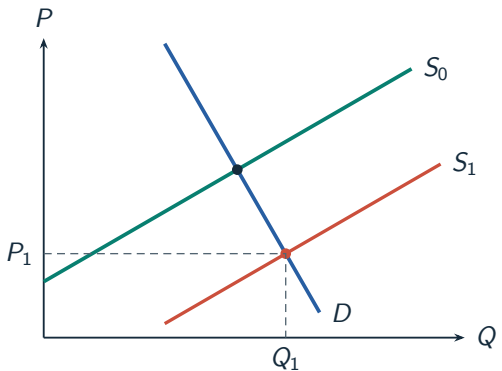
Price and quantity both fall. $TR : 24 \to 15$.

Application: good news for farming



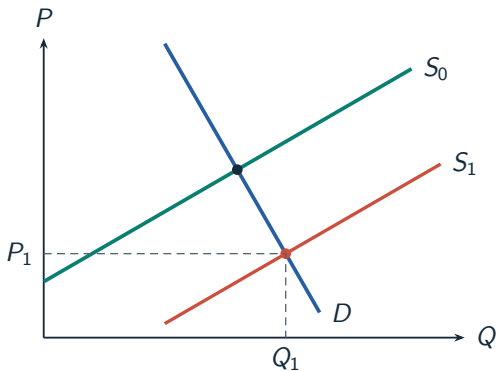
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   A new wheat hybrid raises output per acre by 20%.

Application: good news for farming



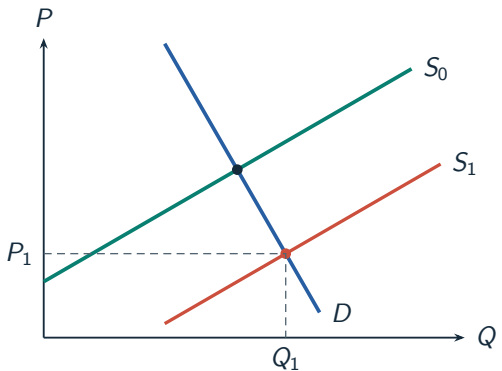
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   A new wheat hybrid raises output per acre by 20%.
2. **Supply shifts right**
   Equilibrium quantity rises and price falls.

Application: good news for farming



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   Quantity rises proportionately less than price falls, so total farm revenue can decrease.

Application: good news for farming



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   A new wheat hybrid raises output per acre by 20%.
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   Equilibrium quantity rises and price falls.
3. **Inelastic demand**
   Quantity rises proportionately less than price falls, so total farm revenue can decrease.

More productive farming can benefit consumers while reducing farmers' total revenue.

Price elasticity of demand: the framework

Framework complete. Connect the parts before moving on.

1 **Definition** ✓
Responsiveness of quantity demanded to price

2 **Calculation methods** ✓
Standard percentage changes and the midpoint method

3 **Determinants** ✓
Substitutes, market definition, necessities/luxuries, and time

4 **Properties and curve types** ✓
Elastic, inelastic, unit elastic, and the two extremes

5 **Along a linear demand curve** ✓
Constant slope, varying elasticity, and the same-point comparison

6 **Total revenue and applications** ✓
Price and quantity effects; drugs and farming

Elasticity: the framework

Our route. The highlighted item is our current focus.

1 **Price elasticity of demand** ✓
Definition, calculation, determinants, curves, and total revenue

2 **Other elasticities of demand** →
Income elasticity and cross-price elasticity

3 **Price elasticity of supply**
Calculation, properties, determinants, and variation along a curve

Income elasticity of demand

$$\varepsilon_Y = \frac{\%\,\Delta Q_D}{\%\,\Delta Y} \qquad \text{where } Y \text{ is income.}$$

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1. **Inferior goods**
   $\varepsilon_Y < 0$: higher income reduces demand.

Income elasticity of demand

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1. **Inferior goods**
   $\varepsilon_Y < 0$: higher income reduces demand.
2. **Normal goods**
   $\varepsilon_Y > 0$: higher income increases demand.

Income elasticity of demand

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1. **Inferior goods**
   $\varepsilon_Y < 0$: higher income reduces demand.
2. **Normal goods**
   $\varepsilon_Y > 0$: higher income increases demand.
3. **Necessities and luxuries**
   Among normal goods, necessities commonly have $0 < \varepsilon_Y < 1$, while luxuries have $\varepsilon_Y > 1$.

Cross-price elasticity of demand

$$\varepsilon_{AB} = \frac{\% \Delta Q_A^D}{\% \Delta P_B}.$$

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Cross-price elasticity of demand

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1. **Substitutes**
   $\varepsilon_{AB} > 0$: a higher price of B increases demand for A.
2. **Complements**
   $\varepsilon_{AB} < 0$: a higher price of B reduces demand for A.
3. **Direction and signs**
   Keep the order of the goods consistent. Own-price elasticity uses the price of A, while this elasticity uses the price of B.

Elasticity: the framework

Our route. The highlighted item is our current focus.

1 **Price elasticity of demand** ✓
Definition, calculation, determinants, curves, and total revenue

2 **Other elasticities of demand** ✓
Income elasticity and cross-price elasticity

3 **Price elasticity of supply** →
Calculation, properties, determinants, and variation along a curve

Price elasticity of supply: the framework

Our route. The highlighted item is our current focus.

1 **Definition and calculation** →
Percentage-change ratio and midpoint method

2 **Properties**
The five elasticity cases and a line through the origin

3 **Determinants**
Ease of adjusting production and the time horizon

4 **Variation and applications**
Capacity limits and the response to a demand increase

Price elasticity of supply

$$\varepsilon_s = \frac{\%\,\Delta Q_S}{\%\,\Delta P} = \frac{dQ_S}{dP}\frac{P}{Q_S}.$$

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1 Definition

How much the quantity supplied of a good responds to a change in the price of that good.

Price elasticity of supply

$$\varepsilon_s = \frac{\%\,\Delta Q_S}{\%\,\Delta P} = \frac{dQ_S}{dP}\frac{P}{Q_S}.$$

1. **Definition**
   How much the quantity supplied of a good responds to a change in the price of that good.
2. **Economic meaning**
   It measures sellers' price-sensitivity: how easily they can change quantity produced when price changes.

Supply elasticity: standard and midpoint methods

1. **Standard percentage changes**

   $\varepsilon_s = \dfrac{(Q_2^S - Q_1^S)/Q_1^S}{(P_2 - P_1)/P_1}$. Use the specified starting values.

Supply elasticity: standard and midpoint methods

1. **Standard percentage changes**

   $\varepsilon_s = \dfrac{(Q_2^S - Q_1^S)/Q_1^S}{(P_2 - P_1)/P_1}$. Use the specified starting values.

2. **Midpoint method**

   Use the average quantity and the average price as denominators.

$$\varepsilon_s = \frac{(Q_2^S - Q_1^S)/[(Q_1^S + Q_2^S)/2]}{(P_2 - P_1)/[(P_1 + P_2)/2]}.$$

Supply elasticity: standard and midpoint methods

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$$\varepsilon_s = \frac{(Q_2^S - Q_1^S)/[(Q_1^S + Q_2^S)/2]}{(P_2 - P_1)/[(P_1 + P_2)/2]}.$$

3. **Interpretation**

   Compare the percentage change in quantity supplied with the percentage change in price.

Price elasticity of supply: the framework

Our route. The highlighted item is our current focus.

1 **Definition and calculation** ✓
Percentage-change ratio and midpoint method

2 **Properties** →
The five elasticity cases and a line through the origin

3 **Determinants**
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Capacity limits and the response to a demand increase

Supply: elastic, unit elastic, and inelastic

1 **Elastic supply**

$\varepsilon_s > 1$: quantity supplied changes proportionately more than price.

Supply: elastic, unit elastic, and inelastic

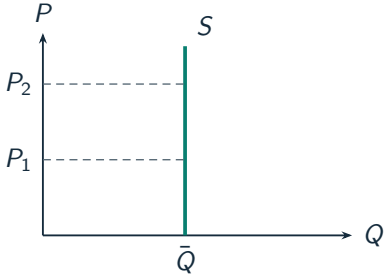
1. **Elastic supply**
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2. **Unit elastic supply**
   $\varepsilon_s = 1$: equal percentage changes in quantity supplied and price.

Supply: elastic, unit elastic, and inelastic

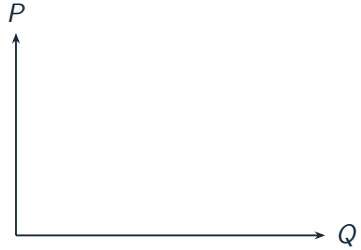
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   $\varepsilon_s = 1$: equal percentage changes in quantity supplied and price.
3. **Inelastic supply**
   $0 < \varepsilon_s < 1$: quantity supplied changes proportionately less than price.

The two extreme supply curves

Perfectly inelastic

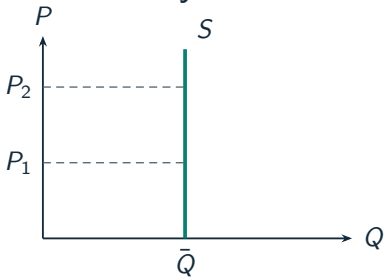


Perfectly elastic

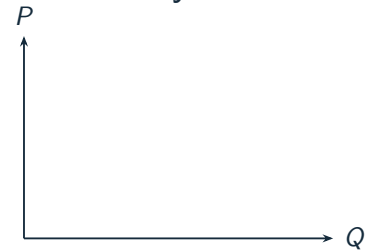


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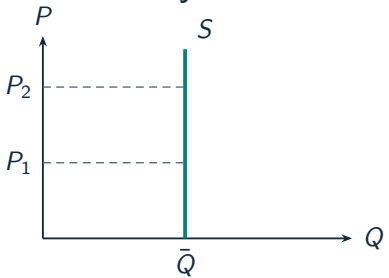
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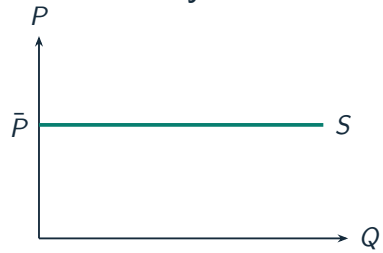
$\varepsilon_s = 0$. Quantity does not respond to price.

The two extreme supply curves

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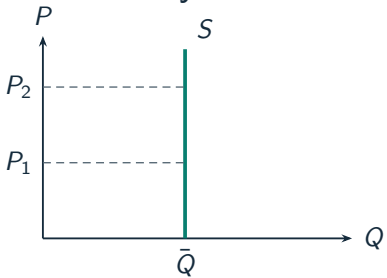
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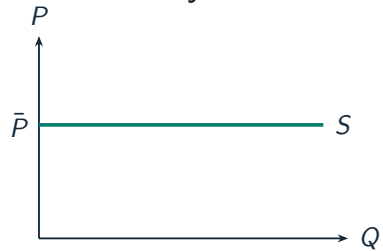
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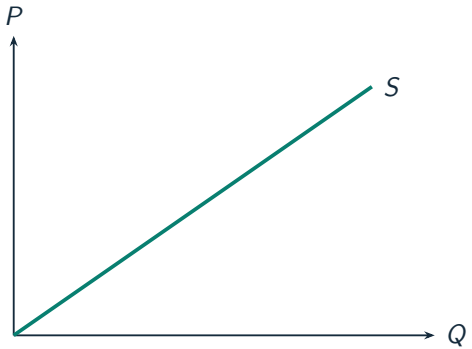
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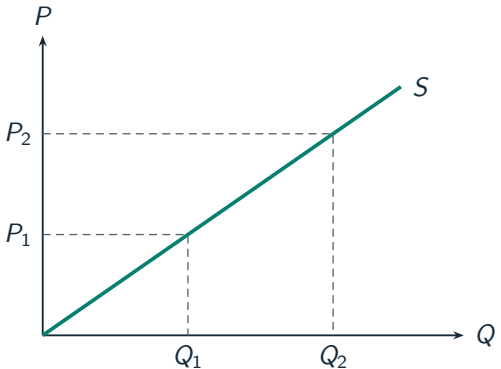
$\varepsilon_s = \infty$. Supply is arbitrarily responsive at the given price.

Linear supply and the origin



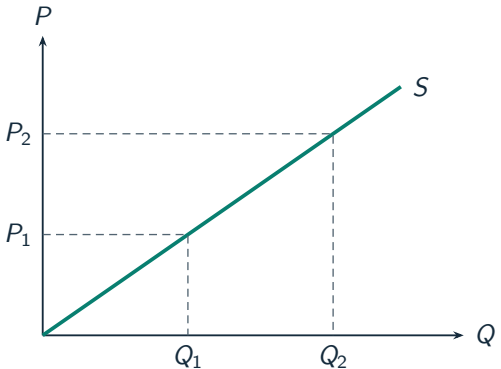
1 **A line through the origin**
For $P = bQ$, the ratio $P/Q = b$ is constant.

Linear supply and the origin



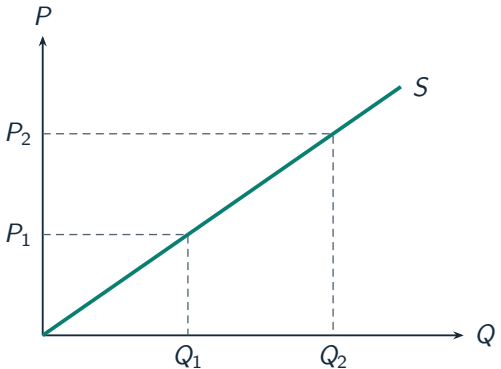
1. **A line through the origin**
   For $P = bQ$, the ratio $P/Q = b$ is constant.
2. **Its elasticity**
   $\varepsilon_s = \frac{1}{b}\frac{P}{Q} = 1$ at every positive point.

Linear supply and the origin



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2. **Its elasticity**
   $\varepsilon_s = \frac{1}{b}\frac{P}{Q} = 1$ at every positive point.
3. **A useful qualification**
   A straight supply curve is unit elastic everywhere only if it passes through the origin.

Linear supply and the origin



1. **A line through the origin**
   For $P = bQ$, the ratio $P/Q = b$ is constant.
2. **Its elasticity**
   $\varepsilon_s = \frac{1}{b}\frac{P}{Q} = 1$ at every positive point.
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   A straight supply curve is unit elastic everywhere only if it passes through the origin.

More generally, for $P = a + bQ$, $\varepsilon_s = P/(P - a)$. The intercept matters.

Price elasticity of supply: the framework

Our route. The highlighted item is our current focus.

1 **Definition and calculation** ✓
Percentage-change ratio and midpoint method

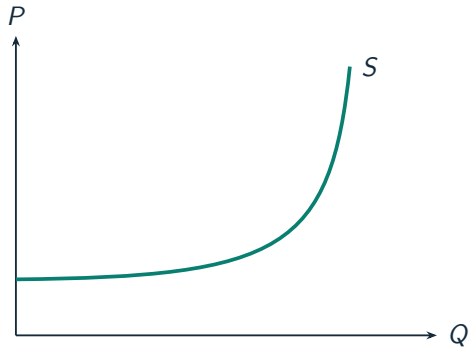
2 **Properties** ✓
The five elasticity cases and a line through the origin

3 **Determinants** →
Ease of adjusting production and the time horizon

4 **Variation and applications**
Capacity limits and the response to a demand increase

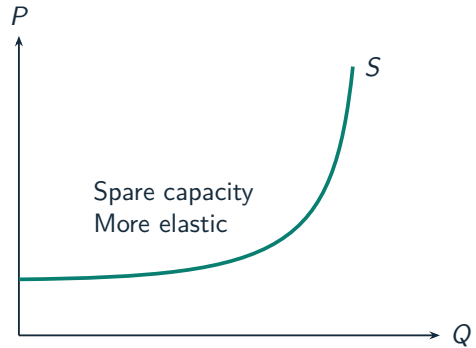
Supply elasticity: time and capacity

1. **Ease of changing production**
   Available inputs, inventories, and spare capacity make supply more responsive.



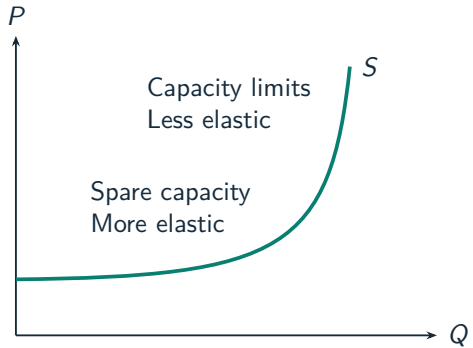
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   Supply is usually more elastic in the long run, when firms can adjust plant and equipment.



Supply elasticity: time and capacity

1. **Ease of changing production**
   Available inputs, inventories, and spare capacity make supply more responsive.
2. **Time horizon**
   Supply is usually more elastic in the long run, when firms can adjust plant and equipment.
3. **Capacity limits**
   At low output, spare capacity can make supply elastic. Near capacity, additional output is harder to produce.



Price elasticity of supply: the framework

Our route. The highlighted item is our current focus.

1 **Definition and calculation** ✓
Percentage-change ratio and midpoint method

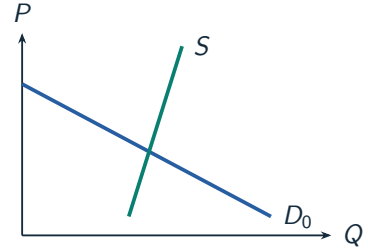
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Ease of adjusting production and the time horizon

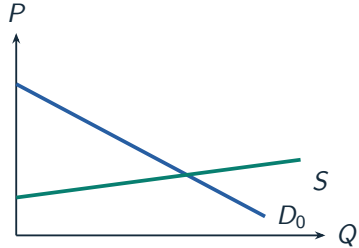
4 **Variation and applications** →
Capacity limits and the response to a demand increase

Supply elasticity and a demand increase

Beachfront property

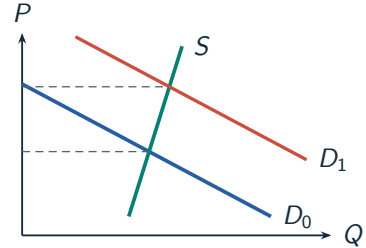


New cars over a longer horizon



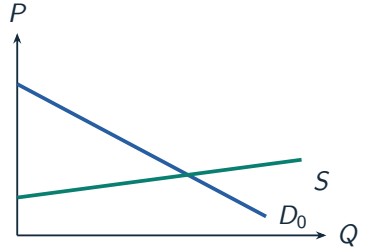
Supply elasticity and a demand increase

Beachfront property



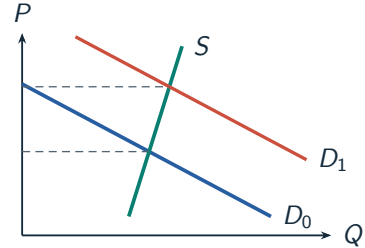
Relatively inelastic supply. A larger price response and a smaller quantity response.

New cars over a longer horizon



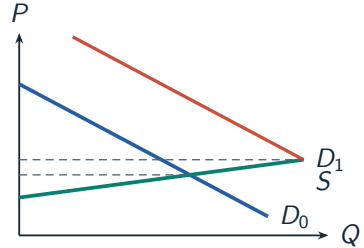
Supply elasticity and a demand increase

Beachfront property



Relatively inelastic supply. A larger price response and a smaller quantity response.

New cars over a longer horizon



More elastic supply. Firms can increase quantity more readily.

Price elasticity of supply: the framework

Framework complete. Connect the parts before moving on.

1 **Definition and calculation** ✓
Percentage-change ratio and midpoint method

2 **Properties** ✓
The five elasticity cases and a line through the origin

3 **Determinants** ✓
Ease of adjusting production and the time horizon

4 **Variation and applications** ✓
Capacity limits and the response to a demand increase

Elasticity: the framework

Framework complete. Connect the parts before moving on.

1. **Price elasticity of demand** ✓
   Definition, calculation, determinants, curves, and total revenue
2. **Other elasticities of demand** ✓
   Income elasticity and cross-price elasticity
3. **Price elasticity of supply** ✓
   Calculation, properties, determinants, and variation along a curve

05

Consumer and producer surplus

Willingness to pay, cost, and the graphical analysis of market efficiency

Consumer and producer surplus: the framework

Our route. The highlighted item is our current focus.

1 **Consumer surplus** →
Willingness to pay, demand, area, and a price change

2 **Producer surplus**
Cost, supply, area, and a price change

3 **Market efficiency**
Total surplus, allocation, equilibrium, and welfare economics

Welfare economics and the allocation of resources

1 **Welfare economics**
The study of how the allocation of resources affects economic well-being.

Welfare economics and the allocation of resources

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   The study of how the allocation of resources affects economic well-being.
2. **Allocation of resources**
   How much of each good is produced, which producers produce it, and which consumers consume it.

Welfare economics and the allocation of resources

1. **Welfare economics**
   The study of how the allocation of resources affects economic well-being.
2. **Allocation of resources**
   How much of each good is produced, which producers produce it, and which consumers consume it.
3. **The question**
   How can society make the benefits received by buyers and sellers as large as possible?

Consumer surplus: the framework

Our route. The highlighted item is our current focus.

1. **Willingness to pay** →
   The maximum amount a buyer will pay
2. **WTP and the demand curve**
   The marginal buyer; step and smooth curves
3. **Compute consumer surplus**
   WTP minus payment; area under demand above price
4. **How a higher price reduces CS**
   Remaining buyers pay more; some buyers leave

Willingness to pay

1 Definition

The maximum amount that a buyer will pay for a good.

Willingness to pay

1. **Definition**
   The maximum amount that a buyer will pay for a good.
2. **Interpretation**
   Willingness to pay measures how much the buyer values the good.

Willingness to pay

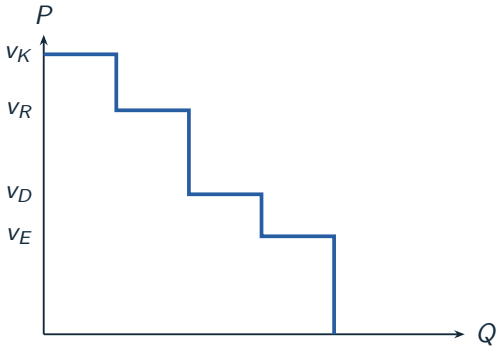
1. **Definition**
   The maximum amount that a buyer will pay for a good.
2. **Interpretation**
   Willingness to pay measures how much the buyer values the good.
3. **The purchase decision**
   A buyer purchases when willingness to pay exceeds the price, and is indifferent when the two are equal.

Consumer surplus: the framework

Our route. The highlighted item is our current focus.

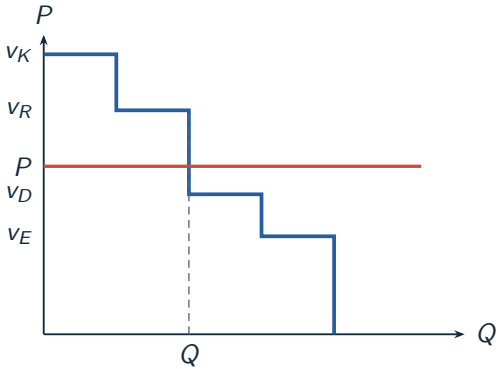
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   WTP minus payment; area under demand above price
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   Remaining buyers pay more; some buyers leave

WTP and the demand curve



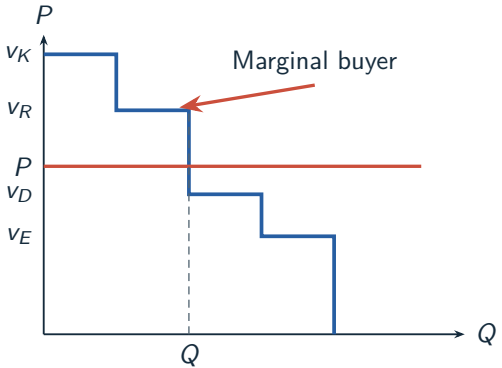
1. **Order buyers by willingness to pay**
   Kenny, Richard, David, and Eric have descending valuations $v_K > v_R > v_D > v_E$.

WTP and the demand curve



1. **Order buyers by willingness to pay**
   Kenny, Richard, David, and Eric have descending valuations $v_K > v_R > v_D > v_E$.
2. **Who buys at price P?**
   The buyers whose valuations exceed the price.

WTP and the demand curve



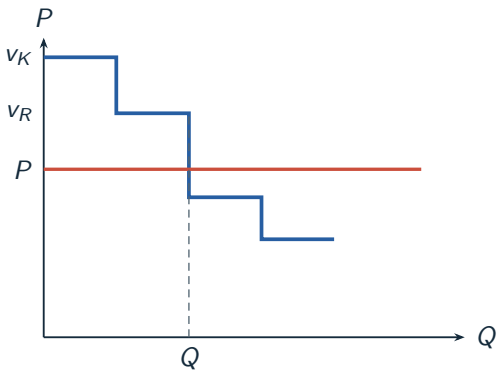
1. **Order buyers by willingness to pay**
   Kenny, Richard, David, and Eric have descending valuations $v_K > v_R > v_D > v_E$.
2. **Who buys at price P?**
   The buyers whose valuations exceed the price.
3. **The marginal buyer**
   At each quantity, demand's height is the WTP of the marginal buyer, who would leave if price exceeded that height.

Consumer surplus: the framework

Our route. The highlighted item is our current focus.

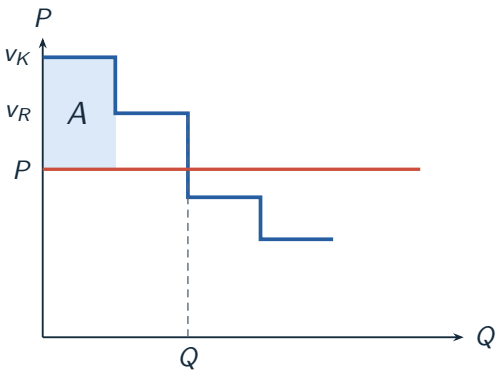
1. **Willingness to pay** ✓
   The maximum amount a buyer will pay
2. **WTP and the demand curve** ✓
   The marginal buyer; step and smooth curves
3. **Compute consumer surplus** →
   WTP minus payment; area under demand above price
4. **How a higher price reduces CS**
   Remaining buyers pay more; some buyers leave

Consumer surplus: the lettered areas



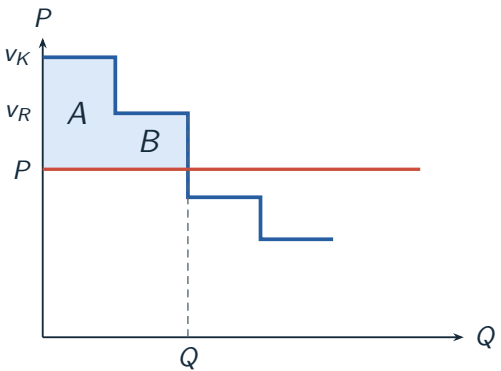
1 **Consumer surplus**
The amount a buyer is willing to pay minus the amount the buyer actually pays.

Consumer surplus: the lettered areas



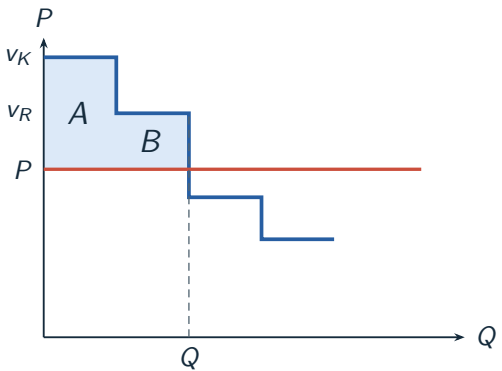
1. **Consumer surplus**
   The amount a buyer is willing to pay minus the amount the buyer actually pays.
2. **Kenny's surplus**
   $CS_K = v_K - P = A$.

Consumer surplus: the lettered areas



1. **Consumer surplus**
   The amount a buyer is willing to pay minus the amount the buyer actually pays.
2. **Kenny's surplus**
   $CS_K = v_K - P = A$.
3. **Richard's surplus**
   $CS_R = v_R - P = B$.

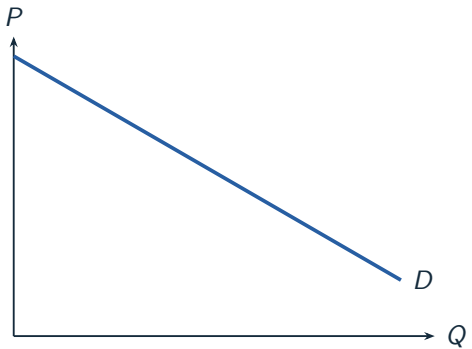
Consumer surplus: the lettered areas



1. **Consumer surplus**
   The amount a buyer is willing to pay minus the amount the buyer actually pays.
2. **Kenny's surplus**
   $CS_K = v_K - P = A$.
3. **Richard's surplus**
   $CS_R = v_R - P = B$.

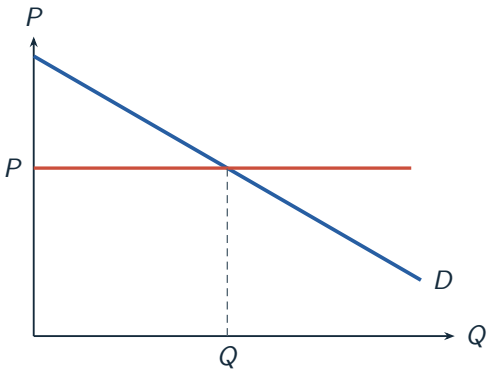
Total $CS = A + B$.

Consumer surplus with a smooth demand curve



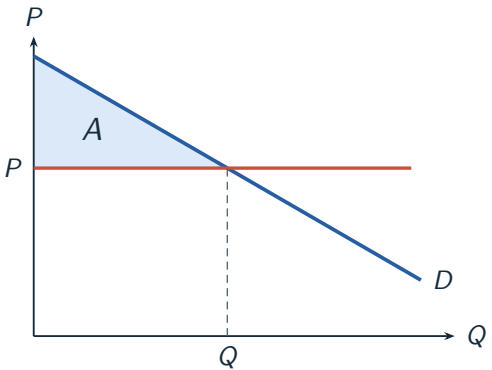
1. **Many buyers**
   A huge number of buyers gives many tiny steps, so demand looks like a smooth curve.

Consumer surplus with a smooth demand curve



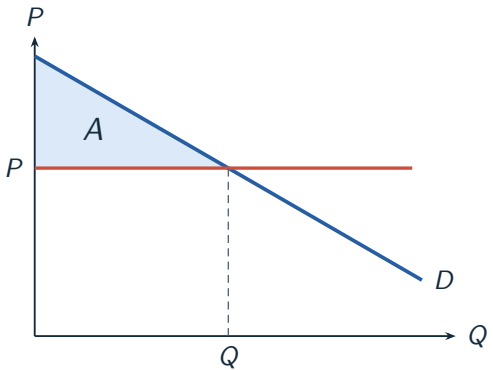
1. **Many buyers**
   A huge number of buyers gives many tiny steps, so demand looks like a smooth curve.
2. **Locate the purchases**
   At price P, buyers purchase quantity Q.

Consumer surplus with a smooth demand curve



1. **Many buyers**
   A huge number of buyers gives many tiny steps, so demand looks like a smooth curve.
2. **Locate the purchases**
   At price P, buyers purchase quantity Q.
3. **Read the area**
   Total CS is the area under demand and above the price, up to Q.

Consumer surplus with a smooth demand curve



1. **Many buyers**
   A huge number of buyers gives many tiny steps, so demand looks like a smooth curve.
2. **Locate the purchases**
   At price P, buyers purchase quantity Q.
3. **Read the area**
   Total CS is the area under demand and above the price, up to Q.

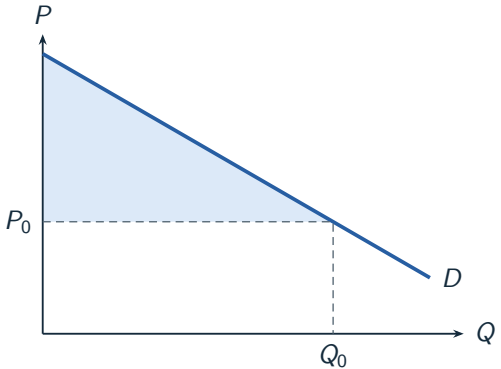
$CS = A$.

Consumer surplus: the framework

Our route. The highlighted item is our current focus.

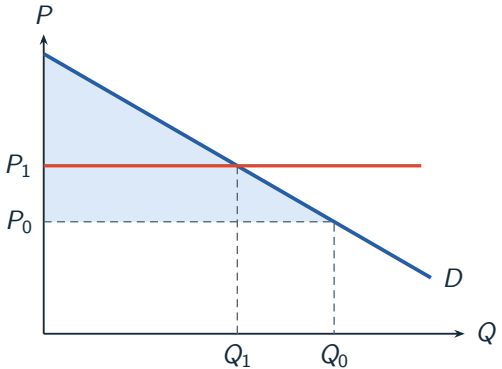
1. **Willingness to pay** ✓
   The maximum amount a buyer will pay
2. **WTP and the demand curve** ✓
   The marginal buyer; step and smooth curves
3. **Compute consumer surplus** ✓
   WTP minus payment; area under demand above price
4. **How a higher price reduces CS** →
   Remaining buyers pay more; some buyers leave

A higher price reduces consumer surplus



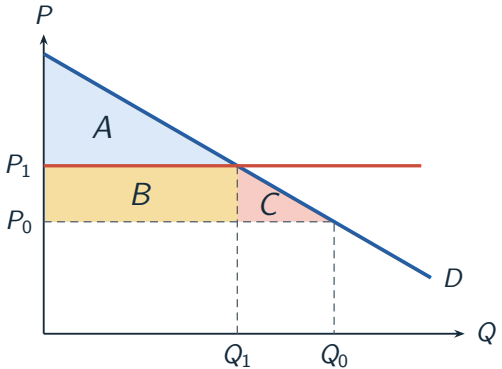
1. **Before the price increase**
   The full initial CS is the large triangle.

A higher price reduces consumer surplus



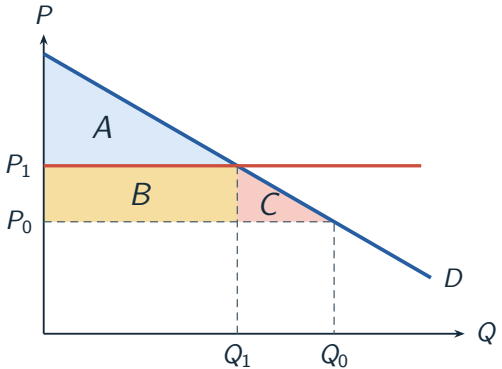
1. **Before the price increase**
   The full initial CS is the large triangle.
2. **A higher price**
   Some buyers remain; other buyers leave the market.

A higher price reduces consumer surplus



1. **Before the price increase**
   The full initial CS is the large triangle.
2. **A higher price**
   Some buyers remain; other buyers leave the market.
3. **Decompose the change**
   $CS_0 = A + B + C$, $CS_1 = A$.
   Remaining buyers pay more: B.
   Exiting buyers lose C.

A higher price reduces consumer surplus



1. **Before the price increase**
   The full initial CS is the large triangle.
2. **A higher price**
   Some buyers remain; other buyers leave the market.
3. **Decompose the change**
   $CS_0 = A + B + C$, $CS_1 = A$.
   Remaining buyers pay more: B.
   Exiting buyers lose C.

$\Delta CS = -(B + C)$.

Consumer surplus: the framework

Framework complete. Connect the parts before moving on.

1. **Willingness to pay** ✓
   The maximum amount a buyer will pay
2. **WTP and the demand curve** ✓
   The marginal buyer; step and smooth curves
3. **Compute consumer surplus** ✓
   WTP minus payment; area under demand above price
4. **How a higher price reduces CS** ✓
   Remaining buyers pay more; some buyers leave

Consumer and producer surplus: the framework

Our route. The highlighted item is our current focus.

1 **Consumer surplus** ✓
Willingness to pay, demand, area, and a price change

2 **Producer surplus** →
Cost, supply, area, and a price change

3 **Market efficiency**
Total surplus, allocation, equilibrium, and welfare economics

Producer surplus: the framework

Our route. The highlighted item is our current focus.

1 **Cost** →
What the seller must give up; willingness to sell

2 **Cost and the supply curve**
The marginal seller; step and smooth curves

3 **Compute producer surplus**
Payment minus cost; area above supply below price

4 **How a lower price reduces PS**
Remaining sellers receive less; some sellers leave

Cost and willingness to sell

1 Cost

The value of everything a seller must give up to produce a good.

Cost and willingness to sell

1. **Cost**
   The value of everything a seller must give up to produce a good.
2. **Willingness to sell**
   Cost measures willingness to sell: a seller produces and sells when the price exceeds cost.

Cost and willingness to sell

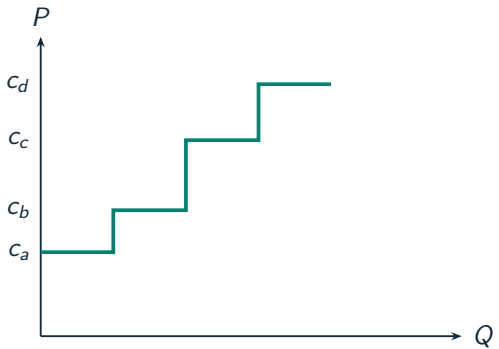
1. **Cost**
   The value of everything a seller must give up to produce a good.
2. **Willingness to sell**
   Cost measures willingness to sell: a seller produces and sells when the price exceeds cost.
3. **The marginal seller**
   At each quantity, the height of supply is the cost of the marginal seller, who would leave if price fell below that cost.

Producer surplus: the framework

Our route. The highlighted item is our current focus.

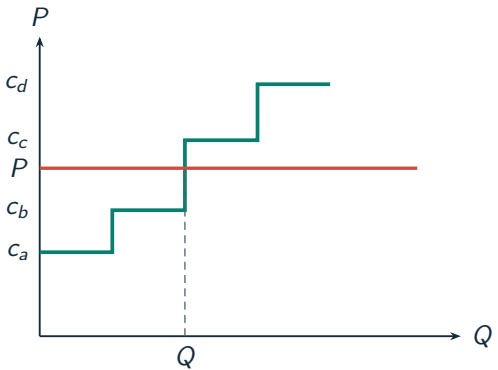
1. **Cost** ✓
   What the seller must give up; willingness to sell
2. **Cost and the supply curve** →
   The marginal seller; step and smooth curves
3. **Compute producer surplus**
   Payment minus cost; area above supply below price
4. **How a lower price reduces PS**
   Remaining sellers receive less; some sellers leave

Cost and the supply curve



1. **Order sellers by cost**
   The costs satisfy
   $c_a < c_b < c_c < c_d$.

Cost and the supply curve



1. **Order sellers by cost**
   The costs satisfy
   $c_a < c_b < c_c < c_d$.
2. **Who sells at price P?**
   The sellers whose costs are below the price.

Cost and the supply curve



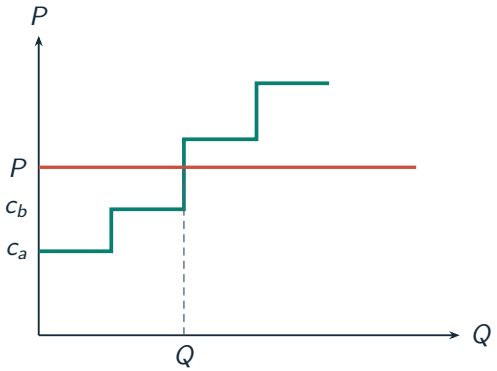
1. **Order sellers by cost**
   The costs satisfy $c_a < c_b < c_c < c_d$.
2. **Who sells at price P?**
   The sellers whose costs are below the price.
3. **Interpret the curve**
   Each step is a seller's cost. With many small sellers, the steps become approximately smooth.

Producer surplus: the framework

Our route. The highlighted item is our current focus.

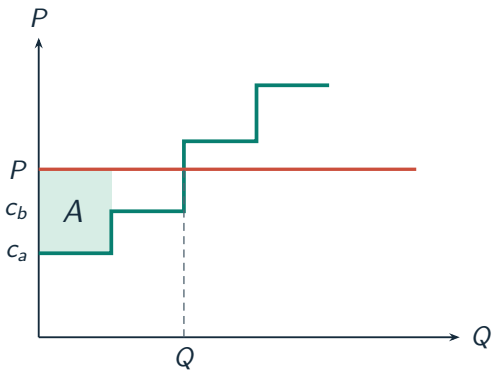
1. **Cost** ✓
   What the seller must give up; willingness to sell
2. **Cost and the supply curve** ✓
   The marginal seller; step and smooth curves
3. **Compute producer surplus** →
   Payment minus cost; area above supply below price
4. **How a lower price reduces PS**
   Remaining sellers receive less; some sellers leave

Producer surplus: the lettered areas



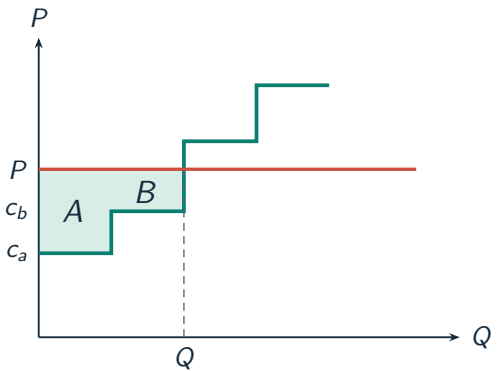
1 **Producer surplus**
The amount a seller is paid for a good minus the seller's cost of providing it.

Producer surplus: the lettered areas



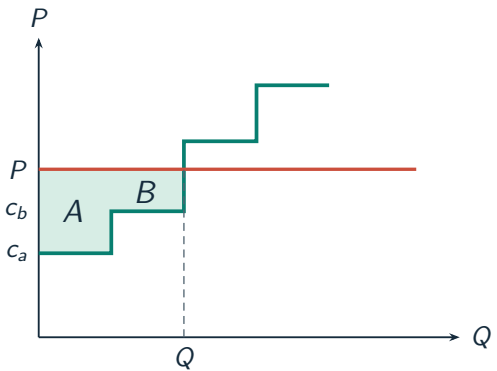
1. **Producer surplus**
   The amount a seller is paid for a good minus the seller's cost of providing it.
2. **Seller a**
   $PS_a = P - c_a = A$.

Producer surplus: the lettered areas



1. **Producer surplus**
   The amount a seller is paid for a good minus the seller's cost of providing it.
2. **Seller a**
   $PS_a = P - c_a = A$.
3. **Seller b**
   $PS_b = P - c_b = B$.

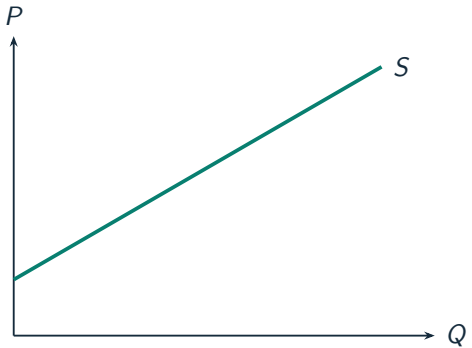
Producer surplus: the lettered areas



1. **Producer surplus**
   The amount a seller is paid for a good minus the seller's cost of providing it.
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3. **Seller b**
   $PS_b = P - c_b = B$.

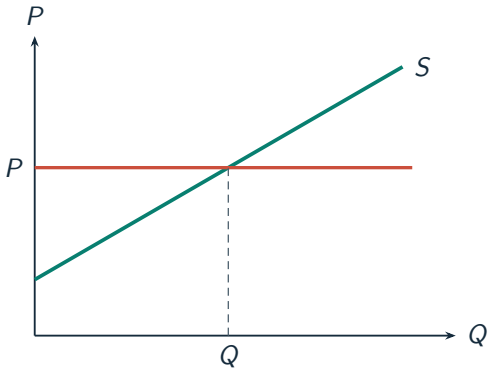
Total $PS = A + B$.

Producer surplus with a smooth supply curve



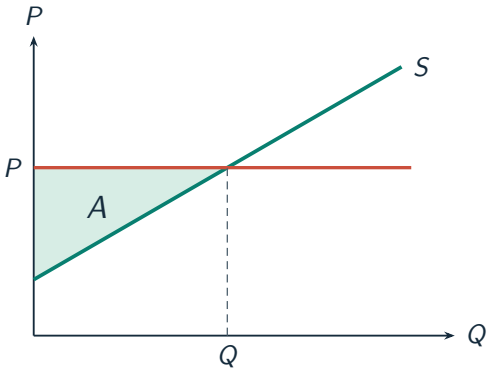
1. **Many sellers**
   With many small sellers, the step supply curve becomes approximately smooth.

Producer surplus with a smooth supply curve



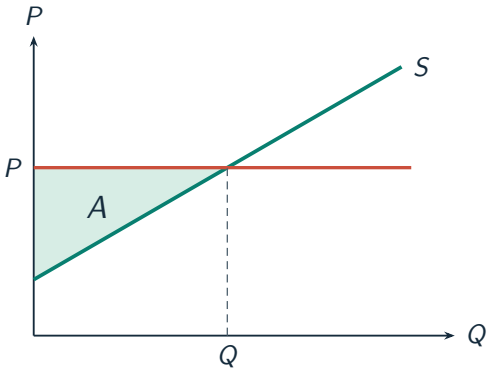
1. **Many sellers**
   With many small sellers, the step supply curve becomes approximately smooth.
2. **Locate the sales**
   At price P, sellers supply quantity Q.

Producer surplus with a smooth supply curve



1. **Many sellers**
   With many small sellers, the step supply curve becomes approximately smooth.
2. **Locate the sales**
   At price P, sellers supply quantity Q.
3. **Read the area**
   Total PS is the area above supply and below the price, up to Q.

Producer surplus with a smooth supply curve



1. **Many sellers**
   With many small sellers, the step supply curve becomes approximately smooth.
2. **Locate the sales**
   At price P, sellers supply quantity Q.
3. **Read the area**
   Total PS is the area above supply and below the price, up to Q.

$PS = A$.

Producer surplus: the framework

Our route. The highlighted item is our current focus.

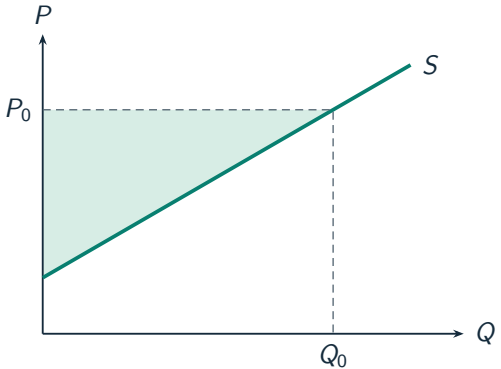
1 **Cost** ✓
What the seller must give up; willingness to sell

2 **Cost and the supply curve** ✓
The marginal seller; step and smooth curves

3 **Compute producer surplus** ✓
Payment minus cost; area above supply below price

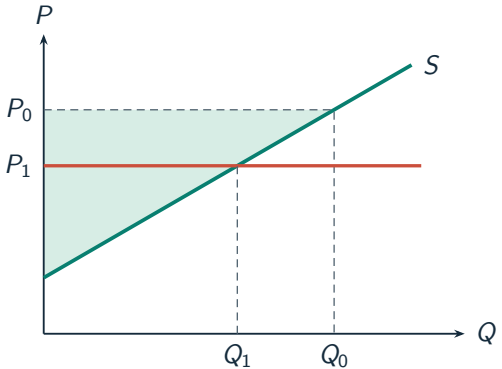
4 **How a lower price reduces PS** →
Remaining sellers receive less; some sellers leave

How a lower price reduces producer surplus



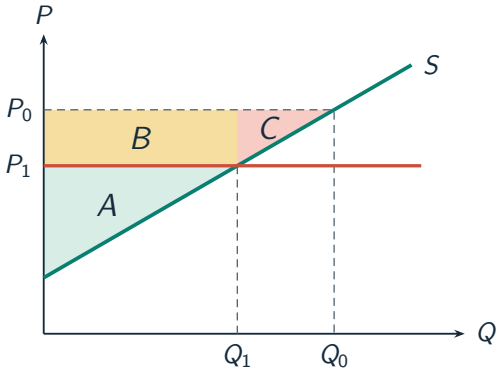
1. **Before the price decrease**
   The full initial PS is the large triangle.

How a lower price reduces producer surplus



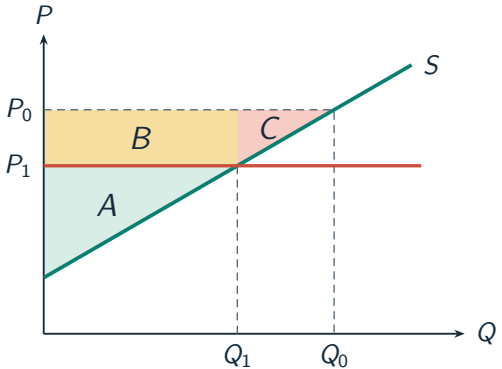
1. **Before the price decrease**
   The full initial PS is the large triangle.
2. **A lower price**
   Some sellers remain; other sellers leave the market.

How a lower price reduces producer surplus



1. **Before the price decrease**
   The full initial PS is the large triangle.
2. **A lower price**
   Some sellers remain; other sellers leave the market.
3. **Decompose the change**
   $PS_0 = A + B + C$, $PS_1 = A$.
   Remaining sellers receive less: B.
   Exiting sellers lose C.

How a lower price reduces producer surplus



1. **Before the price decrease**
   The full initial PS is the large triangle.
2. **A lower price**
   Some sellers remain; other sellers leave the market.
3. **Decompose the change**
   $PS_0 = A + B + C$, $PS_1 = A$.
   Remaining sellers receive less: B.
   Exiting sellers lose C.

$\Delta PS = -(B + C)$.

Producer surplus: the framework

Framework complete. Connect the parts before moving on.

1 **Cost** ✓
What the seller must give up; willingness to sell

2 **Cost and the supply curve** ✓
The marginal seller; step and smooth curves

3 **Compute producer surplus** ✓
Payment minus cost; area above supply below price

4 **How a lower price reduces PS** ✓
Remaining sellers receive less; some sellers leave

Consumer and producer surplus: the framework

Our route. The highlighted item is our current focus.

1 **Consumer surplus** ✓
Willingness to pay, demand, area, and a price change

2 **Producer surplus** ✓
Cost, supply, area, and a price change

3 **Market efficiency** →
Total surplus, allocation, equilibrium, and welfare economics

Market efficiency: the framework

Our route. The highlighted item is our current focus.

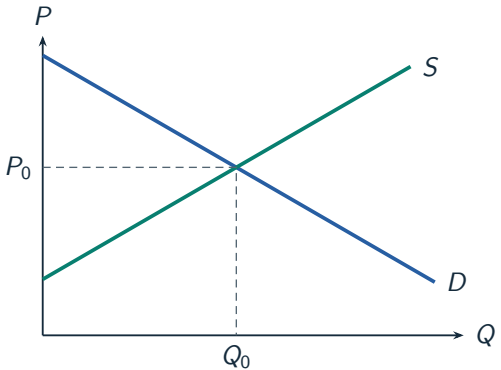
1 **Total surplus** →
Value to buyers minus cost to sellers

2 **Allocation of resources**
Who consumes, who produces, and how much

3 **The equilibrium quantity**
Lost gains from too little or too much production

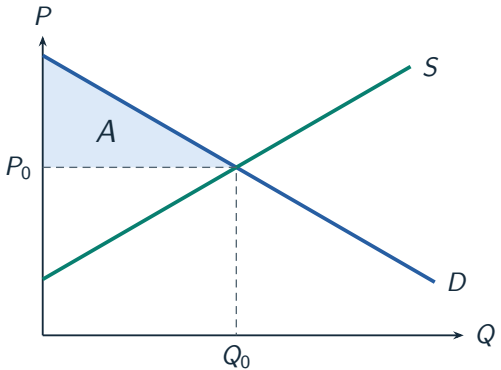
4 **Welfare economics and its assumptions**
Competitive markets, market power, and externalities

Total surplus: value minus cost



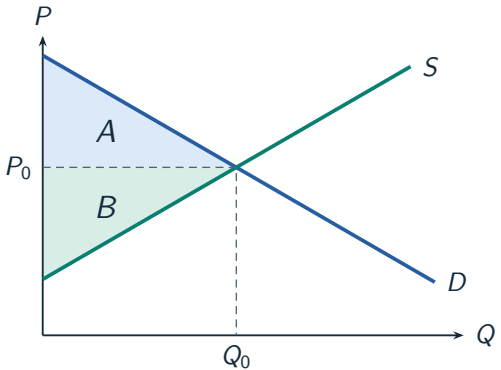
1 **Total surplus**
$TS = CS + PS$.

Total surplus: value minus cost



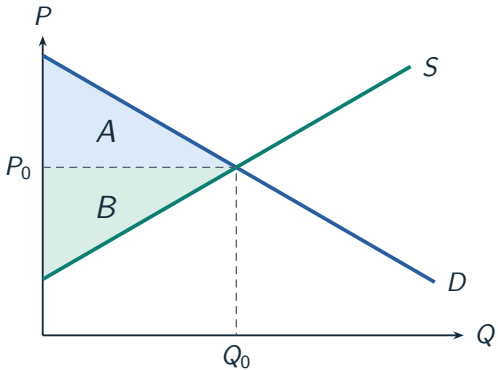
1. **Total surplus**
   $TS = CS + PS$.
2. **Consumer surplus**
   Value to buyers minus amount paid by buyers: $CS = A$.

Total surplus: value minus cost



1. **Total surplus**
   $TS = CS + PS$.
2. **Consumer surplus**
   Value to buyers minus amount paid by buyers: $CS = A$.
3. **Producer surplus**
   Amount received by sellers minus cost to sellers: $PS = B$.

Total surplus: value minus cost



1. **Total surplus**
   $TS = CS + PS$.
2. **Consumer surplus**
   Value to buyers minus amount paid by buyers: $CS = A$.
3. **Producer surplus**
   Amount received by sellers minus cost to sellers: $PS = B$.

Payments cancel.

$TS = A + B$

$=$ **Value to buyers** $-$ **Cost to sellers**.

Market efficiency: the framework

Our route. The highlighted item is our current focus.

1. **Total surplus** ✓
   Value to buyers minus cost to sellers
2. **Allocation of resources** →
   Who consumes, who produces, and how much
3. **The equilibrium quantity**
   Lost gains from too little or too much production
4. **Welfare economics and its assumptions**
   Competitive markets, market power, and externalities

Which allocation maximizes total surplus?

1. **Who consumes?**
   Goods are consumed by the buyers who value them most highly.

Which allocation maximizes total surplus?

1. **Who consumes?**
   Goods are consumed by the buyers who value them most highly.
2. **Who produces?**
   Goods are produced by the producers with the lowest costs.

Which allocation maximizes total surplus?

1. **Who consumes?**
   Goods are consumed by the buyers who value them most highly.
2. **Who produces?**
   Goods are produced by the producers with the lowest costs.
3. **How much?**
   Raising or lowering the quantity would not increase total surplus.

Which allocation maximizes total surplus?

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   Goods are consumed by the buyers who value them most highly.
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   Goods are produced by the producers with the lowest costs.
3. **How much?**
   Raising or lowering the quantity would not increase total surplus.

An efficient allocation of resources maximizes total surplus.

Market efficiency: the framework

Our route. The highlighted item is our current focus.

1 **Total surplus** ✓
Value to buyers minus cost to sellers

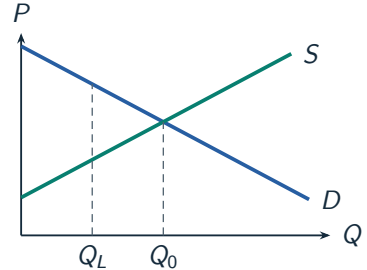
2 **Allocation of resources** ✓
Who consumes, who produces, and how much

3 **The equilibrium quantity** →
Lost gains from too little or too much production

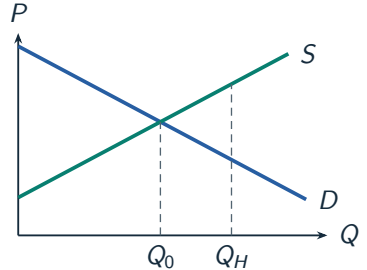
4 **Welfare economics and its assumptions**
Competitive markets, market power, and externalities

Why the competitive quantity is efficient

Too little production

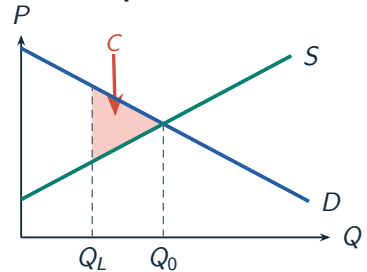


Too much production



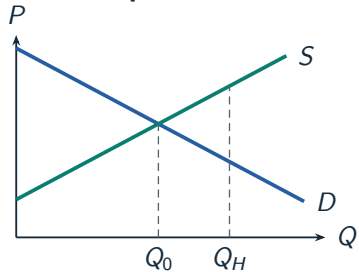
Why the competitive quantity is efficient

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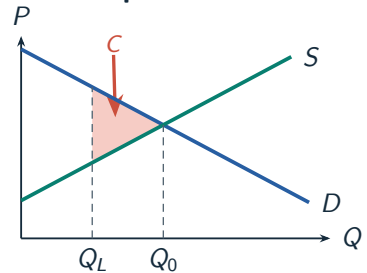
For units between Q_L and Q_0, value exceeds cost. Forgoing them loses surplus: $DWL = C$.

Too much production



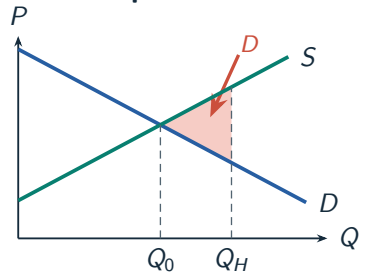
Why the competitive quantity is efficient

Too little production



For units between Q_L and Q_0, value exceeds cost. Forgoing them loses surplus: $DWL = C$.

Too much production



Beyond Q_0, cost exceeds value. Producing these units loses surplus: $DWL = D$.

Market efficiency: the framework

Our route. The highlighted item is our current focus.

1. **Total surplus** ✓
   Value to buyers minus cost to sellers
2. **Allocation of resources** ✓
   Who consumes, who produces, and how much
3. **The equilibrium quantity** ✓
   Lost gains from too little or too much production
4. **Welfare economics and its assumptions** →
   Competitive markets, market power, and externalities

Welfare economics: conclusion and assumptions

1. **Competitive equilibrium**
   The equilibrium quantity maximizes total surplus, with production by the lowest-cost producers and consumption by the highest-value buyers.

Welfare economics: conclusion and assumptions

1. **Competitive equilibrium**
   The equilibrium quantity maximizes total surplus, with production by the lowest-cost producers and consumption by the highest-value buyers.
2. **Deadweight loss**
   The reduction in total surplus relative to the efficient competitive benchmark: a measure of inefficiency.

Welfare economics: conclusion and assumptions

1. **Competitive equilibrium**
   The equilibrium quantity maximizes total surplus, with production by the lowest-cost producers and consumption by the highest-value buyers.
2. **Deadweight loss**
   The reduction in total surplus relative to the efficient competitive benchmark: a measure of inefficiency.
3. **When the conclusion can fail**
   The model assumes perfectly competitive markets and no unpriced external effects. Market power and externalities can prevent efficient allocation.

Market efficiency: the framework

Framework complete. Connect the parts before moving on.

1. **Total surplus** ✓
   Value to buyers minus cost to sellers
2. **Allocation of resources** ✓
   Who consumes, who produces, and how much
3. **The equilibrium quantity** ✓
   Lost gains from too little or too much production
4. **Welfare economics and its assumptions** ✓
   Competitive markets, market power, and externalities

Consumer and producer surplus: the framework

Framework complete. Connect the parts before moving on.

1 **Consumer surplus** ✓
Willingness to pay, demand, area, and a price change

2 **Producer surplus** ✓
Cost, supply, area, and a price change

3 **Market efficiency** ✓
Total surplus, allocation, equilibrium, and welfare economics

06

Price controls and output policies

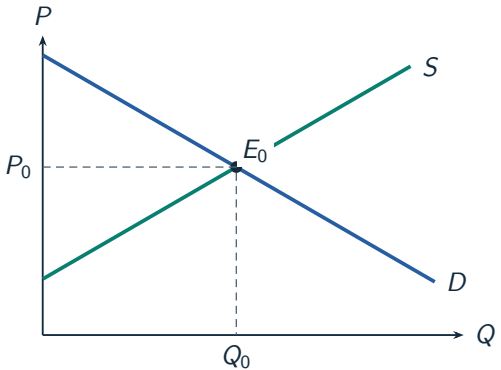
Binding constraints, the quantity actually traded, and the welfare consequences

Price controls and output policies: the framework

Our route. The highlighted item is our current focus.

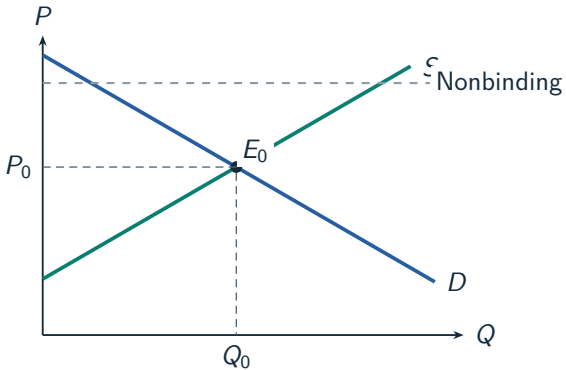
1. **Price ceilings** →
   Nonbinding/binding, rent control, and welfare
2. **Price floors**
   Nonbinding/binding, labor markets, and welfare
3. **Assess price controls**
   Objectives, price signals, rationing, and unsold output
4. **Other ways to maintain a higher price**
   Price supports, production quotas, and incentive programs

A price ceiling is a legal maximum



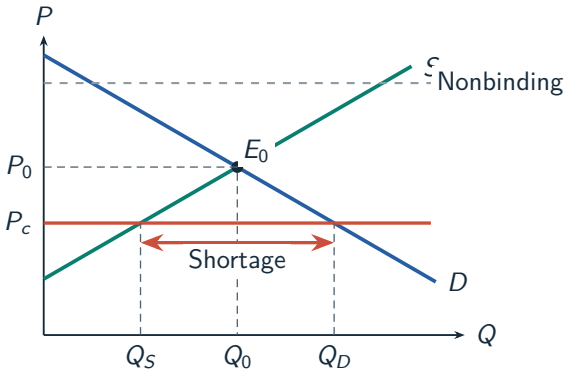
1. **The benchmark**
   Without control, price and quantity are P_0, Q_0.

A price ceiling is a legal maximum



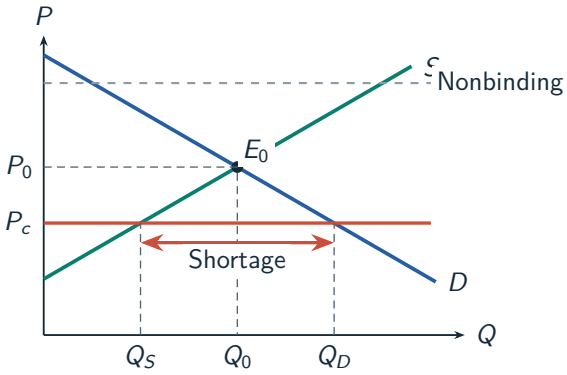
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   A ceiling above P_0 leaves the equilibrium feasible.

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3. **A binding ceiling**
   At $P_c < P_0$, the ceiling creates a shortage of $Q_D - Q_S$.

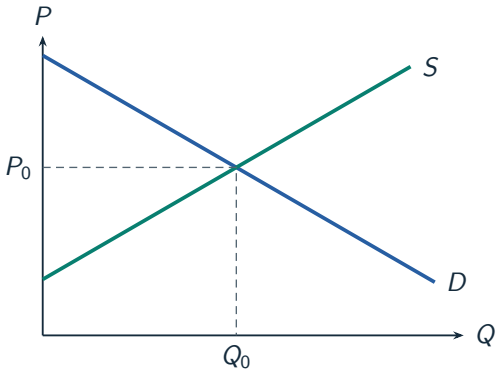
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   At $P_c < P_0$, the ceiling creates a shortage of $Q_D - Q_S$.

With no extra supply, at most Q_S can be traded.

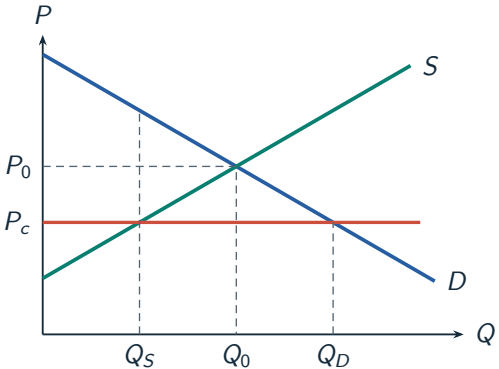
Price ceiling: surplus and deadweight loss



1. **Free-market surplus**
   Trade occurs at P_0, Q_0.

Assumes the limited units go to buyers with the highest willingness to pay. Inefficient rationing or waiting can create additional losses.

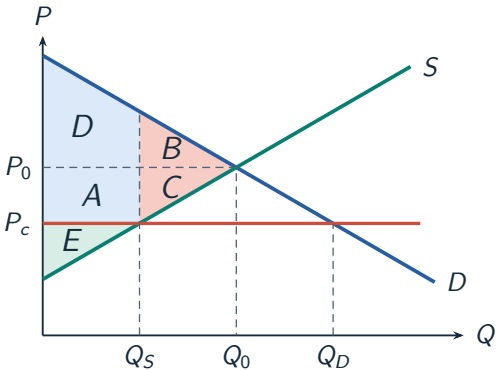
Price ceiling: surplus and deadweight loss



1. **Free-market surplus**
   Trade occurs at P_0, Q_0.
2. **The quantity traded falls**
   At P_c, only Q_S units are available.

Assumes the limited units go to buyers with the highest willingness to pay. Inefficient rationing or waiting can create additional losses.

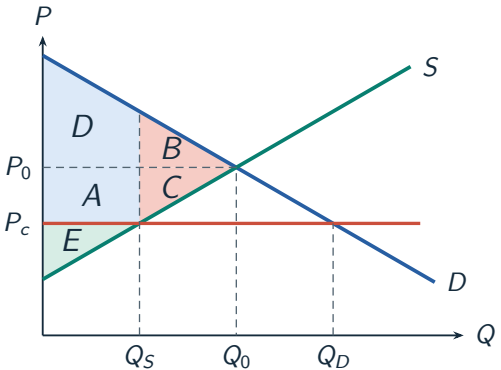
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3. **Surplus after the ceiling**
   $CS' = D + A$, $PS' = E$.
   $TS_0 = A + B + C + D + E$.
   $TS' = D + A + E$.

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Price ceiling: surplus and deadweight loss



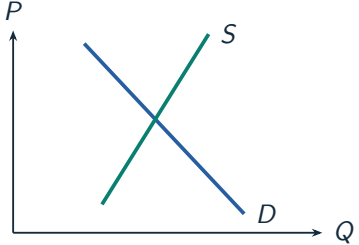
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$DWL = B + C$.

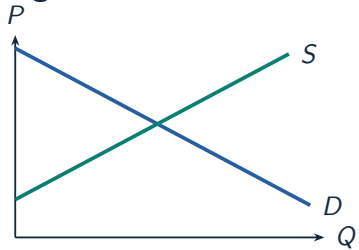
Assumes the limited units go to buyers with the highest willingness to pay. Inefficient rationing or waiting can create additional losses.

Rent control over time

Short run

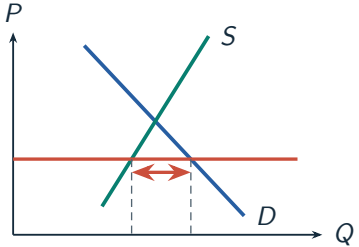


Long run

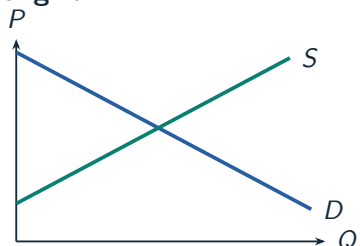


Rent control over time

Short run



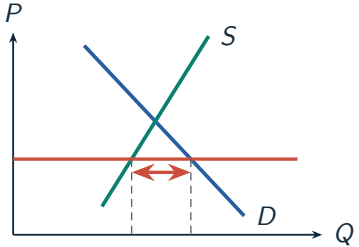
Long run



Supply and demand adjust less. The shortage is smaller.

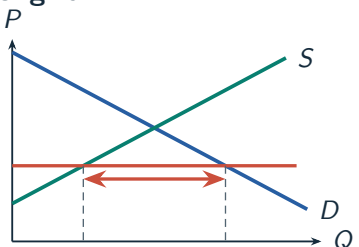
Rent control over time

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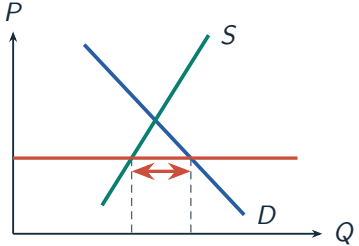
Long run



More elastic responses enlarge the shortage at the same ceiling.

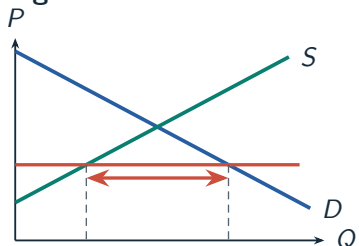
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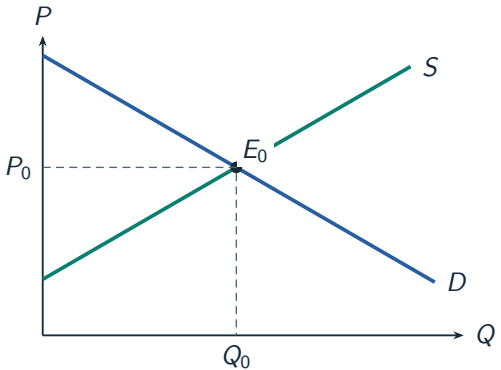
Example mechanisms include housing construction, maintenance, and households' living arrangements.

Price controls and output policies: the framework

Our route. The highlighted item is our current focus.

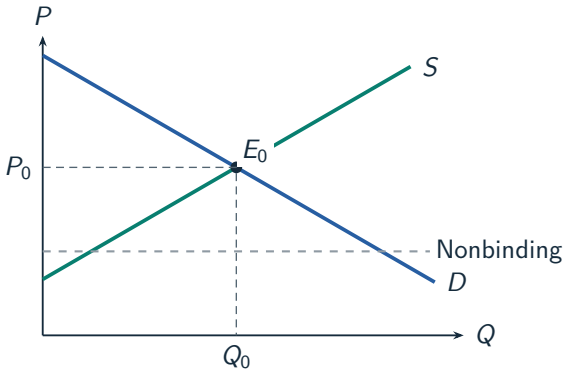
1. **Price ceilings** ✓
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3. **Assess price controls**
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4. **Other ways to maintain a higher price**
   Price supports, production quotas, and incentive programs

A price floor is a legal minimum



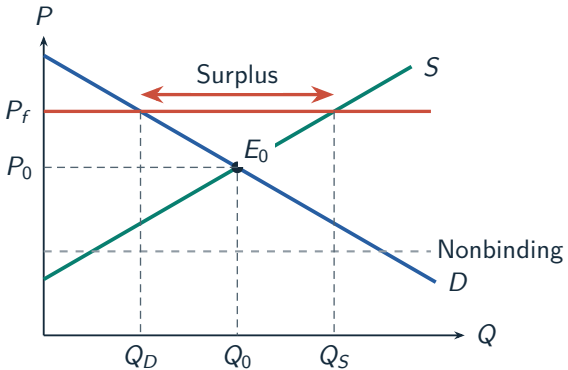
1. **Nonbinding or binding**
   A floor below P_0 is nonbinding. A floor above P_0 is binding.

A price floor is a legal minimum



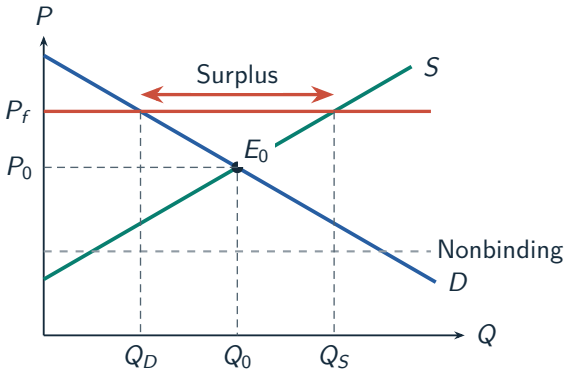
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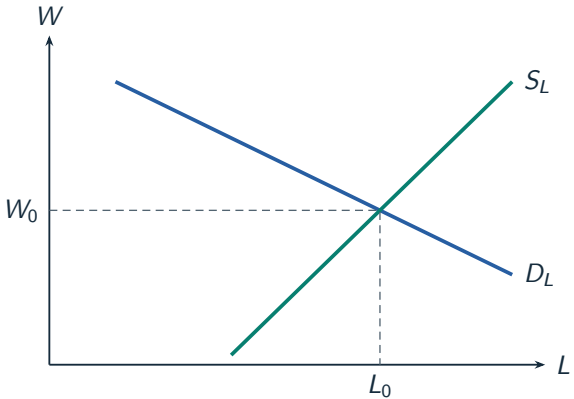
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   At $P_f > P_0$, sellers want to sell Q_S, but buyers want only Q_D.

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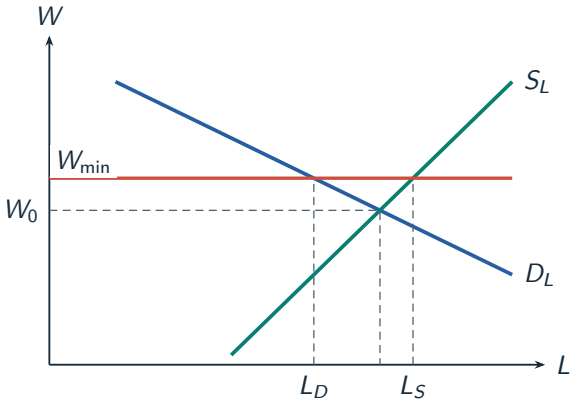
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2. **Excess supply**
   At $P_f > P_0$, sellers want to sell Q_S, but buyers want only Q_D.
3. **The labor-market translation**
   Workers supply labor. Firms demand labor. A binding minimum wage creates excess labor supply in this competitive model.

Workers supply labor, firms demand it



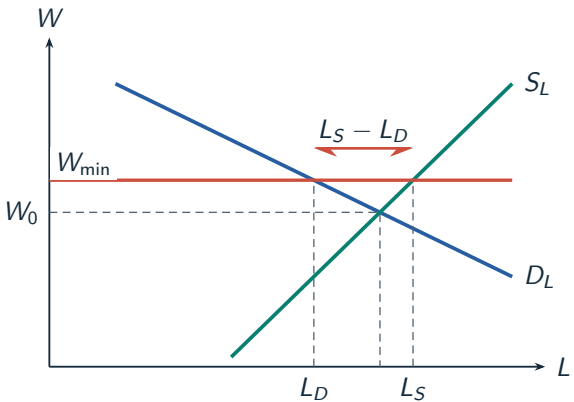
1. **The two sides of the market**
   Workers supply labor services.
   Firms demand them.
   Baseline: W_0, L_0.

Workers supply labor, firms demand it



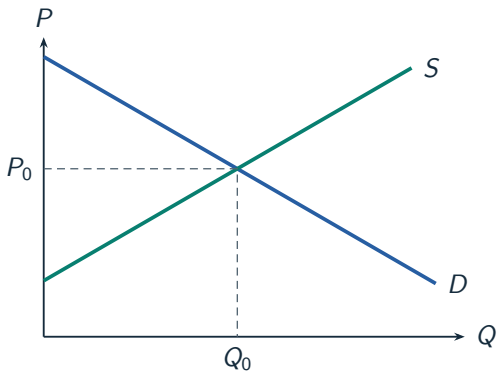
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   At $W_{\min} > W_0$, firms demand L_D, while workers supply L_S.

Workers supply labor, firms demand it



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   At $W_{\min} > W_0$, firms demand L_D, while workers supply L_S.
3. **Excess labor supply**
   $L_S - L_D$ is the unemployment created by the binding minimum wage in this competitive model.

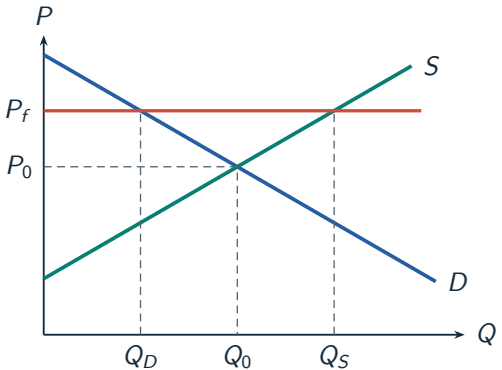
Price floor: surplus and deadweight loss



1. **Free-market surplus**
   Trade occurs at P_0, Q_0.

Assumes the lowest-cost sellers produce the units actually sold. A floor alone does not require production of all the units offered.

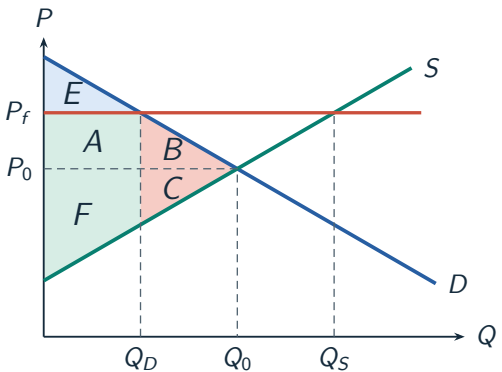
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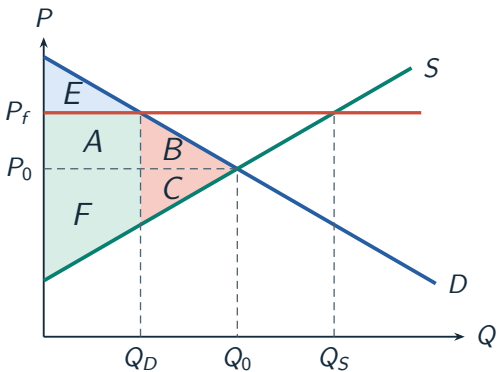
Price floor: surplus and deadweight loss



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3. **Surplus after the floor**
   $CS' = E$, $PS' = A + F$.
   $TS' = E + A + F$.
   $TS_0 = E + A + B + C + F$.

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Price floor: surplus and deadweight loss



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$DWL = B + C$.

Assumes the lowest-cost sellers produce the units actually sold. A floor alone does not require production of all the units offered.

Price controls and output policies: the framework

Our route. The highlighted item is our current focus.

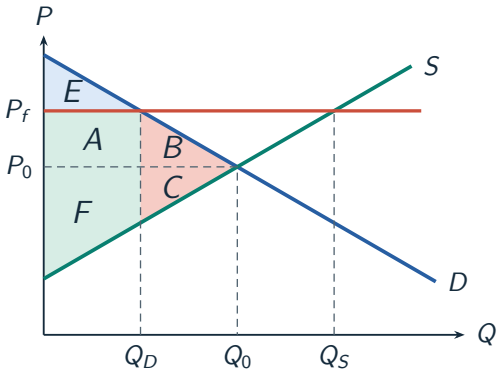
1 **Price ceilings** ✓
Nonbinding/binding, rent control, and welfare

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3 **Assess price controls** →
Objectives, price signals, rationing, and unsold output

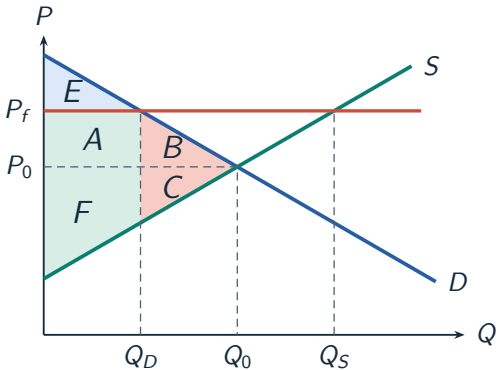
4 **Other ways to maintain a higher price**
Price supports, production quotas, and incentive programs

If sellers produce the unsold units anyway



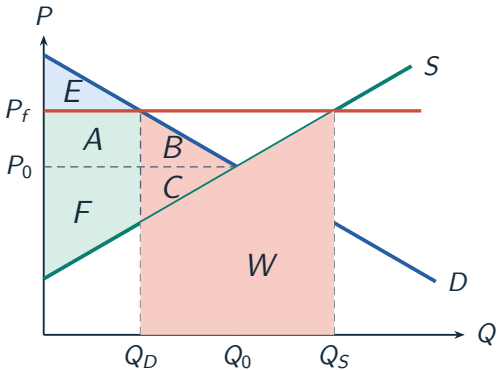
1. **The usual floor outcome**
   $CS' = E$, $PS' = A + F$. Only Q_D is produced and sold.

If sellers produce the unsold units anyway



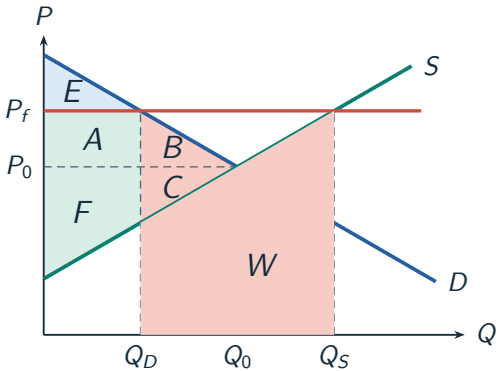
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Price controls: objectives and effects

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4. **Other policy instruments**
   Rent subsidies and wage subsidies can target assistance through different incentives. They also have fiscal costs.

Price controls and output policies: the framework

Our route. The highlighted item is our current focus.

1 **Price ceilings** ✓
Nonbinding/binding, rent control, and welfare

2 **Price floors** ✓
Nonbinding/binding, labor markets, and welfare

3 **Assess price controls** ✓
Objectives, price signals, rationing, and unsold output

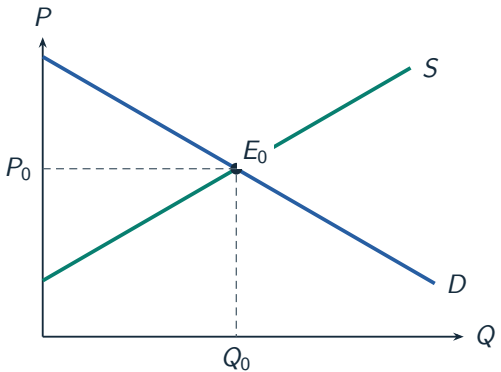
4 **Other ways to maintain a higher price** →
Price supports, production quotas, and incentive programs

Maintaining a higher price: three instruments

Our route. The highlighted item is our current focus.

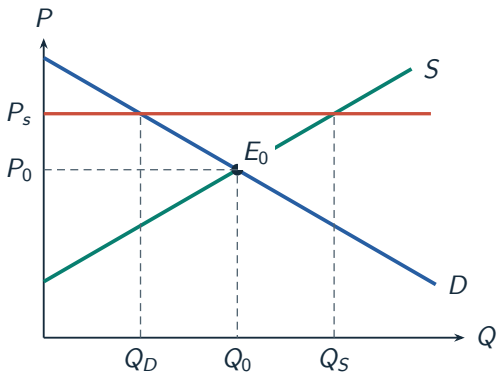
1. **Price supports** →
   Government buys the excess supply
2. **Production quotas**
   Government restricts output directly
3. **Incentive programs**
   Government pays producers to reduce output

Price supports: government purchases the surplus



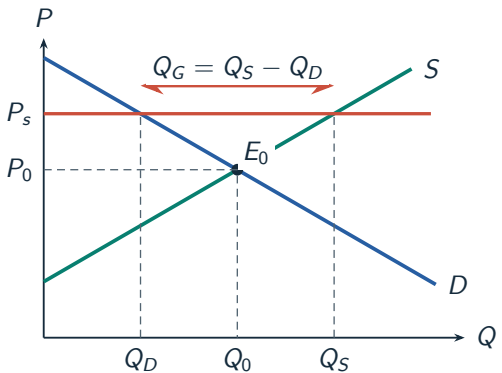
1. **The objective**
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Price supports: government purchases the surplus



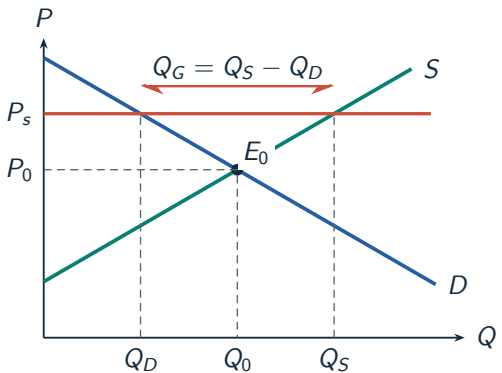
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   $Q_G = Q_S - Q_D$.
   Spending is $P_s(Q_S - Q_D)$.

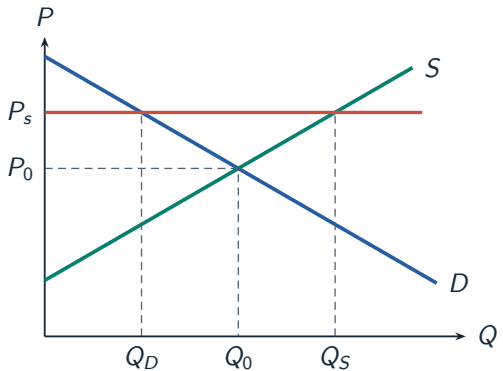
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Government spending belongs in the welfare calculation.

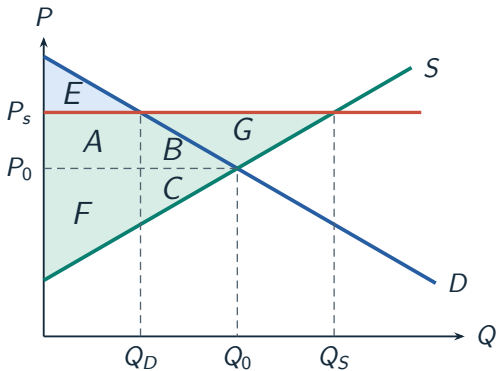
Price supports: the full welfare account



1. **The welfare boundary**
   Count domestic buyers, sellers, and taxpayers. Purchased surplus is assumed to have no social use.

The dashed rectangle is the government purchase bill. Positive public value of purchased stocks would change welfare.

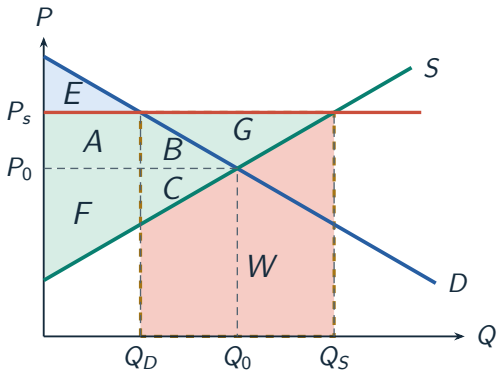
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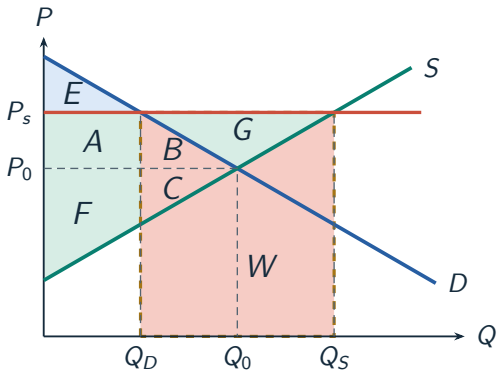
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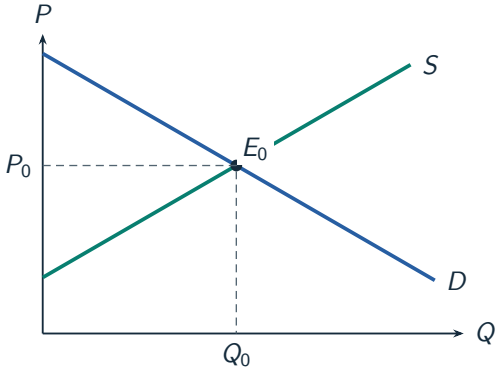
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Maintaining a higher price: three instruments

Our route. The highlighted item is our current focus.

1. **Price supports** ✓
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2. **Production quotas** →
   Government restricts output directly
3. **Incentive programs**
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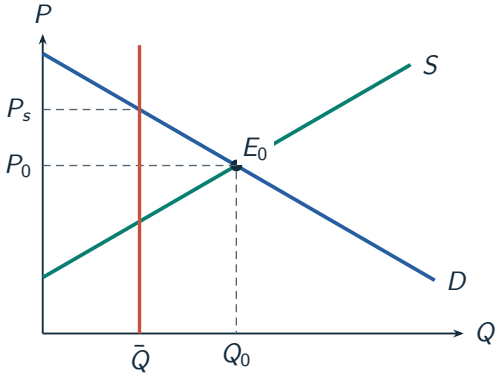
Production quotas restrict output directly



1. **The instrument**
   Restrict output directly.

Assumes lowest-cost production and domestic quota rents.

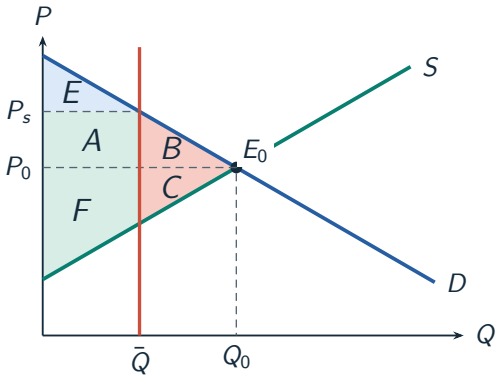
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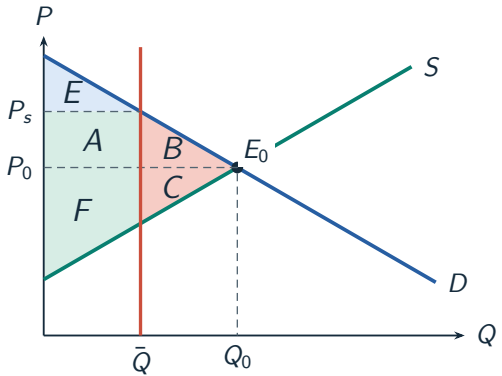
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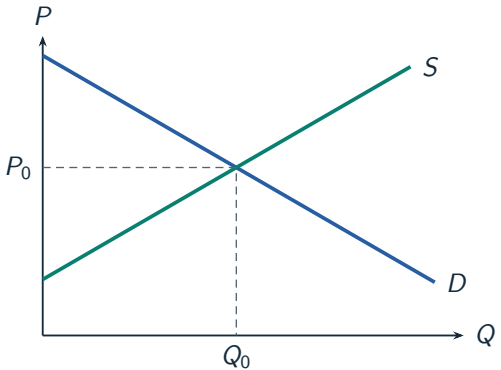
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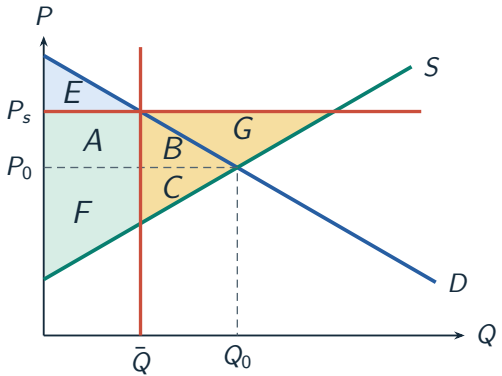
Incentive programs: reduce output



1. **The objective**
   Reduce output to $\bar{Q} < Q_0$, sustaining $P_s > P_0$.

The illustrated payment is a domestic transfer and cancels in total surplus. Example: acreage limitations.

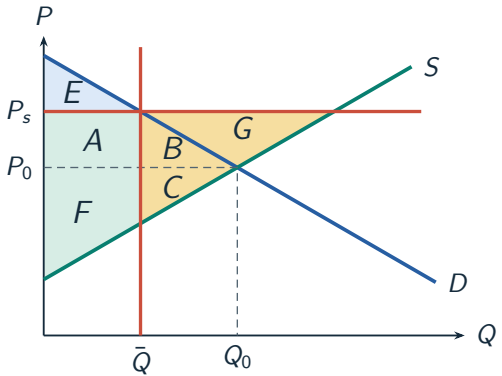
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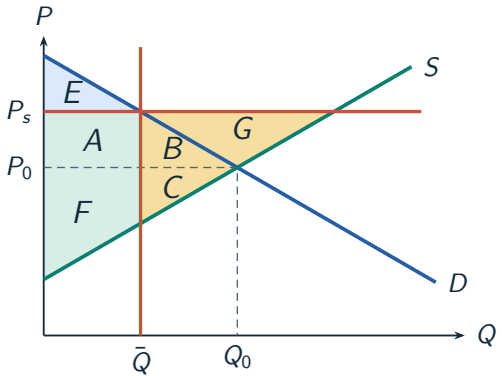
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Maintaining a higher price: three instruments

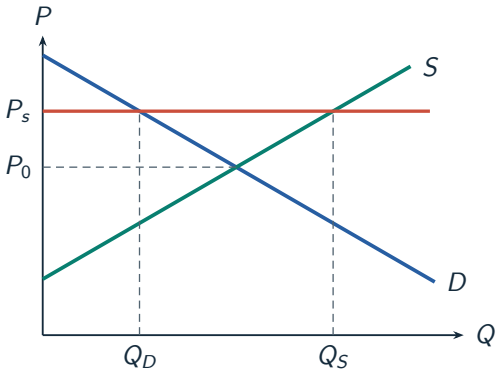
Framework complete. Connect the parts before moving on.

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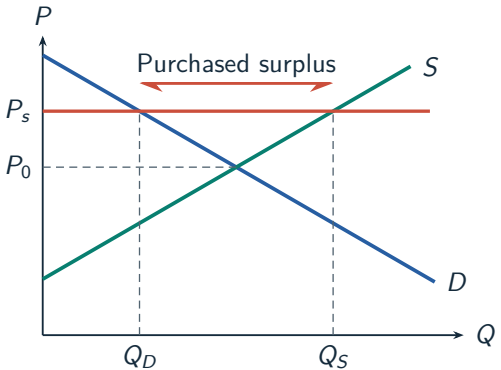
3 **Incentive programs** ✓
Government pays producers to reduce output

One target price, different allocations



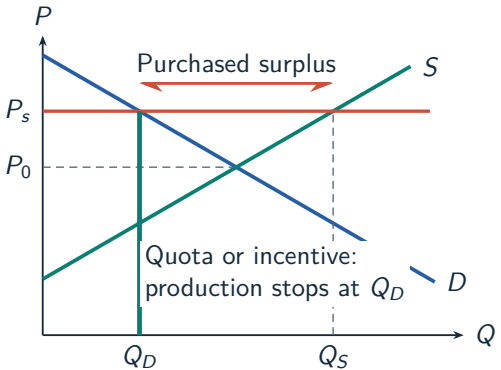
1. **Price alone is not enough**
   Count output, consumption, and the government account.

One target price, different allocations



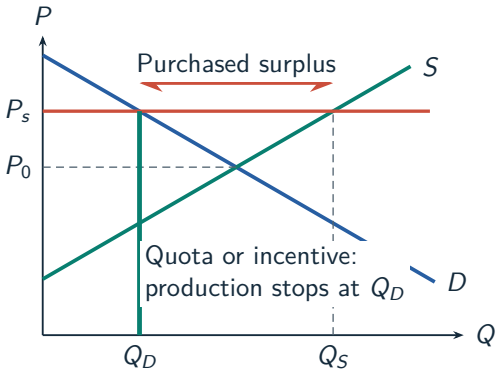
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   Produce Q_S. Government buys $Q_S - Q_D$.

One target price, different allocations



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3. **Quota or incentive program**
   Produce only Q_D. Incentive payments redistribute surplus.

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Count resource costs and fiscal transfers separately.

Price controls and output policies: the framework

Framework complete. Connect the parts before moving on.

1. **Price ceilings** ✓
   Nonbinding/binding, rent control, and welfare
2. **Price floors** ✓
   Nonbinding/binding, labor markets, and welfare
3. **Assess price controls** ✓
   Objectives, price signals, rationing, and unsold output
4. **Other ways to maintain a higher price** ✓
   Price supports, production quotas, and incentive programs

07

Taxes and subsidies

The price wedge, the division of the burden, and the complete welfare account

Taxes and subsidies: the framework

Our route. The highlighted item is our current focus.

1 **A tax wedge and total surplus** →
Buyer/seller prices, fiscal revenue, and deadweight loss

2 **Tax collected from buyers or sellers**
Shift the relevant curve and recover the two prices

3 **Elasticity and tax incidence**
Who bears the burden and why

4 **Subsidy**
The reversed price wedge and the fiscal account

Specific taxes and public revenue

1 **Specific tax**
A tax of a certain amount of money per unit sold.

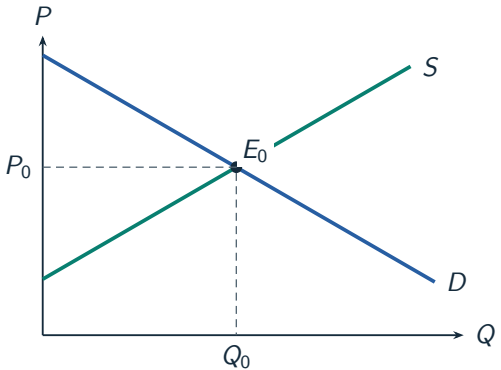
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2. **Role of tax**
   Government uses taxes to raise revenue for public projects.

Specific taxes and public revenue

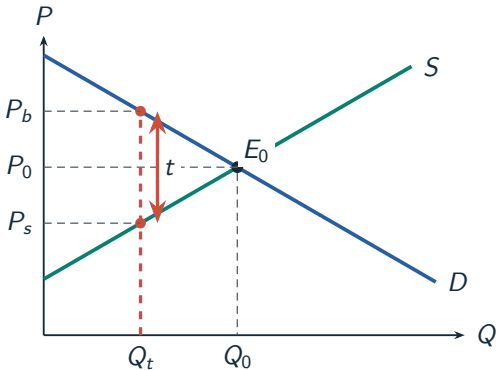
1. **Specific tax**
   A tax of a certain amount of money per unit sold.
2. **Role of tax**
   Government uses taxes to raise revenue for public projects.
3. **Examples**
   Roads, schools, and national defense.

A per-unit tax creates a price wedge



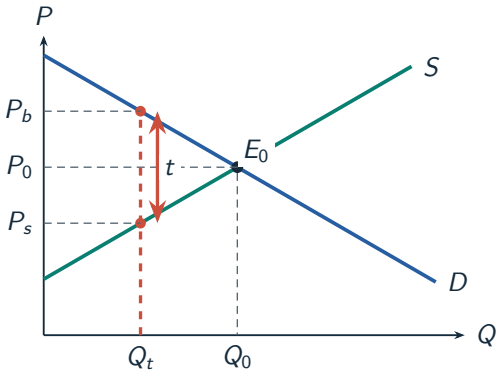
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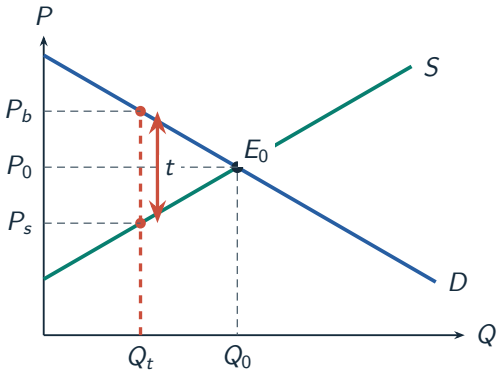
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   Buyers pay P_b. Sellers retain P_s. The difference goes to the government.

A per-unit tax creates a price wedge



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3. **The equilibrium conditions**
   $P_b - P_s = t$.
   $Q_D(P_b) = Q_S(P_s) = Q_t$.

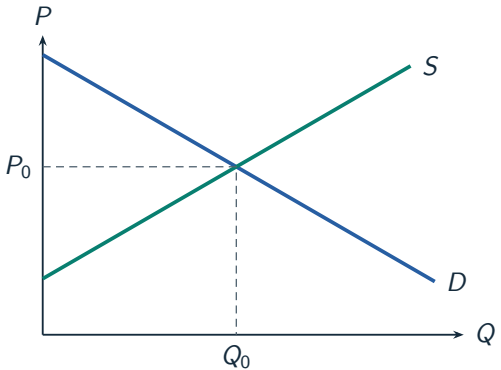
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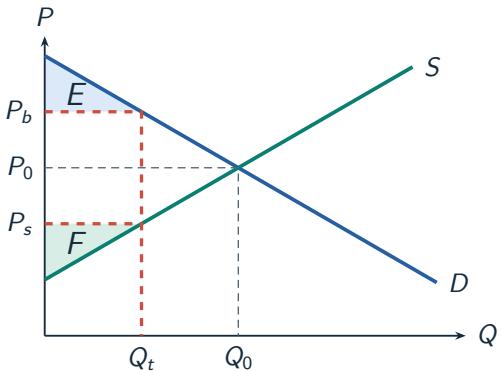
The tax reduces the quantity traded below Q_0.

Taxation and the welfare account



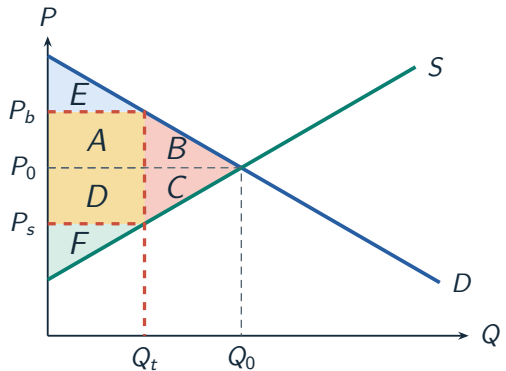
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Taxation and the welfare account



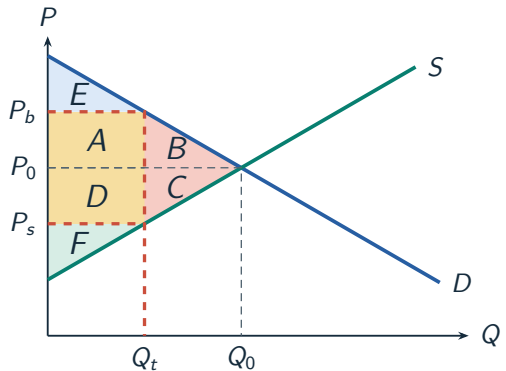
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Taxation and the welfare account



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3. **Government revenue**
   $G' = A + D = tQ_t$.
   $TS' = E + F + A + D$.

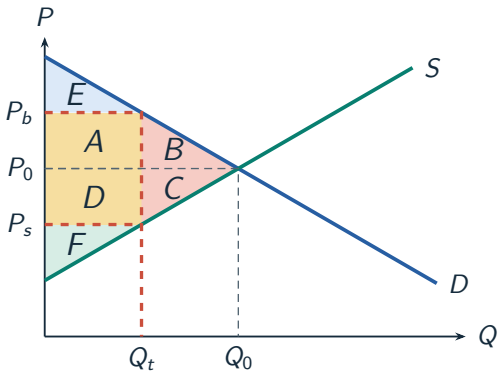
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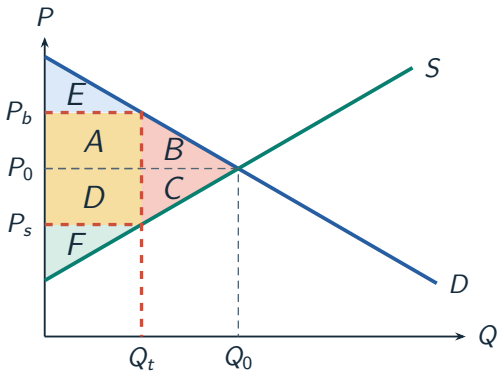
$DWL = B + C$.

Tax revenue and the loss of private surplus



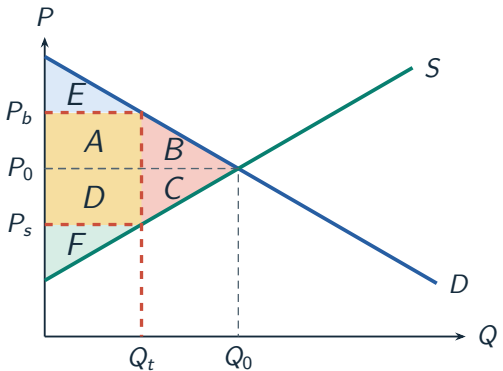
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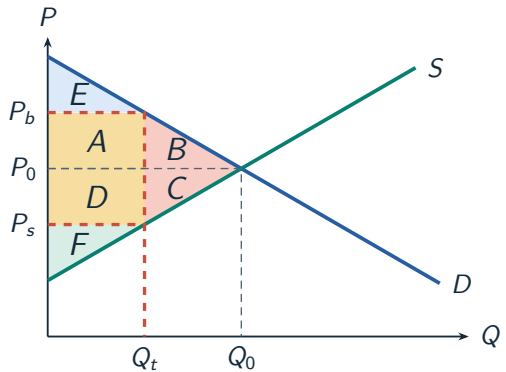
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   $G' = A + D$, retained in the domestic account.

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$DWL = (A + B) + (D + C) - (A + D)$
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Taxes and subsidies: the framework

Our route. The highlighted item is our current focus.

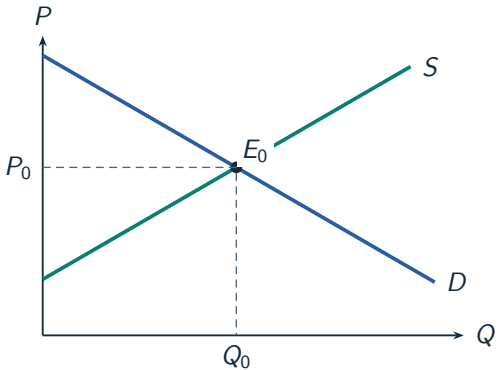
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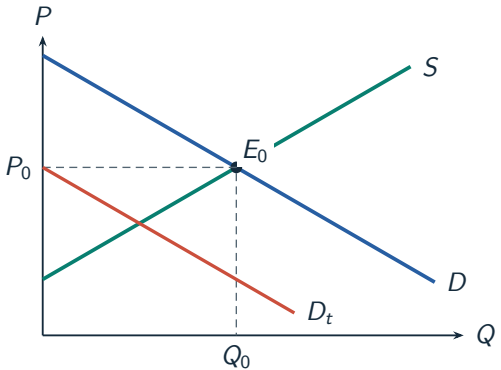
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A tax collected from buyers



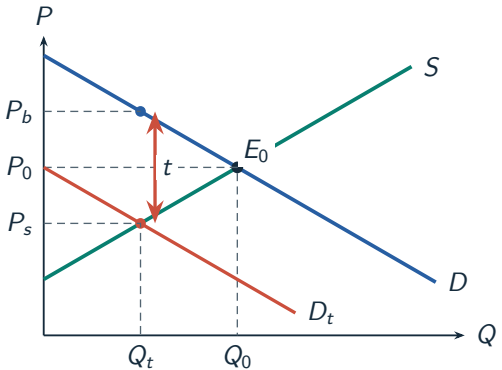
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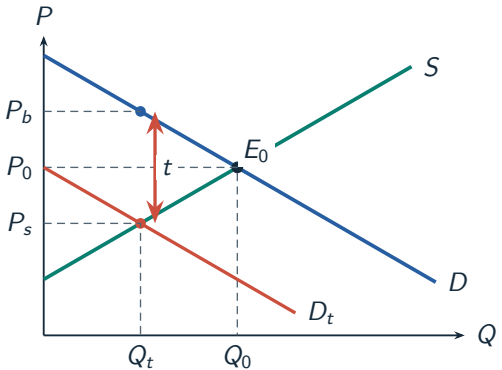
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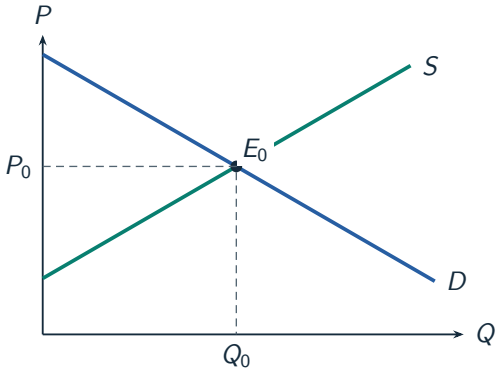
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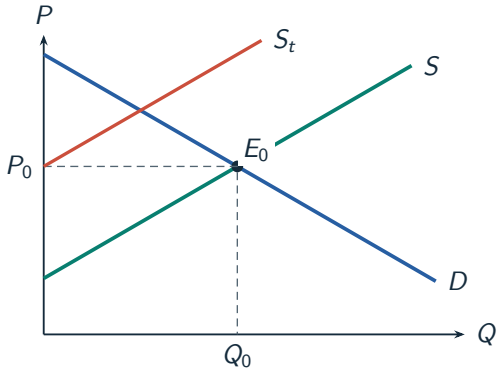
Buyers pay $P_b = P_s + t$; sellers retain P_s.

A tax collected from sellers



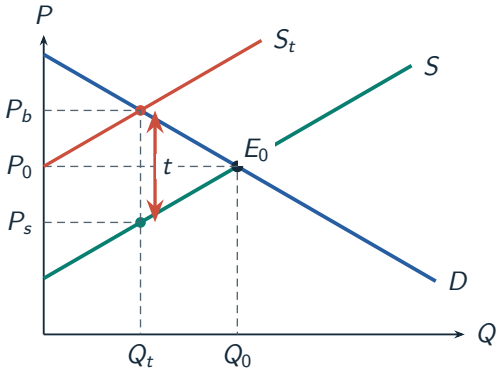
1. **The legal remitter**
   Sellers send the tax to the government.

A tax collected from sellers



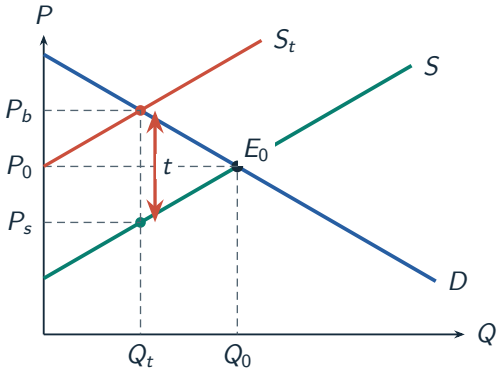
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Sellers retain $P_s = P_b - t$; buyers pay P_b.

Why statutory assignment does not change incidence

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   $P_D(Q) - t = P_S(Q)$.

Why statutory assignment does not change incidence

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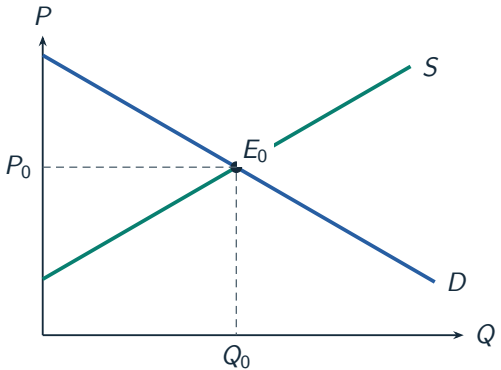
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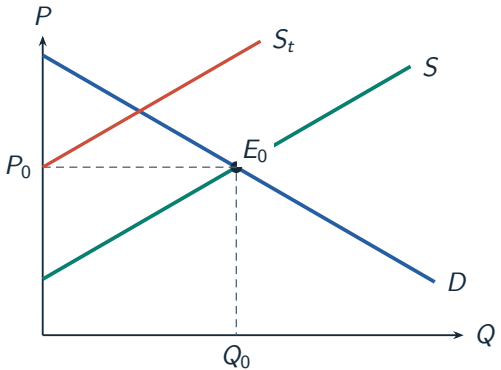
3. **The same economic condition**
   Both legal arrangements produce the same quantity and the same price wedge.
   Economic incidence depends on market responses.

Reading either tax diagram: the sequence



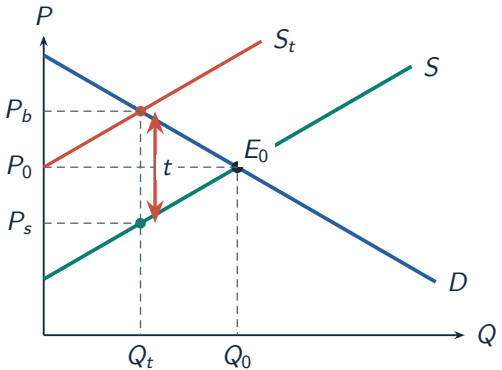
1. **Identify the legal assignment**
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Reading either tax diagram: the sequence



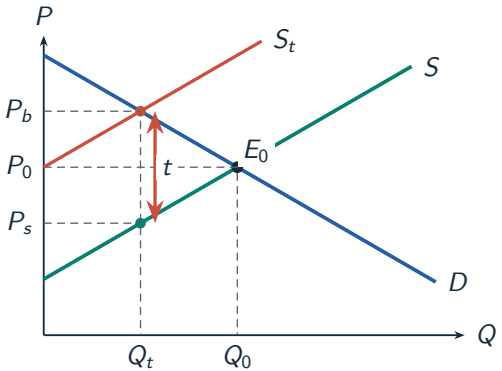
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Buyer's burden: $P_b - P_0$.
Seller's burden: $P_0 - P_s$.

Taxes and subsidies: the framework

Our route. The highlighted item is our current focus.

1 **A tax wedge and total surplus** ✓
Buyer/seller prices, fiscal revenue, and deadweight loss

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Tax incidence: legal assignment and economic burden

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The manner in which the burden of a tax is shared among participants in a market.

Tax incidence: legal assignment and economic burden

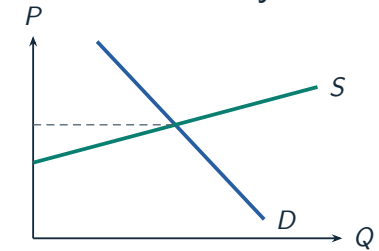
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Tax incidence: legal assignment and economic burden

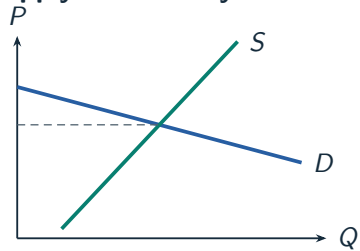
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3. **Economic incidence**
   The changes in buyer and seller prices determine who actually bears the burden. Relative elasticities determine this division.

Relative elasticity determines the tax burden

Demand is relatively inelastic

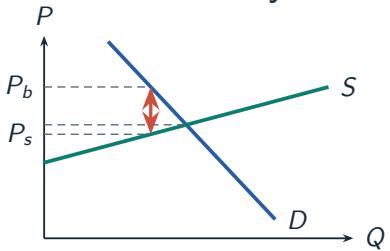


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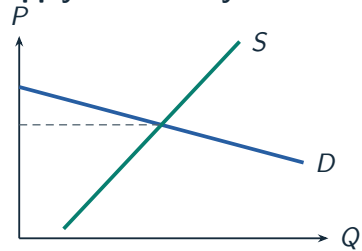


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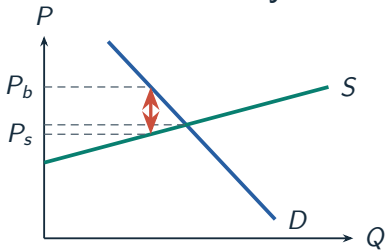
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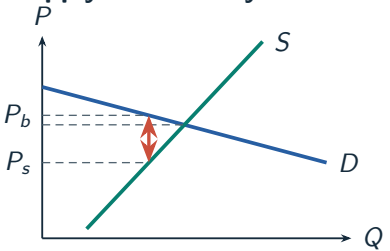
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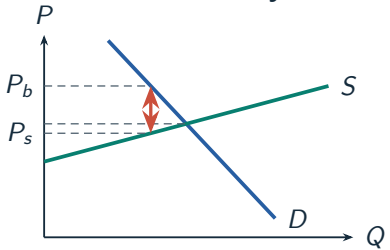
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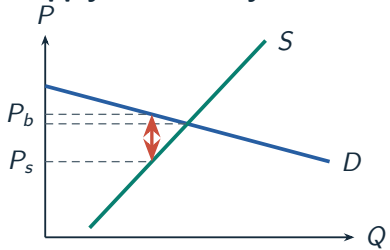
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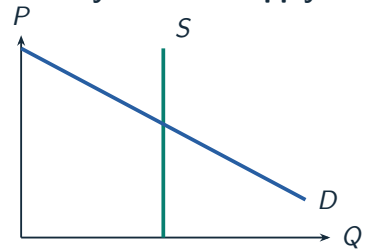


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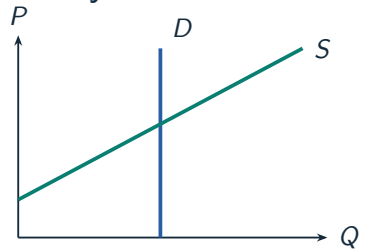
The less elastic side of the market bears the larger burden.

Tax incidence: the two inelastic limits

Perfectly inelastic supply

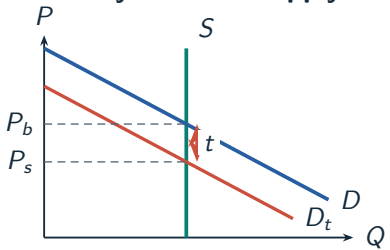


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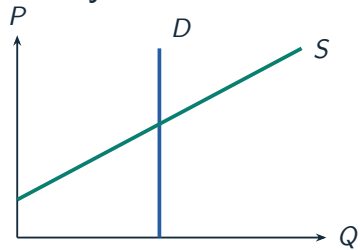
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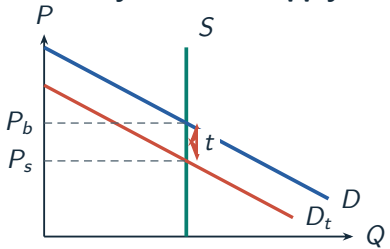
Sellers bear the entire tax. Quantity stays fixed.

Perfectly inelastic demand



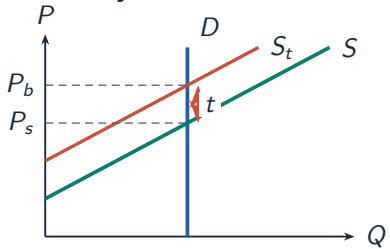
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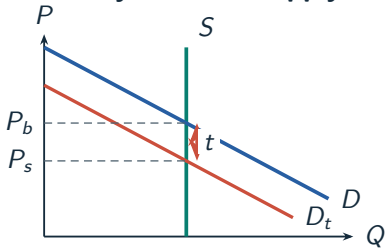
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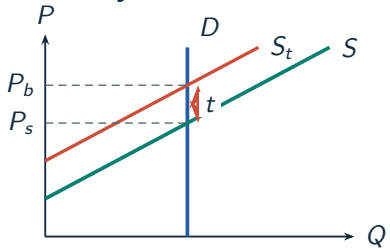
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Buyers bear the entire tax. Quantity stays fixed.

With no quantity distortion in this basic model, there is no lost-trade triangle, although the distribution of surplus changes.

Application: a luxury tax can burden producers

1. **The intended target**
   The course example discusses taxes on luxury items such as yachts. The policy aims to collect revenue from wealthy buyers.

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Workers and producers can bear much of the burden, even when the intended target is the buyer.

Taxes and subsidies: the framework

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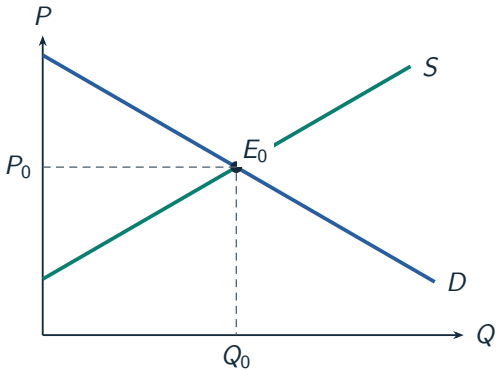
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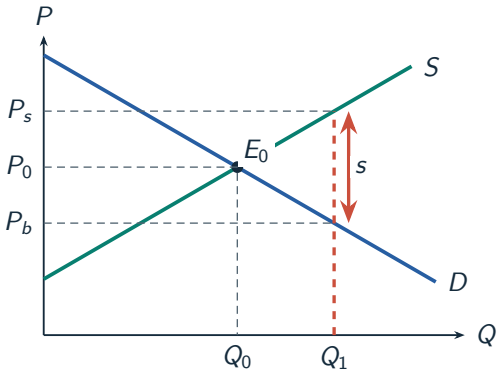
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A subsidy reverses the price wedge



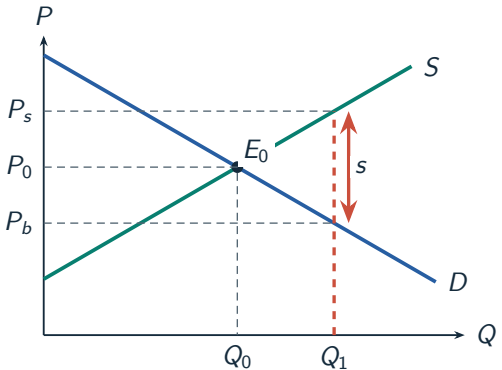
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A subsidy reverses the price wedge



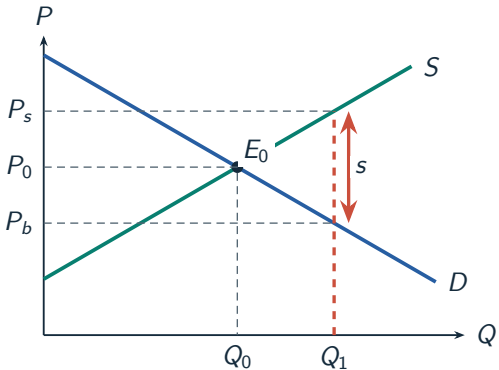
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3. **Equilibrium**
   $Q_D(P_b) = Q_S(P_s)$.
   The subsidy expands quantity above the competitive benchmark.

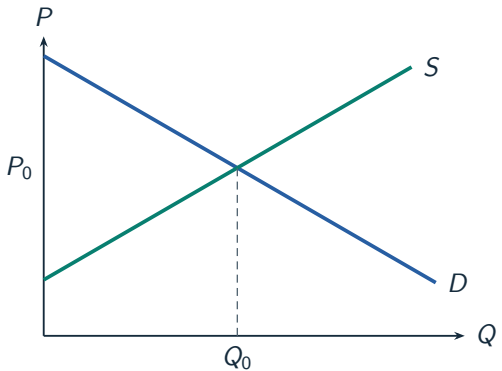
A subsidy reverses the price wedge



1. **The per-unit payment**
   The government pays a subsidy of s per unit traded.
2. **Two prices again**
   $P_s - P_b = s$.
   Sellers receive more than buyers pay.
3. **Equilibrium**
   $Q_D(P_b) = Q_S(P_s)$.
   The subsidy expands quantity above the competitive benchmark.

The subsidy benefit is split between buyers and sellers according to their relative elasticities.

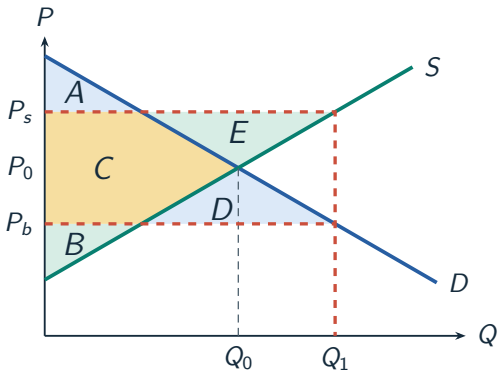
Subsidy: private gains and public spending



1. **Before the subsidy**
   The benchmark is P_0, Q_0.

Region C enters both CS and PS; the subsidy finances this overlap. Assumes no unpriced external benefit.

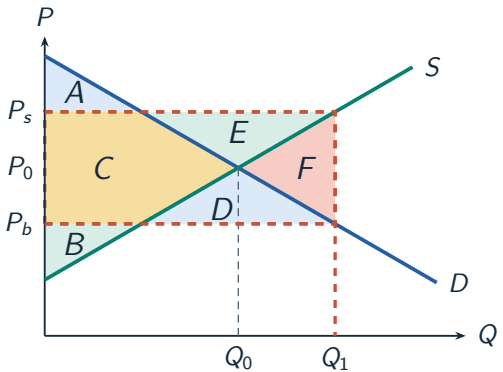
Subsidy: private gains and public spending



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   $CS' = A + C + D$.
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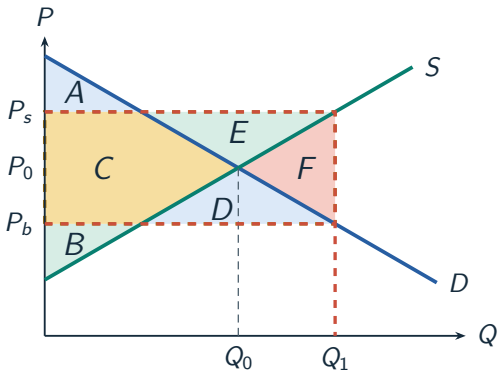
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   The dashed rectangle:
   $sQ_1 = C + D + E + F$.

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   $PS' = B + C + E$.
3. **Government spending**
   The dashed rectangle:
   $sQ_1 = C + D + E + F$.

$TS_0 = A + B + C$.
$TS' = A + B + C - F$.
$DWL = F$.

Region C enters both CS and PS; the subsidy finances this overlap. Assumes no unpriced external benefit.

Taxes and subsidies: the framework

Framework complete. Connect the parts before moving on.

1. **A tax wedge and total surplus** ✓
   Buyer/seller prices, fiscal revenue, and deadweight loss
2. **Tax collected from buyers or sellers** ✓
   Shift the relevant curve and recover the two prices
3. **Elasticity and tax incidence** ✓
   Who bears the burden and why
4. **Subsidy** ✓
   The reversed price wedge and the fiscal account

08
International trade

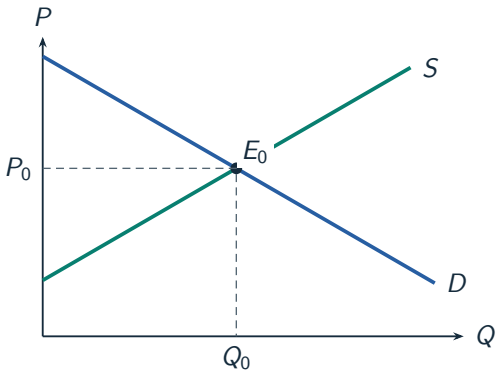
Domestic surplus, import restrictions, and the ownership of policy rents

Import quotas and tariffs: the framework

Our route. The highlighted item is our current focus.

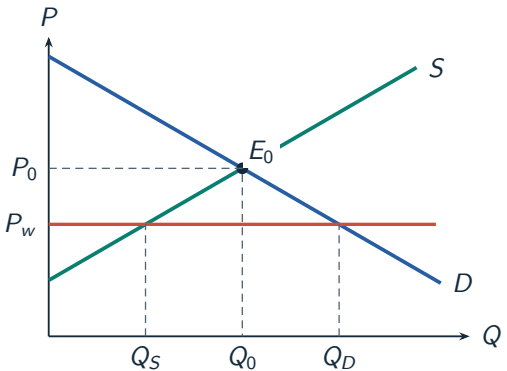
1. **Free trade** →
   Domestic consumption, production, and imports at the world price
2. **Eliminating imports**
   Transfers to producers and the lost gains from trade
3. **A tariff**
   Domestic price, imports, government revenue, and welfare
4. **An import quota**
   The quota rent and who receives it

Free trade at the world price



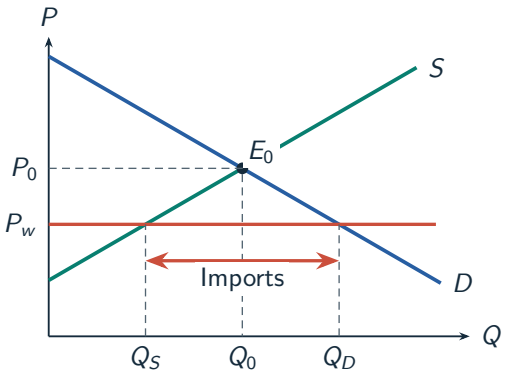
1. **Autarky**
   Without trade, domestic equilibrium is P_0, Q_0.

Free trade at the world price



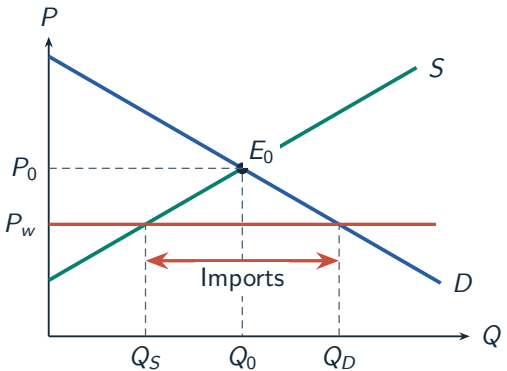
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   With free imports and $P_w < P_0$, domestic production is Q_S and consumption is Q_D.

Free trade at the world price



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3. **Imports fill the gap**
   $M = Q_D - Q_S$.

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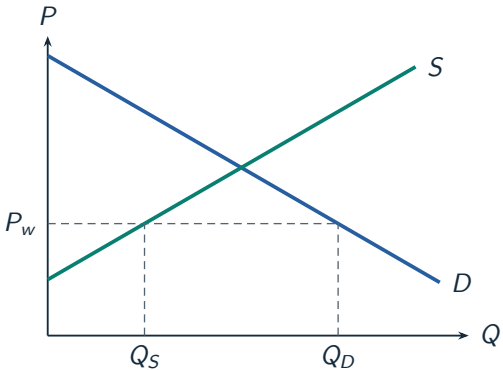
Small-country assumption: the country takes the world price as given. Count domestic welfare unless stated otherwise.

Import quotas and tariffs: the framework

Our route. The highlighted item is our current focus.

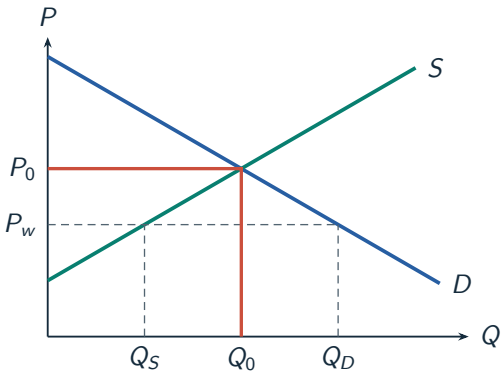
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   The quota rent and who receives it

Eliminating imports: gains and losses



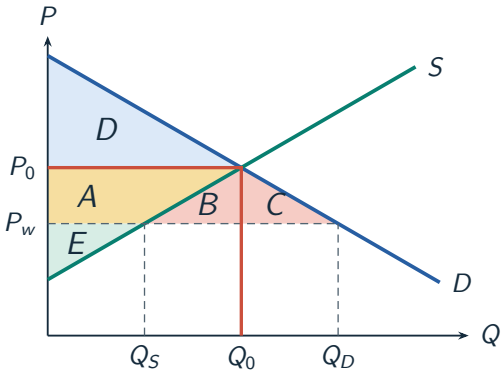
1. **Free trade**
   The domestic price equals the world price P_w.

Eliminating imports: gains and losses



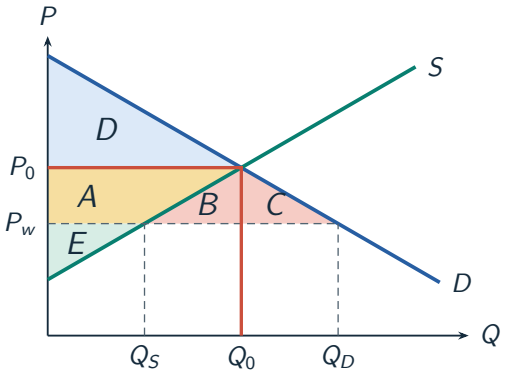
1. **Free trade**
   The domestic price equals the world price P_w.
2. **An import ban**
   Price returns to P_0. Production and consumption both equal Q_0.

Eliminating imports: gains and losses



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   The domestic price equals the world price P_w.
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   Price returns to P_0. Production and consumption both equal Q_0.
3. **Redistribution**
   $CS_0 = D + A + B + C$, $PS_0 = E$.
   $CS' = D$, $PS' = E + A$.
   Consumers lose $A + B + C$.
   Producers gain A.

Eliminating imports: gains and losses



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National $DWL = B + C$**.**

Import quotas and tariffs: the framework

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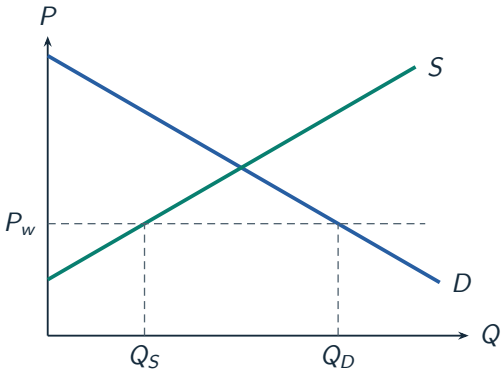
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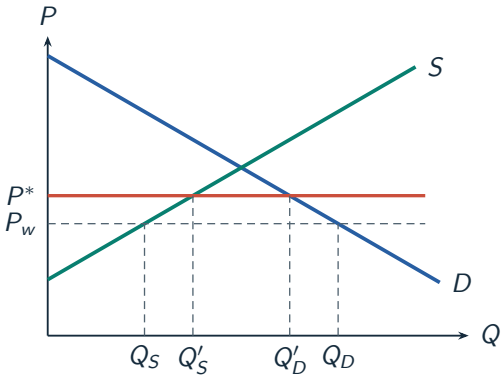
4 **An import quota**
The quota rent and who receives it

A tariff raises the domestic price



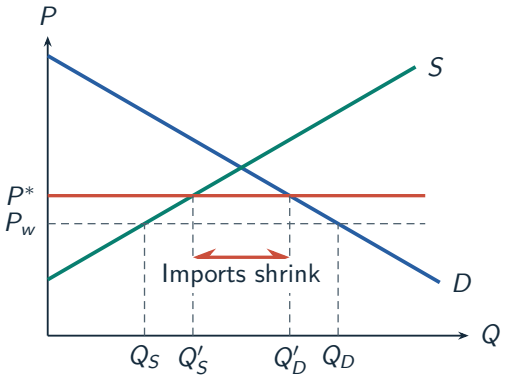
1 **A tariff**
A tax on imported goods.

A tariff raises the domestic price



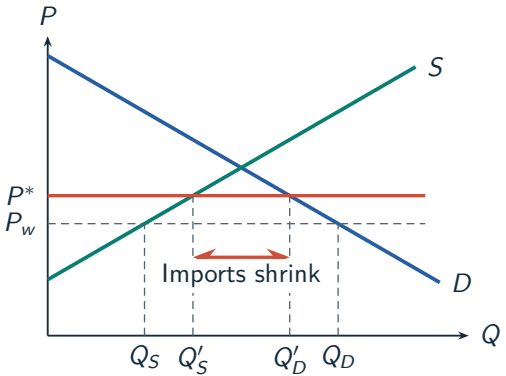
1. **A tariff**
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2. **The new price**
   While imports remain positive, $P^* = P_w + t$.

A tariff raises the domestic price



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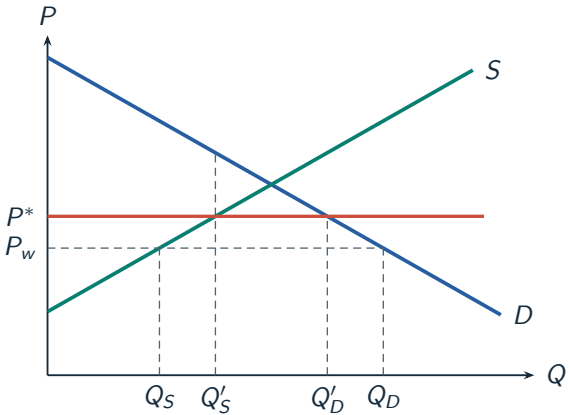
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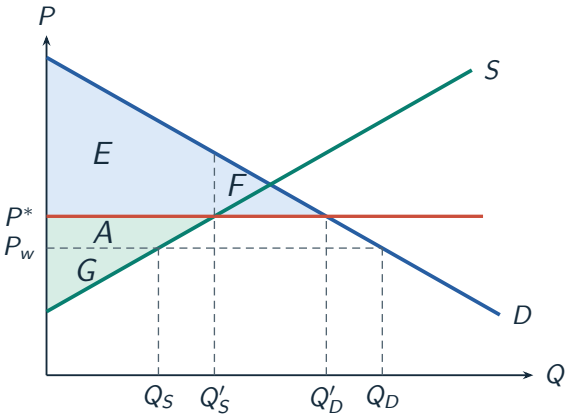
Tariff revenue is $t(Q'_D - Q'_S)$.

Tariff welfare: the board's seven regions



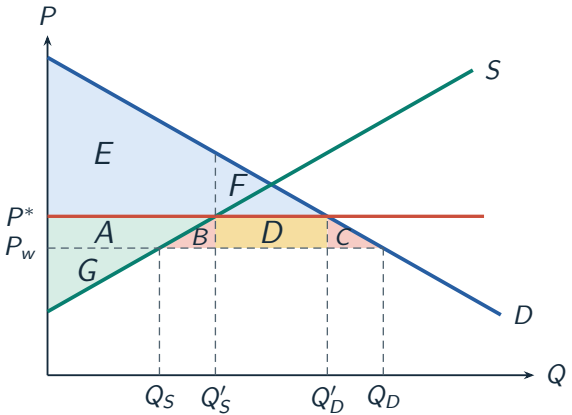
1. **Free-trade benchmark**
   Compare domestic welfare with the free-trade allocation.

Tariff welfare: the board's seven regions



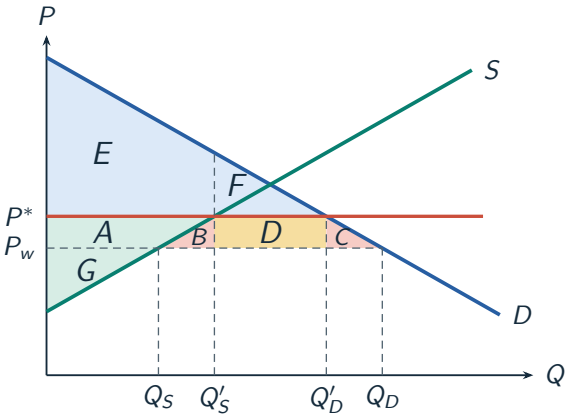
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   $CS' = E + F$.
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   $G' = D$.
   $\Delta CS = -(A + B + C + D)$.
   $TS' = A + D + E + F + G$.

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$DWL = B + C$.

B: **production distortion.**

C: **consumption distortion.**

Import quotas and tariffs: the framework

Our route. The highlighted item is our current focus.

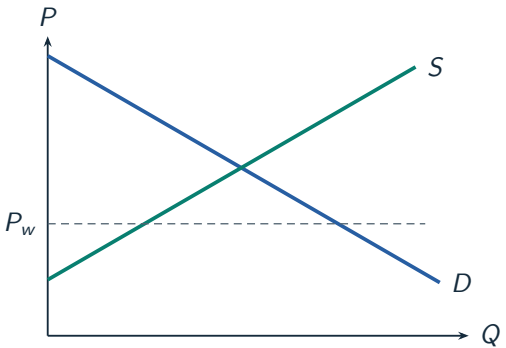
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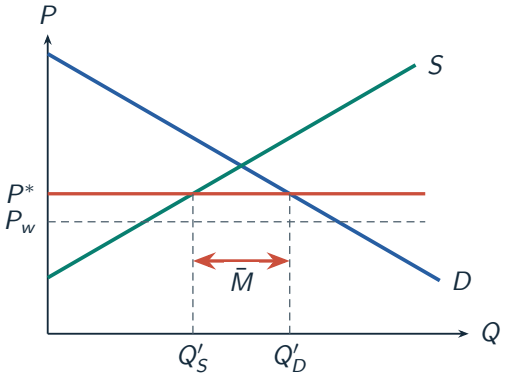
4 **An import quota** →
The quota rent and who receives it

Import quotas and the domestic price



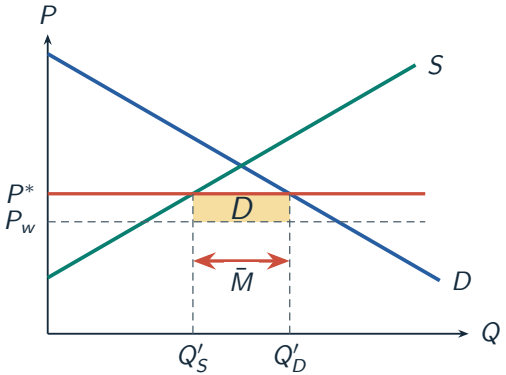
1 **An import quota**
A legal limit on imports.

Import quotas and the domestic price



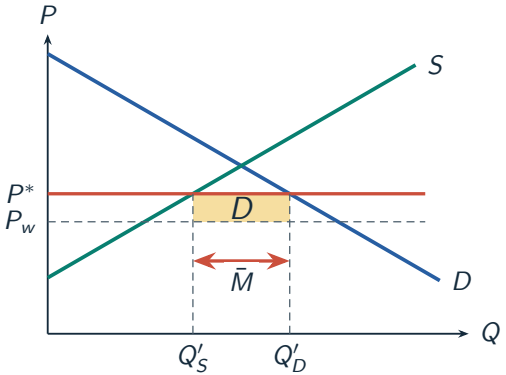
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2. **The market-clearing price**
   For an import quota $\bar{M}$:
   $Q_D(P^*) - Q_S(P^*) = \bar{M}$.

Import quotas and the domestic price



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   Buy abroad at P_w, sell at home at P^*.
   $D = (P^* - P_w)\bar{M}$.

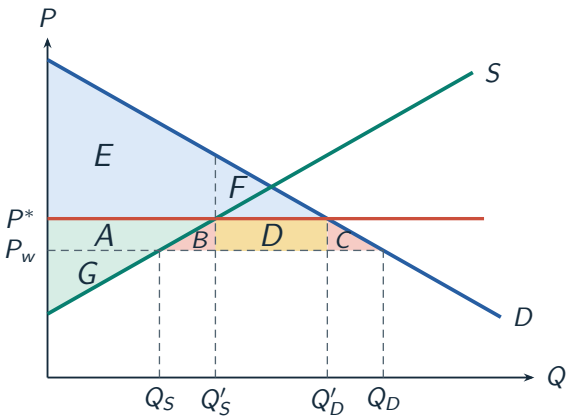
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The rent recipient determines national welfare.

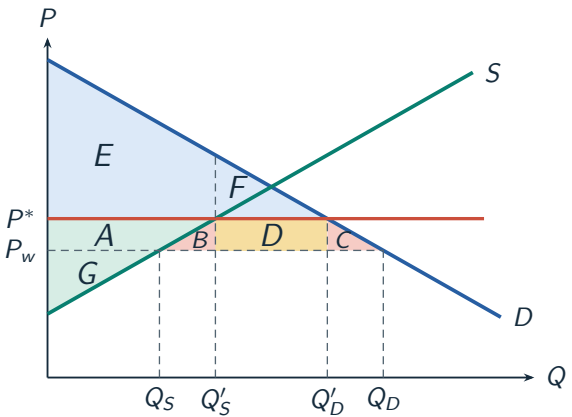
Quota rents: who receives rectangle D?



1. **The same market allocation**
   Equivalent tariff and quota: same P^*, Q'_S, Q'_D. Rent: D.

Real resources used in rent-seeking add to welfare loss. Count the relevant country's welfare.

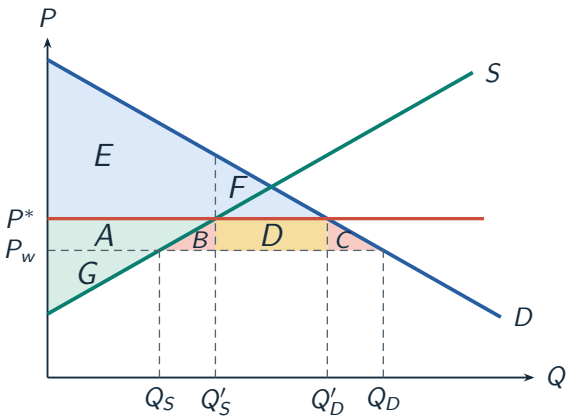
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   Government auctions or domestic license holders retain D. National loss: $B + C$.

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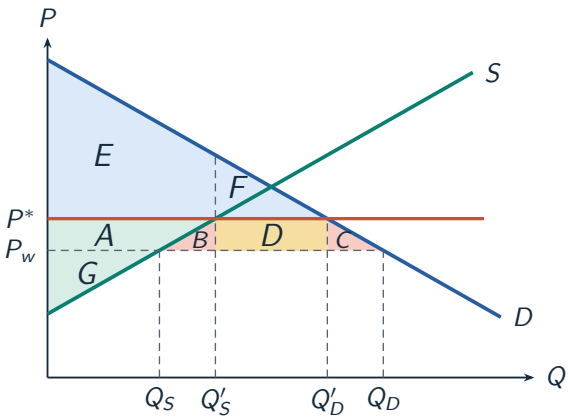
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Quota rents: who receives rectangle D?

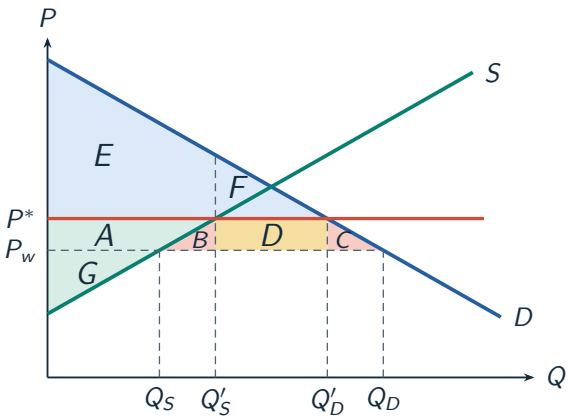


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   Foreign exporters receive D. National loss: $B + C + D$.

The owner of the rent changes national welfare.

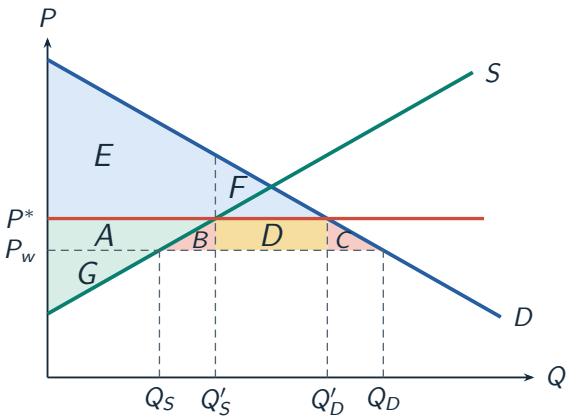
Real resources used in rent-seeking add to welfare loss. Count the relevant country's welfare.

Trade restrictions: return to the welfare account



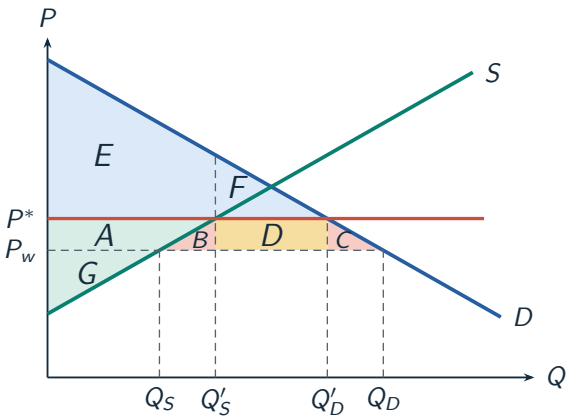
1. **Free trade**
   $TS_0 = A + B + C + D + E + F + G.$

Trade restrictions: return to the welfare account



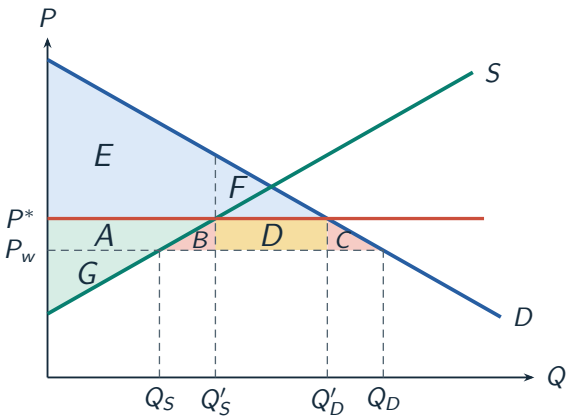
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2. **Tariff or domestically retained quota rent**
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   Loss: $B + C.$

Trade restrictions: return to the welfare account



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Separate transfers from lost gains from trade.

Import quotas and tariffs: the framework

Framework complete. Connect the parts before moving on.

1. **Free trade** ✓
   Domestic consumption, production, and imports at the world price
2. **Eliminating imports** ✓
   Transfers to producers and the lost gains from trade
3. **A tariff** ✓
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A consistent method for every policy question

1. **The original equilibrium**
   Write demand and supply, solve for P_0, Q_0, and calculate the initial surplus.

A consistent method for every policy question

1. **The original equilibrium**
   Write demand and supply, solve for P_0, Q_0, and calculate the initial surplus.
2. **The new constraint or wedge**
   State the ceiling, floor, quota, tax, subsidy, or world price. Calculate the quantities actually produced and consumed.

A consistent method for every policy question

1. **The original equilibrium**
   Write demand and supply, solve for P_0, Q_0, and calculate the initial surplus.
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4. **The comparison**
   $DWL = TS_0 - TS'$, using the relevant efficient benchmark. Check that the lettered regions reconcile across consumer, producer, and government accounts.

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4. **A quota with no auction revenue**
   Is its rent automatically a national loss? **Only if it leaves the domestic welfare boundary or is dissipated.**

Questions and discussion

A clear answer connects the assumption, the diagram, and the economic mechanism.

Victor Chen

ECO2011 Midterm Review

A
Board extensions

Optional connections to empirical work and political economy

Ceteris paribus and exogeneity

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   A demand curve varies the good's price while holding other demand determinants fixed. An equilibrium model can determine price and quantity jointly.

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2. **Endogenous and exogenous variables**
   Endogenous variables are determined within the model. Exogenous factors are taken as given in that model. The classification depends on the model.
3. **The empirical distinction**
   A regression assumption about the error term is a separate question. A diagram alone does not establish that an explanatory variable is statistically exogenous.

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2. **Zero conditional mean**
   $E[u_i \mid X_i] = 0$: the omitted determinants have mean zero at each value of the included regressors.
3. **Mean independence**
   Mean independence means $E[u_i \mid X_i] = E[u_i]$. With $E[u_i] = 0$, it gives the displayed condition. It is weaker than full statistical independence.

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   Rules can restrict participation or concentrate access to opportunities.
3. **The connection to this review**
   Property rights, incentives, and the allocation of resources depend on the institutional environment. This is a wider political-economy question.

Here, "exclusive" describes restricted participation. These concepts provide wider context for property rights and incentives.